
Telescope: Improving Zero-Shot Detection of LLM Generated Content By Measuring Token Repetition Probability

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Abstract

Distinguishing Large Language Model (LLM) generated text from human writing is a critical and difficult challenge. While LLMs are trained to write like humans, we hypothesize that this training leaves an indelible mark. LLMs develop a particularly strong aversion to token repetition very early in training. This bias persists as a “Vestigial Heuristic” (a developmental artifact) that is activated in LLM generated text, separating LLM from human writing. To probe this phenomenon, we introduce Telescope Perplexity, a metric that evaluates the token repetition of the model, $P(s_i | s_{1:i})$. Our empirical investigation reveals that the Telescope Perplexity signature emerges early in pre-training, and Telescope Perplexity effectively enables highly effective zero-shot LLM detection. We show state-of-the-art or competitive performance across diverse datasets (including modern evaluation sets we introduce), reference models, and perturbation schemes with greater efficiency than other methods.

1. Introduction

1.1. Background and Motivations

Large Language Models (LLMs) captured the attention of the general public in 2022 when OpenAI released ChatGPT. This step forward showed the world how well deep learning models could learn to understand and respond to human text (Ouyang et al., 2022). In the time since ChatGPT was released, the industry around Artificial Intelligence has been thrown into the cultural zeitgeist, while the capabilities of large language models have improved significantly. Language models have continued to become smarter (Hoffmann

et al., 2022), faster (Dao et al., 2022), and cheaper (Cai et al., 2024). They are now being used constantly for tasks from real time translation to helping explain difficult topics and problems for students. However, LLMs have also been used for deploying spam bots (Liyanage et al., 2024), disseminating fake news online (Sallami et al., 2024), and writing essays for students (Jelson et al., 2026). Humans simply cannot distinguish between LLM and human written text (Ippolito et al., 2020), so algorithmic methods are needed to help control misuses of LLMs, and ideally these methods will continue to be effective and keep pace with new model releases. For this reason, we focus on **Zero-Shot Detection**. Zero-shot methods perform without the need to continuously update their training data, making them ideal for handling the constant churn of model releases.

1.2. Core Contributions

In this work, we propose and investigate the “Vestigial Heuristics” hypothesis, which presents a novel way to view how early training dynamics can affect the final model’s behaviors. We introduce Telescope Perplexity (Eq. 1) as a simple, conceptually grounded probe specifically targeting one of these hypothesized **Vestigial Heuristics** that lead to a particular and extreme aversion to repetition. Additionally, we provide extensive testing to verify our hypothesis of the “Vestigial Heuristics” and the signature’s early emergence/ locality, which allows us to better understand how Telescope Perplexity works, where it can fail, and how it can be improved in the future. Finally, we provide extensive empirical validation demonstrating Telescope Perplexity’s state-of-the-art or competitive zero-shot detection performance and robustness across diverse scenarios, including on novel evaluation sets using contemporary LLMs (GPT4o Mini, Deepseek-V3). The code required for reproducing all of the experiments of this paper can be found in the following GitHub repository: <https://github.com/ChrisNassif/Telescope>.

2. Related Work

Techniques for detecting whether or not a piece of text is generated by some unknown “target model” (also known as a black-box setting) can largely be grouped into 2 main

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categories: zero-shot detectors or supervised classifiers.

Zero-shot detectors run inference on the text with a “reference model” and then analyze the output tokens of that reference model using some statistical technique. The goal of the statistical analysis is to try to measure some fundamental and interpretable pattern behind LLM generated text that exists for a wide variety of target language models. Additional clarification on the relationship between reference models and target models can be found in the Appendix Section 8.1.

Supervised detectors utilize some trained machine learning model to identify whether or not the text originates from an LLM. These supervised detectors can either use zero-shot detectors as a heuristic input to the model to learn patterns that signal whether a piece of text is LLM generated (Verma et al., 2024) or they can be directly finetuned from another language model to directly operate on the text without using any zero-shot heuristics (Su et al., 2024). The major limitation that these supervised detectors struggle with is that as new target models are released, they may quickly become obsolete and struggle to detect the outputs from new target LLMs. For this reason, we choose to focus this work on just zero-shot methods, and only benchmark against zero-shot methods.

Early attempts at creating a zero-shot detector for LLM generated text consisted of using the log Perplexity or loss of a reference language model to detect text generated by a target language model. As previous work by Xu et al. (2024) has shown, a higher Perplexity score from a reference model means that the piece of text is less likely to be output by the reference model. Higher Perplexity can also help indicate that it has never seen any piece of text in its training similar to the piece of text in inference time, since large language models tend to write in distributions similar to the distributions found during training. Additionally, since large language models are generally trained on very similar, wide ranging datasets, if a reference LLM is perplexed by a piece of text and has never seen anything like it before, that means that other language models likely haven’t seen anything like it in their training data. If other language models are less likely to have seen similar text, they would also be less likely to write the text (Choshen et al., 2022), and this creates a separation between human written and LLM generated text.

Alongside Perplexity based techniques, there also exist rank based techniques, such as DetectLLM LRR (Su et al., 2023), which utilize the log rank of a logits distribution. The rank of a token in a distribution is the index of a token in a logits distribution sorted by their probabilities. The token with the highest probability in a distribution has rank 1, the token with the second highest probability in a distribution has rank 2, etc. The log rank is simply the natural logarithm of the computed rank. Using token rank, Su et al. (2023) devised

a novel detector called LRR (Log Rank Ratio), which is highly accurate at distinguishing LLM generated and human written text.

Su et al. (2023) introduced a second technique named DetectLLM-NPR combining rank information and text perturbation. Perturbation based techniques perturb the text in some way through word swaps or minor rewrites, and then analyze how zero-shot statistics change as you perturb the text. DetectLLM-NPR and similar techniques like DetectGPT (Mitchell et al., 2023) are able to achieve high performance detecting LLM generated text, but at a high computational cost. DetectGPT and DetectLLM-NPR each need to inference the reference model many times to get good performance, which balloons the inference costs tremendously. We choose not to benchmark against perturbation based techniques, since the tremendous computational cost combined with our testing across a wide variety of datasets and reference models would exceed our available compute.

More recent zero-shot work avoids the perturbation cost of DetectGPT and DetectLLM-NPR. Fast-DetectGPT (Bao et al., 2024) replaces perturbation with a single sampling step, scoring text via conditional probability curvature, while DNA-GPT (Yang et al., 2024) regenerates a truncated continuation and measures its n -gram or probability divergence from the original. Both extract a strong signal from few forward passes but do not target repetition aversion specifically. Beyond statistical scoring, Ackerman & Panickssery (2025) show an instruct-tuned Llama3-8B can recognize its own writing via an identifiable residual-stream direction, consistent with our claim that LLM generated text leaves a learned, locally detectable trace. We refer the reader to Wu et al. (2025) for a broader survey. The state of the art in zero-shot black box detectors was achieved using a key refinement to Perplexity. This technique is called the Binoculars Score (Hans et al., 2024), and it attempts to normalize the Perplexity score with a Cross Perplexity score from two models, so that it works better across domains. Computing the Binoculars Score requires two language models instead of one, which doubles the compute requirements relative to simply using Perplexity; however, Hans et al. (2024) show that it is worth the extra cost since the performance gains are very significant. To the best of our knowledge, the Binoculars Score with a Falcon 7B reference model currently stands as the state-of-the-art for zero-shot detection of LLM generated text.

3. Understanding Vestigial Heuristics

It is well understood that neural networks often learn simpler, local patterns before mastering long-range dependencies (Choshen et al., 2022; Belrose et al., 2024). This suggests a compelling question:

Do optimization pressures during early LLM pre-training instill fundamental statistical biases that are never unlearned later in training?

We know from previous work by Belrose et al. (2024) that most language models start out in early training as bigram models, directly predicting the next token from the one or maybe two tokens that precede it. We hypothesize that early in training, while they are developing these bigram models, LLMs learn an important heuristic that aids them in modeling text as a bigram model: tokens very rarely repeat themselves. Therefore, these early stage bigram models will consistently have a uniquely strong aversion to token repetition as they learn that tokens very rarely repeat themselves in natural language.

A couple key results from previous work show that what these bigram models learn persists throughout later stages of training, and that most LLMs will develop the same bigram model as one another even though they are trained through different datasets and architectures (Choshen et al., 2022), (Belrose et al., 2024). This heuristic persists over time even though it is often not entirely necessary to vehemently reject token repetitions at early stages of training. We call this a **Vestigial Heuristic** because it was developed early in training to aid the initial bigram model but is not strictly needed by the model in later stages of training. Both the target and reference model will have a ‘‘Vestigial Heuristic’’ that prevents token repetition, so the text produced by the target model will activate the neurons responsible for the ‘‘Vestigial Heuristic’’ more strongly than in human text, which does not have the same ‘‘Vestigial Heuristic’’.

To investigate this hypothesis and its power in separating human written and LLM generated text, we propose Telescope Perplexity. While the formulation is strikingly simple, we argue its power lies in probing the specific hypothesized aversion to repetition that forms very early in training.

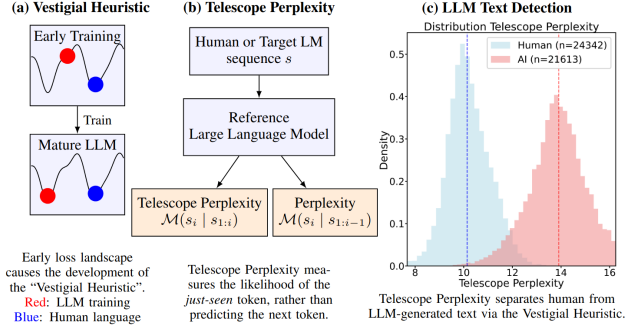


Figure 1. An overview of the ‘‘Vestigial Heuristic’’ hypothesis and measuring the average token repetition probability for the Telescope Perplexity.

Telescope Perplexity measures the reference language model’s learned likelihood of outputting the last token it processed given its context up to that point. For a language model $\mathcal{M}$ and a token sequence $\vec{s} = (s_1, \dots, s_L)$ of length L , the Telescope Perplexity is defined as:

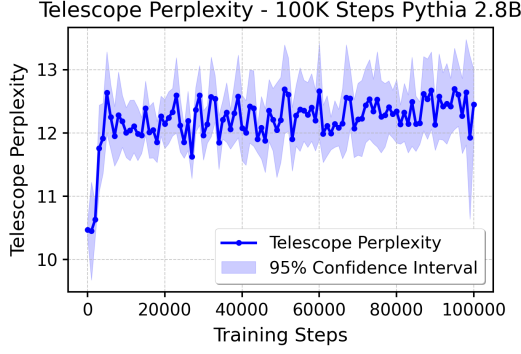
$$\text{Telescope PPL}_{\mathcal{M}}(\vec{s}) = -\frac{1}{L} \sum_{i=1}^L \log \mathcal{M}(s_i | s_{1:i}) \quad (1)$$

$\mathcal{M}(s_i | s_{1:i})$ is the probability assigned by the model $\mathcal{M}$ to token s_i given the sequence $s_1 \dots s_i$. This provides a targeted look into the model’s token repetition likelihood, whereas standard Perplexity works by predicting the *next* token ($\mathcal{M}(s_i | s_{1:i-1})$). Figure 1 provides a conceptual overview of the Telescope Perplexity. Please note that the Telescope Perplexity deviates from standard Perplexity due to the fact that the negative log likelihood is not exponentiated; however, this is more efficient to compute and does not affect any detection results with a fixed threshold since exponentiation is a monotonic function.

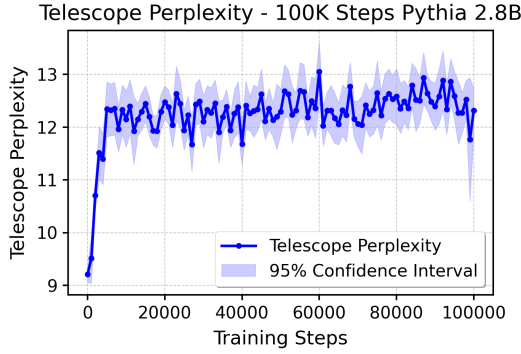
3.1. Training Dynamics

If the statistical biases Telescope Perplexity detects are indeed ‘‘Vestigial Heuristics’’ established early in training, we would expect the signature to emerge relatively quickly and remain stable throughout the later stages of pre-training. To investigate this, we analyzed the Telescope Perplexity of generated text using checkpoints from different stages of LLM pre-training for Amber-7B (Liu et al., 2023) and Pythia (Biderman et al., 2023). Figure 2 shows the evolution of Telescope Perplexity over text generated using the training checkpoints of Pythia-2.8B.

The results in Figure 2 reveal a key characteristic: the Telescope Perplexity rises sharply early in training and then largely plateaus, indicating stability through later training stages. Similar stabilization was observed across various model sizes and architectures in Appendix 8.13. This early



(a) Final Pythia checkpoint as reference model



(b) SmolLM 360M as reference model

Figure 2. Telescope Perplexity evaluated on text generated by Pythia-2.8B model checkpoints throughout training. Note the early stabilization of the Telescope Perplexity.

emergence and subsequent persistence strongly support the hypothesis that Telescope Perplexity captures a “Vestigial” characteristic established during the foundational learning phase, rather than a property that evolves continuously with model capability.

3.2. Signature Locality

The “Vestigial Heuristics” hypothesis posits that the aversion to repetition that we are probing relates to local, statistical patterns learned early in training. If Telescope Perplexity primarily captures behaviors from an early stage bigram model, its effectiveness should largely persist even when the reference model is provided with very limited context. We tested this by computing Telescope Perplexity and standard Perplexity using only the preceding one (bigram context) or two (trigram context) tokens, compared to using the full preceding context available. Table 1 presents the detection performance (AUC) on the HC3 dataset under these conditions.

Table 1 reveals that Telescope Perplexity maintains a remarkably high AUC (0.897) even with purely bigram context, significantly outperforming standard Perplexity under the

same constraint (0.761 AUC). This strong performance with extremely limited context confirms that the signature Telescope Perplexity measures is indeed fundamentally rooted in *local* token relationships, aligning with the hypothesis that it reflects simple pattern-formation biases learned early in training.

Table 1. Detection performance (AUC) on the HC3 dataset using Telescope Perplexity and standard Perplexity with full context versus limited (Bigram, Trigram) context provided to the reference model (SmolLM 360M). Confidence Intervals (95%) are shown in parentheses. Results demonstrate Telescope Perplexity’s strong reliance on local information.

Method	Context	AUC (95% Confidence Interval)
Telescope	Full	0.995 (0.993-0.996)
Telescope	Bigram	0.897 (0.893-0.905)
Telescope	Trigram	0.921 (0.915-0.926)
Perplexity	Full	0.991 (0.990-0.993)
Perplexity	Bigram	0.761 (0.753-0.770)
Perplexity	Trigram	0.925 (0.915-0.926)

3.3. Generality of the Phenomenon

Is the “Vestigial Heuristic” detected by Telescope Perplexity an idiosyncrasy of specific models or a more general artifact? Our experiments provide two lines of evidence for generality. First, as detailed in Section 5 (Table 2), Telescope Perplexity demonstrates strong detection performance when using a wide variety of reference models, spanning different architectures (Gemma, Llama, Falcon, SmolLM, etc) and parameter counts (135M to 9B). This suggests the underlying signature is not confined to one model family.

Second, using the SmolLM 360M model and its distinct training corpus components FineWeb (human written data) and Cosmopedia V2 (synthetic LLM data) (Ben Allal et al., 2024), we found that Telescope Perplexity could distinguish text originating from these different internal training sources with very high accuracy (AUC 0.996, F1 0.987). See Appendix 8.3 for additional details. This confirms that models **do** internalize fine-grained statistical properties related to their training experience, lending credence to the idea that persistent, detectable artifacts like “Vestigial Heuristics” can indeed be generally learned. Importantly, this demonstrates that we are not measuring the text’s similarity to the training data, but a property learned in training!

Taken together, these results suggest that the persistent local biases probed by Telescope Perplexity are likely a relatively general characteristic associated with current deep learning approaches to language modeling.

4. Experimental Setup

We formulate LLM generated text detection as a **binary classification problem**: given a passage $\vec{s}$, decide whether it was produced by a large language model or written by a human. Each detector we study outputs a single real-valued score $f(\vec{s})$ that is intended to be larger (or smaller, depending on sign convention) on LLM generated text than on human text. A classifier is obtained by thresholding: predict “LLM generated” iff $f(\vec{s}) > \tau$ for some threshold τ , and “human written” otherwise.

This framing motivates the two complementary metrics we report in Section 5. **AUROC** measures the detector’s ability to rank LLM generated text above human written text *without committing to any particular τ* ; it is the most informative single summary of a score’s discriminative power and lets us compare detectors without considering how the threshold is set. **F1-Score**, by contrast, requires a concrete τ and so reflects classification performance at an operating point. We report (i) the maximum F1 achievable on each test set, which characterises the best-case behaviour of the score, and (ii) a *Transferability F1* that fixes τ from a logistic regression fit on all *other* datasets, so as to characterise how well a threshold chosen without access to the target distribution generalises. Together these answer two practically relevant questions: “how separable are the two classes under this score?” (AUROC) and “how well does a threshold calibrated elsewhere transfer to a new domain?” (Transferability F1).

To empirically validate the effectiveness of Telescope Perplexity in detecting the hypothesized “Vestigial Heuristics” and compare it against existing methods, we designed a comprehensive experimental setup covering diverse datasets, contemporary language models, and evaluation procedures.

4.1. Datasets

We gathered an array of reputable datasets that distinguish between AI-generated text and human-generated text. These datasets include HC3 (Guo et al., 2023), HC3 Plus (Su et al., 2024), a popular Kaggle competition dataset named LLM - Detect AI Generated Text (King et al., 2023), another, more difficult, Kaggle dataset named AI Vs Human Text (Gerami, 2024), and the set of Ghostbusters paper datasets (Verma et al., 2024), whose names will be hereafter preceded by “GB” to identify them easily.

4.1.1. NOVEL DATASETS WITH STRONGER TARGET MODELS

One of the key limitations of the datasets in the literature is that they are generated using rather old language models, which are not commonly in use today; some have even been deprecated. This makes many of the publicly available

datasets somewhat unrepresentative for modern LLM detection applications. Therefore, in addition to our existing suite of datasets, we also generated several novel datasets using the GPT4o Mini (OpenAI et al., 2024) and Deepseek V3 models (DeepSeek-AI et al., 2025). These models are regarded as highly capable by the research community, placing near or at the top in a variety of benchmarks, and importantly, the free tiers for both GPT4o Mini and Deepseek are quite generous for users and allow anyone to access them easily. Many of the datasets that we used did not provide the prompts given to the language models to generate the data; however, the essays and creative writing portions of the Ghostbusters dataset did contain the prompts used in their creation. Using these prompts, we generated the GB Essays GPT4o Mini, GB Creative Writing Deepseek, GB Creative Writing GPT4o Mini, and GB Essays Deepseek datasets.

4.1.2. ENGLISH AS A SECOND LANGUAGE TEXT

To evaluate our model’s robustness to ESL (English as a Second Language) text, which is a known failure mode for AI-detection schemes (Liang et al., 2023), we introduce the ESL GPT4o Mini dataset. The human responses for this dataset are adapted from an existing corpus of ESL student essays (Franklin et al., 2022). We then tasked GPT4o Mini to rewrite the essays to improve clarity and structure. Rewriting is a difficult environment for AI detection models since entire sentences and ideas are often re-used from the human written text, possibly obfuscating the statistical fingerprints.

4.1.3. ADVERSARIAL PERTURBATIONS

A student asking an LLM to write their essay may submit a rephrased version of an LLM’s text. We wish to evaluate each detector’s robustness to such adversarial attacks, using the “perturbations” datasets by Verma et al. (2024), which contains attacks that can range from rephrasing a word or sentence to changing the ordering of paragraphs. This dataset also contains varying levels of each perturbation, which allows us to test how each detector performs with varying levels of perturbations. We additionally test each method with an adversarial “humanizer” system described in Section 8.18.

4.2. Detection Methods and Reference Models

We evaluate our proposed Telescope Perplexity method (Eq. 1) against several established zero-shot baselines representing diverse approaches. These include standard Perplexity based on next-token prediction ($P(s_i | s_{1:i-1})$), the rank-based DetectLLM LRR, which utilizes log rank (Su et al., 2023), Binoculars, which utilizes Cross-Perplexity between two models (Hans et al., 2024), and Fast-DetectGPT,

which uses conditional probability curvature (Bao et al., 2024). To assess the generality and robustness of detection methods with respect to the underlying inference engine, we employed a diverse set of 12 reference models varying in size, architecture, and origin: Gemma 2 2B, Gemma 2 9B, Llama 3.1 8B, Falcon 7B, SmoLLM 135M, SmoLLM 360M, SmoLLM 1.7B, SmoLLM2 135M, SmoLLM2 360M, SmoLLM2 1.7B, GPT-J 6B, GPT-Neo 2.7B. When a single reference model is needed, we use the instruct variant, and in Binoculars, which requires two reference models, we use the instruct and pretrained variant.

4.3. Evaluation Metrics and Procedure

We primarily evaluate detection performance using the Area Under the Receiver Operating Curve (AUROC) score. We also report the maximum F1-Score achievable by finding the optimal threshold on the test set itself. To assess practical threshold robustness across domains, we employ a Transferability Test: for each target dataset, an optimal threshold is determined via logistic regression trained on scores from all *other* datasets, and the resulting F1-Score on the held out target dataset is reported. Unless otherwise stated, reported AUROC and Transferability F1-Scores are averaged across all 12 reference models. We do not include true positive rate at very low false positive rates due to the fact that the confidence intervals are often inconclusive from these metrics when using a small false positive rate value like 0.05%. See Appendix Section 8.12 for more information. Larger fixed false positive rate values such as 5% offer more substantial confidence intervals, but are limited in how useful they actually are, since LLM content detectors that flag content as LLM generated when they are not 5% of the time are generally not very useful. For this reason we include all of the experiment results with true positive rate at 5% false positive rate in Appendix Section 8.20.

5. Experimental Results

We now present the empirical results evaluating Telescope Perplexity’s effectiveness as a zero-shot detector, its robustness under various conditions, and its practical limitations.

5.1. Detection Performance

First, we assess the overall detection performance of Telescope Perplexity compared to baselines across our suite of evaluation sets. Table 2 summarizes the average AUROC scores across all reference models for each dataset. Full results per reference model are available in Appendix Section 8.20.

The results demonstrate that Telescope Perplexity consistently achieves high AUROC scores, often outperforming the current state-of-the-art method, Binoculars, as well as

other strong baselines like Perplexity and DetectLLM LRR, when averaged across reference models and on each reference model. Notably, Telescope Perplexity shows exceptional performance on our novel evaluation sets featuring contemporary models like GPT4o Mini and Deepseek-V3 (e.g., GB Essay GPT4o Mini, ESL GPT4o Mini). This validates that probing the hypothesized “Vestigial Heuristic” via Telescope Perplexity provides a powerful and broadly effective signal for zero-shot LLM detection, even against modern target models.

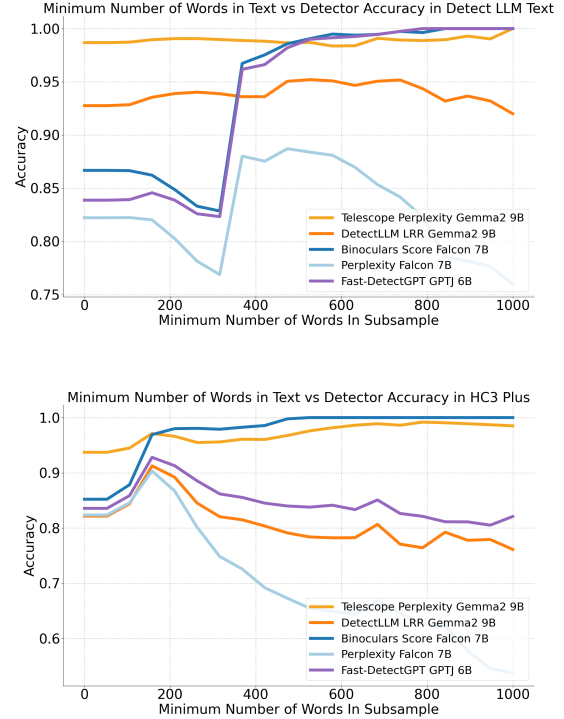


Figure 3. Impact of minimum text length on the AUROC performance of several detectors on the Detect LLM Text dataset (top) and the HC3 Plus dataset (bottom).

5.2. Robustness

Figure 3 demonstrates the impact of minimum text length on detector performance. While longer texts generally provide more evidence, Telescope Perplexity maintains strong performance even on relatively shorter texts (e.g. < 100 or < 200 words), suggesting the underlying local bias aggregates reliably. Its performance degrades less sharply with decreasing length compared to some baselines in certain datasets. For more examples, see Appendix Section 8.19.

To assess resilience to simple obfuscation, we evaluated performance on texts where words were increasingly replaced by synonyms and where sentences were increasingly paraphrased. Figure 4 illustrates that while performance degrades for all detectors as perturbations increase, Tele-

Table 2. Detection performance (Average AUROC across 12 reference models) of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across diverse datasets. Bold indicates the best performance per dataset. Since we test on so many different variations of SmolLM, our averaging will be inherently biased and overvalue performance on the SmolLM architecture. The headline numbers in this table also hide variation that matters operationally. On Detect LLM Text, Telescope Perplexity attains 0.99219 AUROC against 0.89307 for standard Perplexity, which corresponds to error rates of 6.8×10^{-3} and 1.0×10^{-1} respectively, a roughly $15\times$ reduction. On ESL GPT4o Mini, the gap from 0.9999 to 0.82523 corresponds to an error-rate reduction of more than three orders of magnitude. Averaging across reference models also obscures the largest per-model gains. Readers are strongly encouraged to view our full results with the performance of each reference model on each dataset with each detection technique in Appendix Section 8.20.

Dataset	AUROC				
	Telescope (ours)	Binoculars	Perplexity	DetectLLM	Fast-DetectGPT
Detect LLM Text	0.99219	0.76588	0.89307	0.92981	0.70085
AI vs Human	0.95143	0.86297	0.90743	0.90316	0.75608
HC3	0.99155	0.99441	0.99471	0.98436	0.95584
HC3 Plus	0.98451	0.88510	0.90999	0.87758	0.83575
ESL GPT4o Mini	0.99983	0.79637	0.82523	0.69051	0.60603
GB Essay ChatGPT	0.98628	0.88434	0.99810	0.99730	0.55624
GB News ChatGPT	0.90480	0.98773	0.98817	0.99050	0.91940
GB Creative ChatGPT	0.99397	0.91846	0.94990	0.91852	0.52336
GB Essay GPT4o	0.98136	0.85505	0.99365	0.99163	0.51477
GB Creative GPT4o	0.99271	0.91276	0.92303	0.87374	0.63813
GB News Claude	0.88038	0.89263	0.87211	0.86317	0.77787
GB Creative Claude	0.96604	0.82929	0.89304	0.87449	0.60276
GB Essay Claude	0.94223	0.77288	0.94310	0.95988	0.61633
GB Essay Deepseek V3	0.98484	0.99225	0.99881	0.99680	0.82763
GB Creative Deepseek V3	0.98199	0.99569	0.98852	0.96391	0.90439

specter Perplexity maintains relatively high AUROC compared to baselines, indicating the local bias it measures is not solely dependent on exact lexical choice. Further evaluation on dedicated perturbation datasets by Verma et al. (2024) confirms Telescope Perplexity’s robustness against various character, word, and paragraph-level modifications, which can be found in Appendix Section 8.19. Our tests of sophisticated “AI Humanizer” attacks show only small degradations in performance (see Appendix Section 8.18).

These results suggest the “Vestigial Heuristic” signature captured by Telescope Perplexity is reasonably robust to variations in text length and common perturbation strategies.

5.3. Domain Sensitivity

While generally effective, Telescope Perplexity’s performance varies across domains. As seen in Table 2, Telescope Perplexity achieves near-perfect scores when detecting ESL rewriting with an average 0.99983 AUROC across every reference model tested. We hypothesize this is a result of the ESL writing triggering the “Vestigial Heuristic” even more weakly than standard human text.

Conversely, Telescope Perplexity’s performance, while still often strong, is comparatively lower on some news writing datasets (e.g., Ghostbusters News Claude with an average AUROC of 0.88038 across every reference model tested). Additionally, Telescope Perplexity consistently misclassi-

fied AI-generated poetry as human written, as seen in Appendix Section 8.7. This suggests that genres governed by rigid lexical constraints or highly stylized registers (such as poetry or specific journalistic styles) might obscure the typical “Vestigial Heuristic” signature, which defines limitations for the current probe. Even though Telescope Perplexity seems to struggle with news writing, it also seems to perform better on essay writing than creative writing. Since essay writing is generally more formulaic than creative writing, this suggests that there isn’t a clear direct correlation between detection performance and formulaicity.

5.4. Practical Considerations and Recommendations for Deployment

Finally, we examine practical aspects relevant to deployment. Table 3 presents the results of our Transferability Test, evaluating how well a threshold tuned on N-1 datasets generalizes to the held-out Nth dataset.

While Telescope Perplexity often performs well even in this challenging scenario, its F1 score sometimes drops compared to its potential maximum F1 (achieved when tuning on the test set itself). This indicates that while the “Vestigial Heuristic” signature is generally present, its baseline level (and thus the optimal threshold) can vary slightly depending on the domain, style, or target model.

Like most other zero-shot detectors we tested, Telescope Perplexity’s raw output scores are poorly calibrated and should not be interpreted as true probabilities. They serve as

Table 3. Transferability of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT when tuned on every other dataset and tested on a specific dataset. We report the F1-Score of an algorithm’s performance on the test dataset averaged across all of the reference models tested. Similarly to Table 2, we also strongly recommend that readers view our full results in Appendix Section 8.20.

Test Dataset	F1 Score				
	Telescope (ours)	Binoculars	Perplexity	DetectLLM	Fast-DetectGPT
GB Essay ChatGPT	0.94317	0.84594	0.93620	0.94249	0.76713
GB News ChatGPT	0.86115	0.93763	0.81355	0.87625	0.63212
GB Creative ChatGPT	0.96255	0.85858	0.89799	0.85935	0.72314
GB Essay GPT4o	0.93922	0.82922	0.93493	0.93341	0.73008
GB Creative GPT4o	0.96269	0.85953	0.73004	0.74011	0.72738
GB News Claude	0.69429	0.79833	0.76009	0.74903	0.60027
GB Creative Claude	0.89028	0.76001	0.69536	0.75085	0.62287
GB Essay Claude	0.87984	0.74559	0.86685	0.88829	0.69177
GB Essay Deepseek V3	0.94006	0.94689	0.93718	0.94351	0.71785
GB Creative Deepseek V3	0.93721	0.90241	0.95466	0.90851	0.69879

effective discriminative scores for classification but exhibit overconfidence as shown in Appendix Section 8.5.

Our experiments suggest several concrete recommendations for anyone deploying Telescope Perplexity in practice.

Reference-model selection. Per-reference-model results in Appendix 8.20 show that no single reference model is uniformly best, and the spread across reference models on a fixed dataset can exceed the spread between detectors on the same row. We therefore recommend evaluating candidate reference models on a small held-out portion of the target domain before committing to one. When inference budget allows, taking the mean or median Telescope Perplexity over a small ensemble of reference models from different families (e.g. one Gemma, one Llama, one SmolLM) trades a constant-factor compute cost for substantially reduced variance across domains.

Threshold calibration. The Transferability results in Table 3 indicate that thresholds tuned on unrelated datasets degrade, sometimes substantially, when transferred to a new domain. We therefore recommend the following calibration procedure for any new deployment:

1. Collect a small labeled validation set (~ 200 – 500 examples, balanced between LLM generated and human written) drawn from the same distribution the detector will see in deployment.
2. Score every example with the chosen reference model and detector.
3. Choose τ to maximize F1 in this validation set, or, if a fixed false positive budget is required (e.g., academic-misconduct settings where false positives carry severe cost), choose τ as the smallest threshold whose validation set false positive rate is below the operational budget.
4. Re-calibrate periodically as the target-model and writing distributions drift.

For settings where labeled validation data is genuinely unavailable, the transferability scores in Table 3 give a reasonable lower bound on what to expect from an externally-tuned threshold.

5.5. Failure Modes and Misclassification Analysis

To complement the aggregate metrics above, we examine where Telescope Perplexity errs and how confidently. Across all reference models the expected distance from the decision threshold for a misclassified example is approximately one standard deviation of the score distribution, with human written texts misclassified as LLM generated typically sitting *closer* to the threshold (i.e. low-confidence errors) and LLM generated texts misclassified as human written typically sitting *further* from the threshold (high-confidence errors). Full per-model error-distance plots are deferred to Appendix Section 8.7; here we highlight the two qualitative patterns that matter most for deployment.

Stylised and formulaic text. The most confident false negatives we observed were in stylized or formulaic genres. Figure 10 (deferred to Appendix Section 8.7 for space) shows an LLM generated pirate-themed poem in the GB Creative ChatGPT split that Telescope Perplexity scored at 12.03 — well below the dataset decision boundary of 13.36. Stylized constraints (rhyme, meter, register) appear to suppress exactly the local repetition aversion that our score probes, in part by forcing end-line repetition (“Ahoy, me hearties ...” echoed across stanzas) that human written verse would also exhibit. This is consistent with the weaker performance on highly formulaic news writing (e.g. GB News Claude, average AUROC 0.880) observed in Section 5.3.

Direction of error and operational implications. The asymmetry described above has a concrete deployment implication: the most damaging error mode for an academic-integrity use case is the high-confidence false negative (LLM

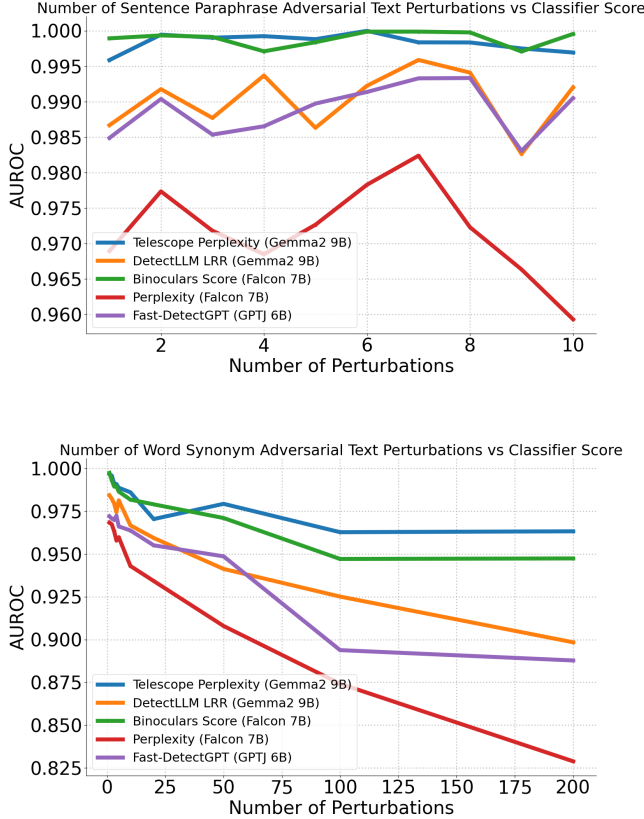


Figure 4. Impact of paraphrasing sentences (top) and swapping words with synonyms (bottom) on detector AUROC performance on the Ghostbusters dataset.

generated text passed off as human). Practitioners who care about this asymmetry should set thresholds against a target false-negative rate on stylized genres specifically, rather than against the data set-averaged decision boundary used for our reported maximum F1.

A more detailed dataset-by-dataset error decomposition, including representative misclassified samples and per-model error-distance distributions, is provided in Appendix Section 8.8 and the surrounding subsections.

6. Limitations

While we have attempted to rigorously test Telescope Perplexity against other LLM text detectors, there are still a couple of limitations to our experimentation methods due to compute constraints and practical limitations.

1) Perturbation-Based Detectors: We do not benchmark on DetectLLM NPR, DetectGPT or other perturbation based detection algorithms because of their massive computational cost. Preliminary testing showed worse performance on NPR compared to LRR, and there have been studies that

show that DetectGPT heavily underperforms compared to other modern techniques (Li et al., 2024).

2) Datasets Representative of Deployment: We attempt to address data limitations in AI detection by introducing a suite of new datasets. Although these datasets use modern LLMs as target models, they do not accurately reflect the data distribution in deployment environments, where students may employ multiple adversarial methods to avoid detection alongside newer, more powerful language models.

7. Conclusion

In this work, we present the Telescope Perplexity metric, which we hypothesize probes an underlying behavior that forms early in training which discourages token repetition with a “Vestigial Heuristic”. We demonstrate the local nature of this “Vestigial Heuristic” and track its development in LLM training. Through rigorous experimentation, we empirically compare our method with baselines and show that Telescope Perplexity is effective at discriminating between LLM generated and human written text and is particularly effective when tuned to a particular domain, while being robust to the choice of reference model.

Impact Statement

Advances in the detection of synthetic text, if adopted in high-stakes settings (e.g., academic integrity), carry significant risks alongside their intended benefits. The primary benefit of improved detection capabilities is defensive, weakening the value proposition of misuses of language models such as automated disinformation and academic dishonesty. More effective detection is also expected to have stronger negative impacts on the authors of misclassified texts. As more reliable techniques, like Telescope Perplexity, become available in non-lab settings, we caution against the exclusive use of automated single-point-of-failure systems in favor of using such systems as a first line of defense. In addition, we note that on our ESL evaluation set Telescope Perplexity is less susceptible to the non-native-speaker bias documented in prior work (Liang et al., 2023); (Section 5.3).

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8. Appendices

8.1. Clarifications on Vocabulary

Neural networks can become arbitrarily good at a task with enough data (Cybenko, 1989), and we can apply them to classify whether or not a piece of text has been generated by LLMs; however they do have some very major drawbacks. Simply using a neural network to classify text generated by one language model may not perform well on language models it hasn't been trained on. In a world where new language models are released constantly, it is uniquely valuable to create an LLM generated text detector that will work on any language model trained in the future. In addition, data for these tasks on a wide variety of current language models in a wide variety of tasks is often difficult to find which can limit these detectors' reliability. This is why in the literature, there is a distinction between zero-shot methods and non-zero-shot methods; zero-shot methods promise to correctly classify LLM generated text without training on the specific language model that may have generated the text, while non-zero-shot methods must be trained on specific language models and may not be able to generalize to newly released language models. Since consumers will switch to whichever language model performs the best at a given point in time, zero-shot methods are generally thought to be more robust but are more difficult to design and make accurate.

Because of their desirable properties, many researchers have focused on developing and improving the performance of these zero-shot detectors. Zero-shot detection is a difficult problem since it requires directly analyzing the high dimensional space of text, so in an attempt to compress this space many of the zero-shot detectors in the literature opt to use a reference model to help detect the text generated by a given target model. A target model is simply the model that may have generated the text under question, while the reference model is a model that a specific AI generated text detector may use to produce the probabilities for the next token for every word in the sentence. For example, in the sentence "The quick brown fox jumped over the lazy dog", the reference language model will produce the probability for any word to come after "The", "The quick", "The quick brown", etc. These probabilities can then be analyzed using metrics such as the average Perplexity of each word that the model predicted. It turns out that depending on the reference model used, the logits distributions can often convey very interesting information for whether the text under question was generated by some target language model. The target model is generally unknown, since we generally don't have a good idea which language models may have been used to generate text out in the wild; however, the reference model is chosen by the algorithm designer, and different reference models may have different properties that make them better or worse at detecting AI generated content from a variety of

target models. The relationship between reference models and target models is illustrated in Figure 5. In this paper, we mainly focus on zero-shot methods and attempt to push the boundaries of the accuracy that can be achieved with them while still being able to easily generalize to any language model.

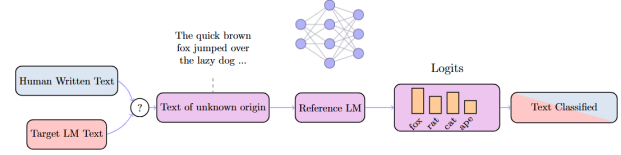


Figure 5. A graphical illustration of the relationship between the reference language model and the target language model.

8.2. Analysis of Frequency Components of the Telescope Score

We now consider the single token Telescope Perplexity:

$$\text{Single Token Log Telescope PPL}_{\mathcal{M}}(s_i) = -\log \mathcal{M}(s_i | s_{1:i}) \quad (2)$$

We can use this to plot our per-token Telescope Perplexity as in Figure 6. Readers familiar with signal processing may notice that the Telescope Perplexity over the token position resembles a signal. Interrogating this similarity, we found that the only truly relevant component of this signal is its mean. Future work applying sequence models in this domain is needed to fully evaluate if only the mean of this signal is truly important; our analysis applying signal processing yielded no results.



Figure 6. Per-token Telescope Perplexity, Perplexity, and Binoculars Score over the sequence

8.3. Training Data Separability

Here we present additional results on using SmolLM 360M to perform detection on a 10k subsample of its synthetic and human written training data. We find that, interestingly, standard Perplexity is able to somewhat effectively separate its own training data into synthetic and human data, imply-

ing Perplexity’s effectiveness in detecting LLM generated text needs further study. The Binoculars Score does not effectively delineate between AI and human training data.

Method	AUC	F1 Score
Binoculars	0.7692	0.7327
Perplexity	0.9523	0.9056
Telescope Perplexity	0.9956	0.9874

Table 4. F1 and AUC of SmolLM 360M on its own training data.

8.4. Error Independence Analysis

We analyzed error independence across all detection methods to identify potential ensemble opportunities. For each classifier pair, we computed Cohen’s Kappa, Yule’s Q-statistic, and Normalized Mutual Information (NMI) to measure how independently they make errors. The Q-statistic, calculated as $Q = \frac{n_{11} \cdot n_{00} - n_{10} \cdot n_{01}}{n_{11} \cdot n_{00} + n_{10} \cdot n_{01}}$, ranges from -1 to 1, with lower values indicating greater error independence between classifiers.

Table 5 shows aggregated statistics across all datasets, sorted by Q-statistic. DetectLLM using Falcon 7B exhibited the highest independence ($Q = 0.017$), making errors on almost completely different examples than other methods. Telescope Perplexity and standard Perplexity from the same model showed moderate dependence ($Q = 0.7$ - 0.8), while Binoculars showed lower dependence due to its Cross-Perplexity normalization.

These results suggest promising ensemble combinations: DetectLLM (especially with Falcon 7B) paired with any Perplexity-based method, different model sizes within the same architecture, and Telescope Perplexity combined with Binoculars. The low mutual information values (mostly < 0.2) indicate substantial opportunities for ensemble improvements through complementary error patterns.

8.5. The Calibration of Zero-Shot Detection Methods

Both Binoculars and Telescope Perplexity are natively very poorly calibrated, meaning the scores produced by these models do not accurately reflect the real probabilities of a given text being LLM generated. The calibration behavior of both detectors is shown in Figure 7 with one particularly poorly calibrated example shown in Figure 8.

8.6. Datasets

For each dataset, we filter out samples that are less than 100 words long since when text is shorter, it may be statistically impossible to determine whether the text is written by a large language model. Shorter pieces of text have fewer different ways to write them and because of this, the way that a human is likely to write the text may be very similar to the

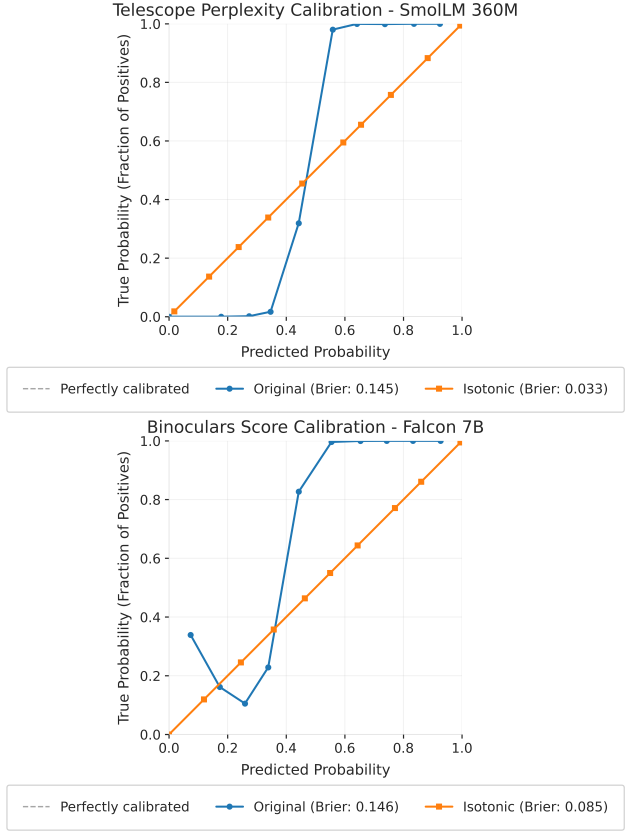


Figure 7. Calibration comparison between SmolLM 360M Telescope Perplexity (top) and Falcon Binoculars method (bottom)

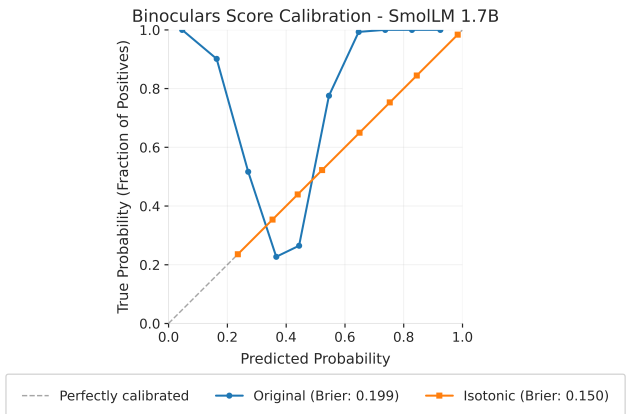


Figure 8. An extremely poorly calibrated Binoculars example using SmolLM 1.7B as a reference model.

Detector	Reference Model	Kappa	Q-stat	MI
LRR	Falcon 7B	0.059	0.081	0.023
Fast-DetectGPT	SmolLM 1.7B	—	0.504	0.058
Fast-DetectGPT	Llama3 8B	0.339	0.518	0.057
Fast-DetectGPT	SmolLM 360M	—	0.583	0.059
Fast-DetectGPT	SmolLM 135M	0.309	0.590	0.062
Binoculars	SmolLM 1.7B	0.420	0.605	0.079
Fast-DetectGPT	SmolLM2 135M	0.410	0.637	0.086
Fast-DetectGPT	SmolLM2 1.7B	0.405	0.639	0.070
Fast-DetectGPT	SmolLM2 360M	0.440	0.678	0.082
Binoculars	SmolLM 360M	0.480	0.683	0.085
Binoculars	SmolLM 135M	0.539	0.748	0.086
Binoculars	SmolLM2 1.7B	0.604	0.818	0.083
Telescope	Falcon 7B	0.640	0.830	0.054
Fast-DetectGPT	Gemma2 2B	0.620	0.832	0.105
Binoculars	Llama3 8B	0.626	0.835	0.082
LRR	SmolLM2 135M	0.630	0.836	0.129
Telescope	SmolLM2 135M	0.677	0.850	0.075
LRR	SmolLM2 360M	0.671	0.853	0.149
Fast-DetectGPT	Gemma2 9B	0.647	0.855	0.105
LRR	Llama3 8B	0.684	0.859	0.138
LRR	SmolLM 135M	0.671	0.860	0.126
Telescope	SmolLM 135M	0.686	0.860	0.078
Telescope	SmolLM2 360M	0.693	0.861	0.090
Telescope	Gemma2 2B	0.681	0.861	0.055
LRR	SmolLM2 1.7B	0.684	0.861	0.154
Fast-DetectGPT	Falcon 7B	0.667	0.862	0.086
LRR	Gemma2 2B	0.686	0.863	0.130
Binoculars	SmolLM2 360M	0.669	0.864	0.097
Telescope	SmolLM2 1.7B	0.704	0.868	0.082
Telescope	SmolLM 360M	0.704	0.870	0.086
Binoculars	SmolLM2 135M	0.683	0.871	0.106
Telescope	SmolLM 1.7B	0.705	0.871	0.082
Perplexity	SmolLM2 135M	0.695	0.872	0.183
Perplexity	Gemma2 9B	0.705	0.872	0.160
Perplexity	SmolLM 135M	0.702	0.873	0.181
LRR	SmolLM 360M	0.697	0.874	0.138
Telescope	Llama3 8B	0.707	0.874	0.080
Telescope	Gemma2 9B	0.711	0.876	0.057
LRR	SmolLM 1.7B	0.705	0.878	0.130
LRR	Gemma2 9B	0.705	0.879	0.111
Perplexity	Falcon 7B	0.709	0.880	0.181
Perplexity	SmolLM 360M	0.715	0.880	0.192
Perplexity	SmolLM 1.7B	0.718	0.881	0.196
Binoculars	Gemma2 2B	0.705	0.881	0.107
Perplexity	SmolLM2 360M	0.718	0.883	0.197
Binoculars	Falcon 7B	0.715	0.884	0.090
Binoculars	Gemma2 9B	0.707	0.885	0.097
Perplexity	SmolLM2 1.7B	0.726	0.887	0.192

Table 5. Error independence statistics (average Kappa, average Q-statistic, and average Mutual Information (MI)) aggregated across all datasets. Lower values for the Kappa, Q-statistic, and Mutual Information (MI) indicate greater error independence.

way that an LLM would have written the text. Intuitively, this is similar to trying to make a decision based off of too little evidence. Including these short samples introduces unwanted variance to our measurements since the best the models can do some of the time is simply guess. To the best of our knowledge, there isn’t any research specifically targeting how long the input to these detectors should be to minimize the data-points that are statistically impossible or difficult to classify for a theoretically perfect classifier, so we choose to filter out all text samples with fewer than 100 words from our datasets. A popular commercial detector called GPTZero also recommends that text be longer than

100 words for accurate classification. Other than following the lead of GPTZero, this decision is fairly arbitrary, and we decided on it at the beginning of testing and have not changed it since. In addition, to save on computation cost, we also filter out any samples that are larger than 5000 words, since some of the LLM generated samples were incredibly long and would often induce out of memory errors. To reduce out of memory errors on our limited compute resources, we decided to simply take out the samples which were long enough to cause errors. For extremely long datasets with hundreds of thousands of samples, we only use the first 10,000 samples to save on compute.

Figure 9 visualizes pairwise error independence, using a dendrogram with less error independent models being closer in distance.

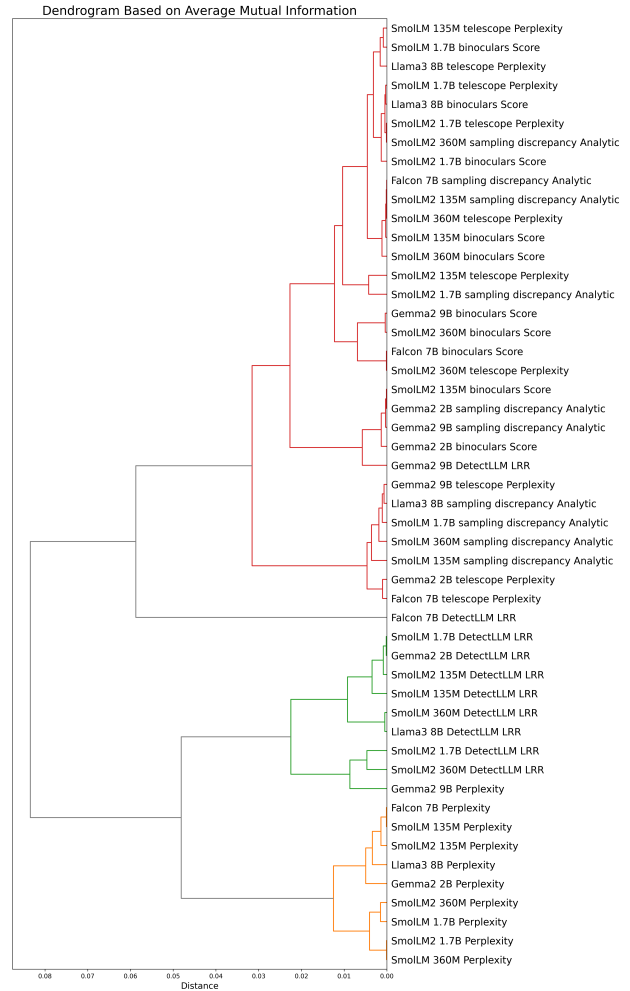


Figure 9. MI for aggregated datasets with different colors showing model groups that are error independent with other groups.

8.7. Example of Misclassification on Poetry/ Stylized Text

In this section, we show an example of an extremely misclassified AI generated text sample in the GB Creative Writing ChatGPT dataset that Telescope Perplexity very confidently classified as human written. This text sample is stylized to be pirate-like, which may indicate that this is a way to avoid detection (although this may be a rather impractical way to avoid detection in many circumstances).

Text:

Ahoy, me hearties, gather round, I'll spin for ye a tale profound, Of an island in the distant deep, Where dragons fly and secrets keep. 'Twas a land surrounded by azure seas, Where the dragon isles danced with the breeze, Guarded fiercely by three women bold, Their stories, legends yet untold. The first was Rosie, with flaming hair, A tempest rage, none could compare, With fiery spirit, she stood her ground, To protect the island, her dragons, renowned. Then there was Lily, with eyes so keen, A huntress born, her aim, unseen, With bow in hand and arrow true, No enemy could escape her view. Lastly was Bella, the wise and serene, Her words like gold, her heart a queen, She bore the knowledge of ancient roam, And in her presence, wisdom will bloom. Together they sailed amidst the tide, With dragon wings, they took their stride, The island shrouded in mystery rare, Its secrets guarded with utmost care. Dragons gleamed in the sun's warm glow, With scales that shimmered, a mighty show, In harmony, they danced and soared, Their fiery breath, their mighty hoard. But one day came a pirate crew, With greed and darkness to pursue, To snatch the dragons, to rule with might, And claim the island, lost from sight. But Rosie, Lily, and Bella stood, United as one, they knew they should, With swords unsheathed and hearts aflame, They fought for the dragons they held no blame. As the battle raged with thunderous sound, The pirates fled, their treasure unfound, The island stood proud, the dragons remained, A testament of the women unchained. So hear me shanty, a tale of courage bold, Of an island guarded against pirates untold, Through legends sung, let their bravery live, Three fierce women, their souls will thrive. Ahoy, me hearties, let our voices ring, Of the island where dragons took wing, And when the sea sings this tale divine, May their bravery echo throughout all time.

Telescope Perplexity: 12.03

Label: LLM Generated

Decision Boundary for This Dataset: 13.36

Figure 10. Example of Misclassification

8.8. Misclassifications Analysis

Understanding misclassifications is important in any high impact application. We find that for Telescope Perplexity the expected distance from the threshold for a classification made in error is ≈ 1 standard deviation, with human written texts misclassified as AI being less than that on average and AI text misclassified as human typically being more than a single standard deviation from the threshold. Figure 11 shows a couple of examples of the distribution of misclassifications with Deepseek-v3 being the only model in the entire suite of reference models to have a lower σ for the AI generated misclassifications than human ones.

8.9. Statistical Significance and Confidence Intervals

To compute our confidence intervals, we use a python package by Gildenblat (2023), which contains functions to compute the 2 standard deviation confidence intervals of an AUROC score using the Delong method (DeLong et al., 1988), (Sun & Xu, 2014) and the 2 standard deviation confidence intervals of the F1-Score using the Takahashi method (Takahashi et al., 2022). The 2 standard deviation confidence intervals for the true positive rate at 5% false positive rate score was found simply by bootstrapping.

8.10. Poor Transfer Examples

If only shown a specific target model on a specific type of text, Telescope Perplexity can have poor threshold transfer capabilities. An illustrative example of poor threshold transfer for Telescope Perplexity is shown in Figure 12. This highlights the importance of using a diverse set of data to tune Telescope Perplexity's threshold.

Even with this single example of poor transfer, it is important to note that Telescope Perplexity's transfer performance is still generally better than other approaches as seen in Appendix Section 8.20.

8.11. Computational Resources And Execution Time

The majority of these experiments were performed on Nvidia L40S GPUs with 48 GB of VRAM. Every model was run using 8 bit quantization using the BitsAndBytes¹ library to ensure that there was minimal loss of precision when benchmarking each detector, while also allowing us to run two models at once on single GPU. A subset of the ghostbusters datasets with large models such as Gemma2 9B were prone to out of memory exceptions, so for this reason, we utilized A100 GPUs with 80 GB of VRAM instead when necessary. Compute resource constraints did not allow us to test on reference models larger than 9 billion parameters. We have shown that in this range there is no

¹<https://huggingface.co/docs/bitsandbytes/en/index>

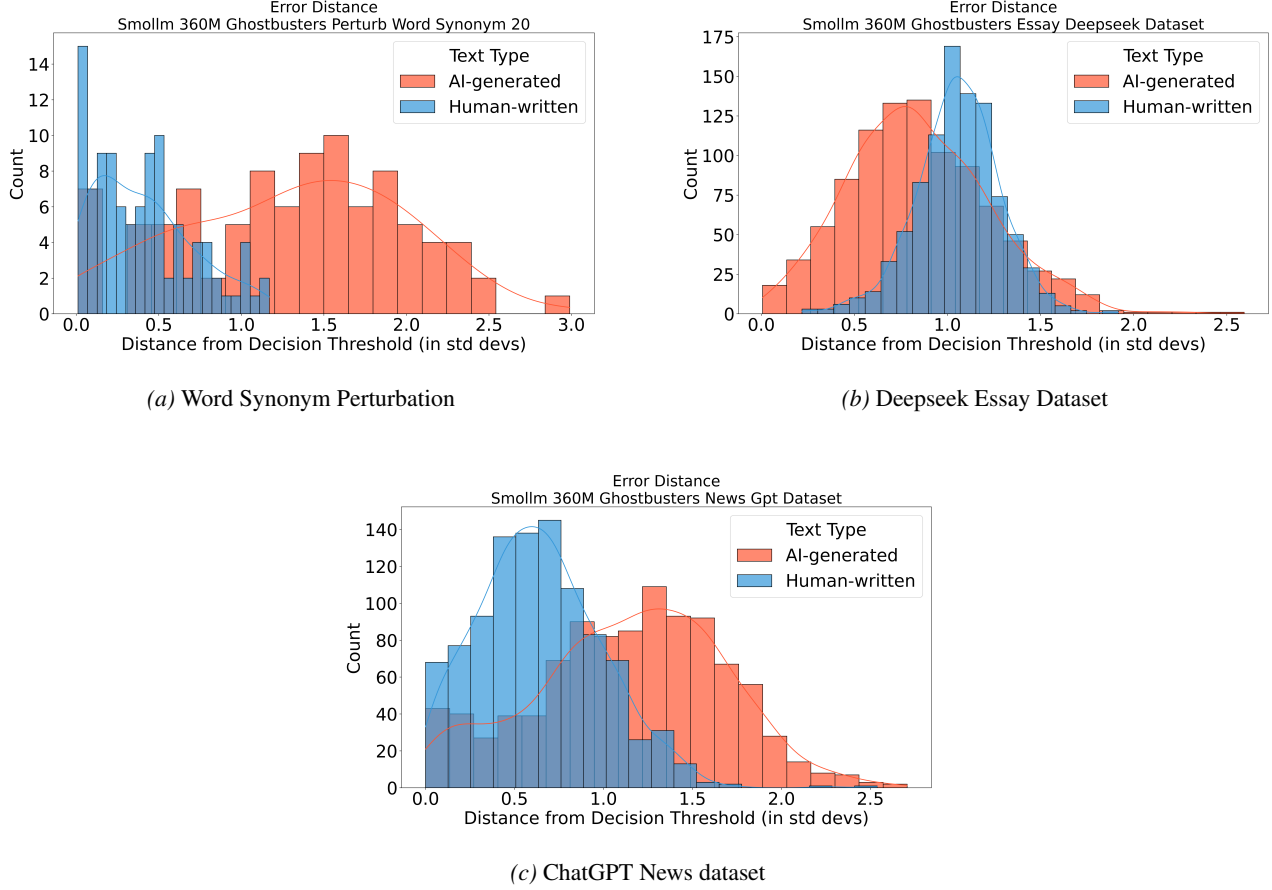
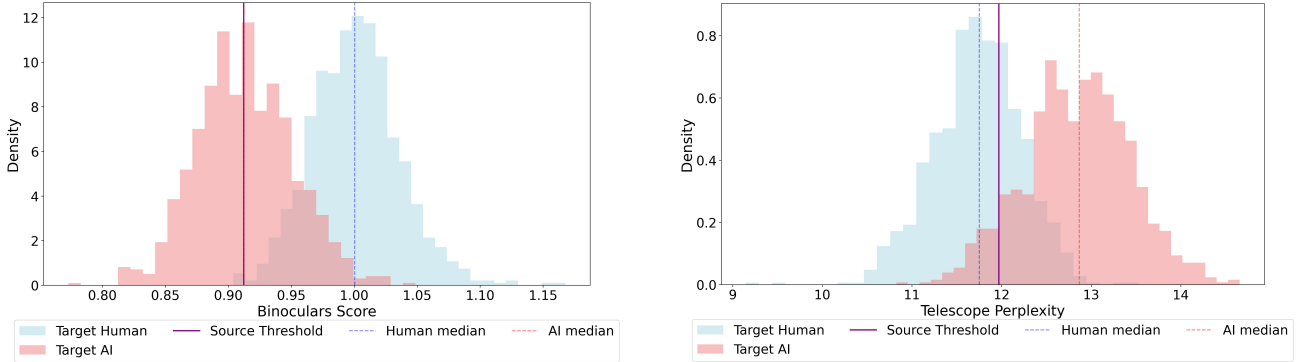


Figure 11. Error distance metrics for SmolLM 360M across different datasets and perturbation types.



direct correlation between model size and detector accuracy, but this relationship may not hold at larger model sizes.

The execution time of the experiments were highly dependent on the reference model used, the number of reference models a detection algorithm used, and the dataset. Different datasets contain different numbers of samples while different reference models may have more or less active parameters. In order to save time and effort while run-

ning these experiments, many of these experiments were run while batching multiple reference models and detection techniques in a single run, which further complicates matters, and makes it difficult to ascertain the run-time of a single experiment. Therefore in Table 6 we report the total runtime of experiments on an L40S GPU on the Detect LLM Text dataset (10,000 samples). Knowing these execution times should provide a rough sense of how long

all of the other experiments in this paper would take to run individually.

Table 6. Execution times of each technique used in this study on the Detect LLM Text dataset (10,000 samples) with L40S GPUs.

Reference Model	Execution Time (minutes)				
	Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
Gemma2 2B	63.43	72.18	63.54	35.67	33.05
Gemma2 9B	73.63	88.69	73.91	44.60	42.84
Llama3 8B	53.24	64.04	52.80	42.12	37.16
Falcon 7B	44.95	65.62	45.86	45.43	37.02
GPT-Neo 2.7B	45.51	65.42	45.52	47.70	40.93
GPT-J 6B	64.98	105.00	65.21	69.37	61.10
SmolLM 135M	31.98	35.39	32.34	35.76	24.81
SmolLM 360M	31.86	36.08	31.15	34.66	28.74
SmolLM 1.7B	30.03	33.39	30.25	32.24	24.80
SmolLM2 135M	28.58	31.65	28.58	31.18	24.92
SmolLM2 360M	28.29	31.70	28.24	30.65	24.77
SmolLM2 1.7B	30.66	34.33	30.62	31.12	25.06

8.12. Unused Performance Metrics

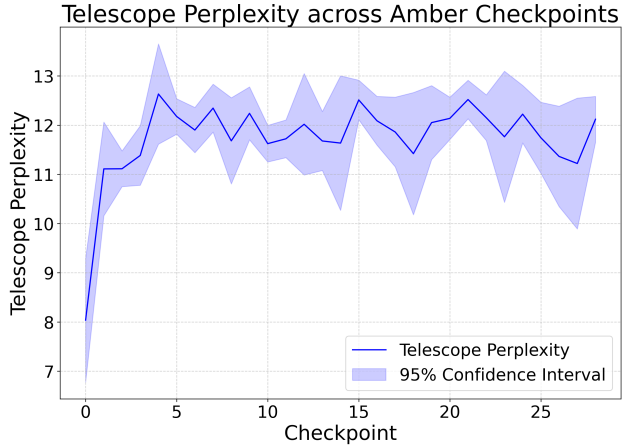
Unlike Binoculars, we decide to completely disregard using the TPR at ultra low FPR score, such as 0.05% FPR since we simply do not have enough data in our datasets to make accurate and reliable measurements of behavior that only happens once in every 10,000 samples. We initially attempted to use these metrics, but during testing, we noticed substantial jumps of around 40% near the end of testing a dataset, which helps corroborate that using these TPR at ultra low FPR scores as performance metrics to rate classifiers is unreliable and high variance for the number of samples that we have from each dataset. Other works by Tufts et al. (2025) show that the confidence intervals for this metric are massive and inconclusive.

8.13. Additional Training Dynamics

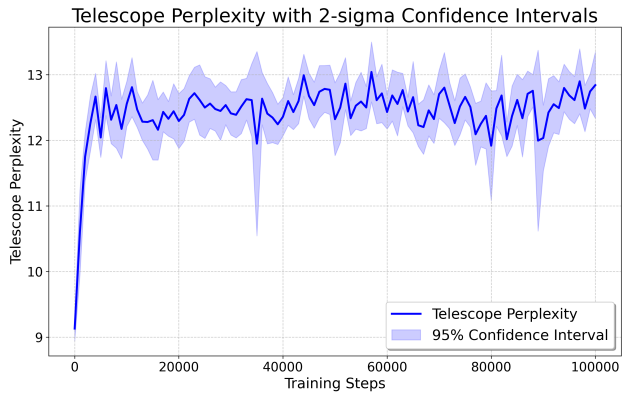
Liu et al. (2023) provide checkpoints throughout the training of their 7 billion parameter model Amber, these are taken regularly but the number of steps per checkpoint is not given. Compute constraints make evaluating all the Amber-7B model checkpoints infeasible so we instead focus on the early training steps to validate our hypothesis. We also present a chart showing the emergence of the “Vestigial Heuristic” in a much smaller model (Pythia-160M). As shown in Figure 13 we observe Telescope Perplexity rising early in training before stabilizing across both model architectures and across a few model sizes.

8.14. Multilingual Ablation

The Telescope Perplexity probe is designed to apply to English-language models. For completeness, we include limited multilingual testing on Wang et al. (2024)’s M4 dataset using Riviere et al. (2024)’s Gemma2-2b. Variance in the original data sources across languages is large enough to make comparison between languages difficult. Additionally, Riviere et al. (2024)’s models are “not trained specifi-



(a) Early training of Liu et al. (2023)’s Amber-7B



(b) First 100k training steps of Biderman et al. (2023)’s Pythia-160M model

Figure 13. Training curves for language models

cally for state-of-the-art multilingual capabilities,” making the performance of each technique quite remarkable.

Table 7. Performance Metrics by Language and Method with Gemma2-2B as reference model

Language	Metric	AUC	Confidence Interval
Chinese	Telescope Perplexity	0.94944	(0.94354, 0.95533)
	Binoculars Score	0.98895	(0.98677, 0.99114)
	Perplexity	0.96257	(0.95722, 0.96792)
	LRR	0.95358	(0.94848, 0.95868)
	Fast-DetectGPT	0.94745	(0.94208, 0.95282)
Russian	Telescope Perplexity	0.66543	(0.65152, 0.67935)
	Binoculars Score	0.89772	(0.88992, 0.90551)
	Perplexity	0.70169	(0.68836, 0.71502)
	LRR	0.63027	(0.61578, 0.64476)
	Fast-DetectGPT	0.78909	(0.77760, 0.80057)
Urdu	Telescope Perplexity	0.93901	(0.93226, 0.94576)
	Binoculars Score	0.99996	(0.99991, 1.00000)
	Perplexity	0.97867	(0.97517, 0.98217)
	LRR	0.98020	(0.97694, 0.98346)
	Fast-DetectGPT	0.99975	(0.99962, 0.99988)
Arabic	Telescope Perplexity	0.94478	(0.93926, 0.95030)
	Binoculars Score	0.97650	(0.97324, 0.97977)
	Perplexity	0.92182	(0.91478, 0.92887)
	LRR	0.90266	(0.89520, 0.91012)
	Fast-DetectGPT	0.93667	(0.93080, 0.94254)

8.15. Adversarial Temperature and Prompt Attacks

In order to gauge how the temperature of the target model affects Telescope Perplexity’s performance on detecting text from that target model, we have decided to provide a small ablation study where we generate new datasets using different target model sampling temperatures. To conduct this ablation, we regenerate the Ghostbusters Essay dataset with the GPT4o target model with sampling temperatures of 0, 1 (the default for the OpenAI API), and 1.2 (the highest temperature that was also stable). Using a reference model of SmolLM 360M, we find that with a temperature of 0 corresponds to an AUROC of 0.999992, a temperature of 1 corresponds to an AUROC of 0.99993, and a temperature of 1.2 corresponds to an AUROC of 0.9988. Overall, we see a trend that a lower sampling temperature leads to better detection performance, which makes sense, since as the temperature of the target language model decreases, the chance for the language model to accidentally mimic human text via random sampling also decreases.

In addition, we also wish to study how prompt attacks affect the performance of the Telescope Perplexity. One simple prompting attack that people may employ to dodge detection is asking the target model to attempt to repeat words, which may interfere with the token repetition signal. To accomplish this, we regenerated the Ghostbusters Essay dataset with the GPT4o target model, but instead of using the system prompt “You are a helpful assistant.”, we use the system

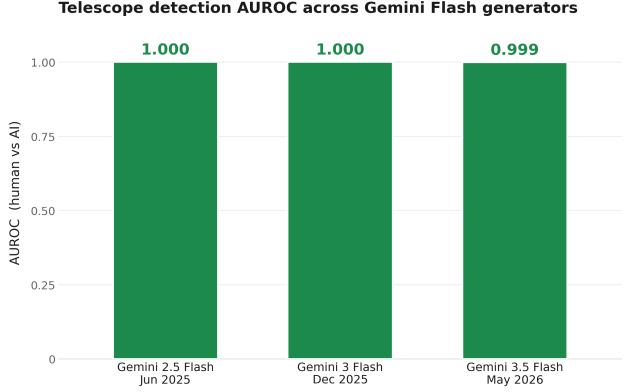


Figure 14. Telescope Perplexity maintains its high AUC across many model releases.

prompt “You are a helpful assistant. Try to repeat key words one after another while still following the prompt”. In doing so, we found that the AUROC on the dataset with the reference model of SmolLM 360M degraded from 0.99993 to 0.996. This suggests a slight but statistically significant degradation; however, this is not as bad as one may expect from these types of adversarial prompting attacks.

8.16. LLM-as-classifier

We performed a simple statistical test using Gemini 3.5-Flash as a classifier. This regime fails to beat chance ($p = 0.184 > 0.05$) in our simple 50 sample test drawn from Verma et al. (2024)’s GPT essay dataset.

8.17. Performance Drift

We measure Telescope Perplexity’s performance on the last years slate of Gemini flash model releases (2.5, 3.0, 3.5). We find no statistically significant change in model performance over these model releases as shown in Figure 14.

8.18. Adversarial AI Humanizers

To assess detection robustness under adversarial conditions, we employ a BERT-based humanization system that iteratively modifies text to minimize detection scores. The algorithm operates as follows:

Given the input text T and the human baseline score μ_H , the system randomly masks the tokens with probability $p_m = 0.08$. For each masked position, a BERT model generates $k = 2$ candidate replacements from the top- k most probable tokens. Candidates are filtered by semantic similarity $\text{sim}(T, T') \geq 0.98$ using sentence embeddings. Among valid candidates, the system selects the replacement that minimally decreases the detection score relative to μ_H , accepting changes only if they move its score closer to the human baseline.

This process iterates for 500 steps, generating $n = 4$ variants per input and selecting the optimal variant by detection score. Notably, we choose minimally destructive settings for the humanizer where semantic meaning is strongly preserved. In practice, more destructive settings can decrease detector accuracy at the cost of the text becoming incomprehensible compared to the original. Telescope maintains strong performance with degradation concentrated in datasets where it’s performance is typically weaker (see Tables 8 and 9).

8.19. Additional Ablation Results

Figures 15, 16, 17, and 18 show model performance over text length. Figures 19a, 19b, 20a, 20b, and 20c show model performance over perturbations for a number of different perturbation regimes described in Verma et al. (2024).

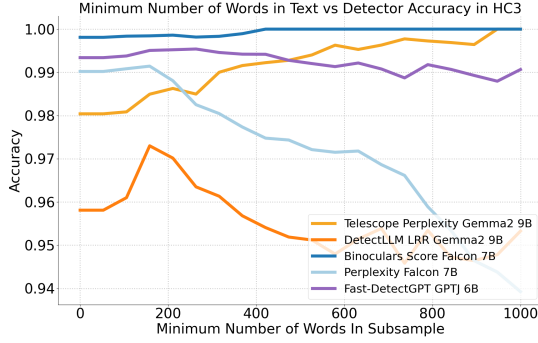


Figure 15. Impact of text length on detector performance (HC3). We filter out samples below a minimum word count and report accuracy on the remaining subset.

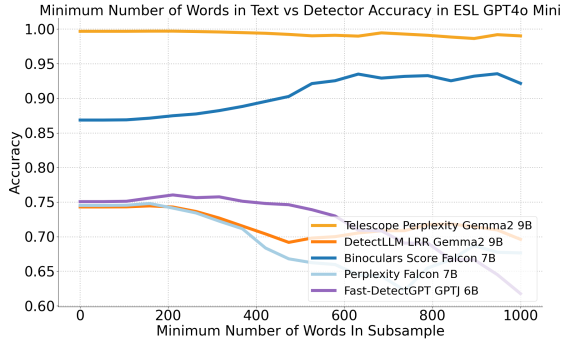


Figure 16. Impact of text length on detector performance (ESL GPT-4o Mini). We filter out samples below a minimum word count and report accuracy on the remaining subset.

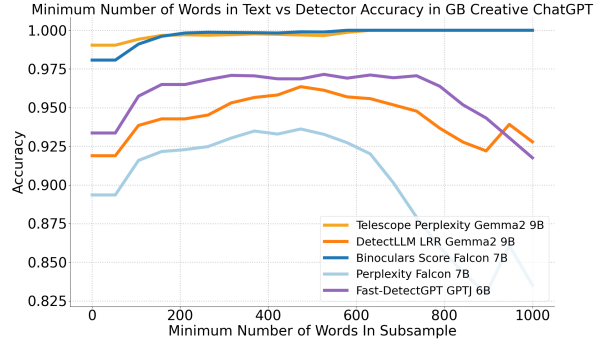


Figure 17. Impact of text length on detector performance (GB Creative GPT). We filter out samples below a minimum word count and report accuracy on the remaining subset.

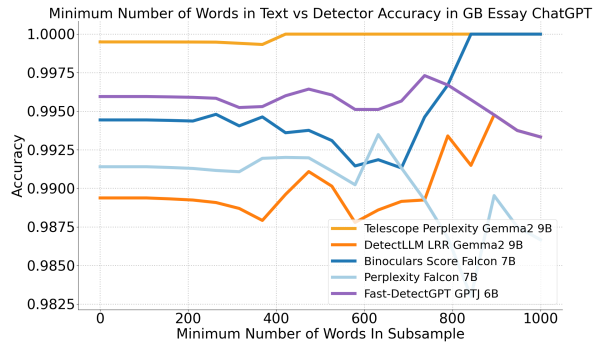


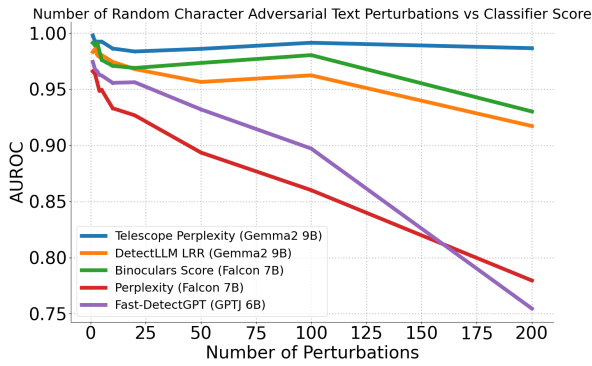
Figure 18. Impact of text length on detector performance (GB Essay GPT). We filter out samples below a minimum word count and report accuracy on the remaining subset.

Table 8. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across humanizer-perturbed datasets using SmolLM 360M as reference model. We report the AUROC of each detection technique. The best performance on each dataset is bolded.

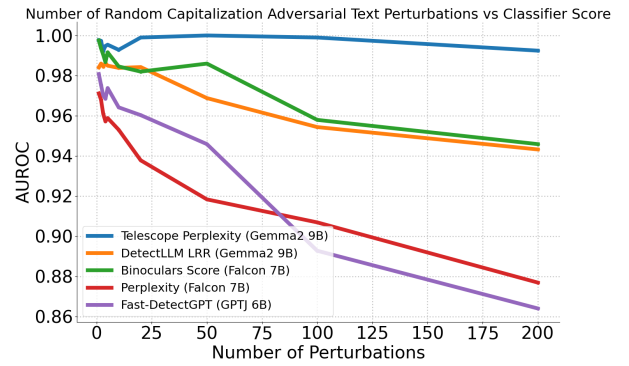
Dataset	Reference Model	AUROC				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Creative Claude	SmolLM 360M	0.99507	0.88834	0.95074	0.93432	0.69787
GB Creative DeepSeek	SmolLM 360M	0.90452	0.98675	0.97280	0.94070	0.91483
GB Creative GPT-4o Adversarial Prompt	SmolLM 360M	0.98217	0.92634	0.97159	0.96413	0.77258
GB Creative GPT-4o	SmolLM 360M	0.98831	0.94116	0.93884	0.90649	0.76582
GB Creative ChatGPT	SmolLM 360M	1.00000	0.81408	0.98507	0.98102	0.50416
GB Essay Claude	SmolLM 360M	0.97759	0.77987	0.93455	0.93838	0.63550
GB Essay DeepSeek	SmolLM 360M	0.92191	0.90351	0.99880	0.99453	0.72387
GB Essay GPT-4o Adversarial Prompt	SmolLM 360M	0.84277	0.72826	0.97662	0.98000	0.59864
GB Essay GPT-4o	SmolLM 360M	0.94656	0.76053	0.98266	0.98733	0.58917
GB Essay ChatGPT	SmolLM 360M	0.99186	0.70428	0.99373	0.99498	0.57298
GB News Claude	SmolLM 360M	0.89180	0.91674	0.95341	0.90736	0.87278
GB News ChatGPT	SmolLM 360M	0.98826	0.98662	0.99836	0.99618	0.96857

Table 9. AUROC confidence intervals (95%) for detection methods across humanizer-perturbed datasets using SmolLM 360M. The method with the narrowest confidence interval for the highest AUROC is bolded.

Dataset	Reference Model	AUROC Confidence Intervals				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Creative Claude	SmolLM 360M	(0.98407, 1.00608)	(0.77655, 1.00013)	(0.89862, 1.00286)	(0.86970, 0.99894)	(0.54258, 0.85315)
GB Creative DeepSeek	SmolLM 360M	(0.86856, 0.94047)	(0.97454, 0.99896)	(0.95688, 0.98871)	(0.91289, 0.96851)	(0.88246, 0.94720)
GB Creative GPT-4o Adversarial Prompt	SmolLM 360M	(0.96587, 0.99848)	(0.89467, 0.95802)	(0.95159, 0.99160)	(0.94331, 0.98494)	(0.71800, 0.82716)
GB Creative GPT-4o	SmolLM 360M	(0.97945, 0.99717)	(0.91345, 0.96886)	(0.91276, 0.96493)	(0.87393, 0.93905)	(0.71163, 0.82001)
GB Creative ChatGPT	SmolLM 360M	(1.00000, 1.00000)	(0.76492, 0.86325)	(0.97252, 0.99761)	(0.96723, 0.99481)	(0.43794, 0.57037)
GB Essay Claude	SmolLM 360M	(0.96418, 0.99100)	(0.72729, 0.83245)	(0.90634, 0.96276)	(0.91165, 0.96510)	(0.57316, 0.69784)
GB Essay DeepSeek	SmolLM 360M	(0.89131, 0.95251)	(0.86729, 0.93974)	(0.99704, 1.00056)	(0.98894, 1.00012)	(0.66676, 0.78098)
GB Essay GPT-4o Adversarial Prompt	SmolLM 360M	(0.79225, 0.89329)	(0.66925, 0.78727)	(0.96163, 0.99160)	(0.96819, 0.99181)	(0.53448, 0.66280)
GB Essay GPT-4o	SmolLM 360M	(0.92164, 0.97148)	(0.70457, 0.81649)	(0.96896, 0.99636)	(0.97669, 0.99797)	(0.52477, 0.65357)
GB Essay ChatGPT	SmolLM 360M	(0.97902, 1.00470)	(0.64294, 0.76562)	(0.98717, 1.00029)	(0.98940, 1.00055)	(0.50794, 0.63803)
GB News Claude	SmolLM 360M	(0.85601, 0.92760)	(0.88525, 0.94823)	(0.93007, 0.97676)	(0.87373, 0.94099)	(0.83372, 0.91184)
GB News ChatGPT	SmolLM 360M	(0.98050, 0.99603)	(0.97451, 0.99873)	(0.99623, 1.00049)	(0.99279, 0.99956)	(0.94890, 0.98825)



(a) Random characters



(b) Random capitalization

Figure 19. Impact of random character and random capitalization perturbations on the AUROC of each detector.

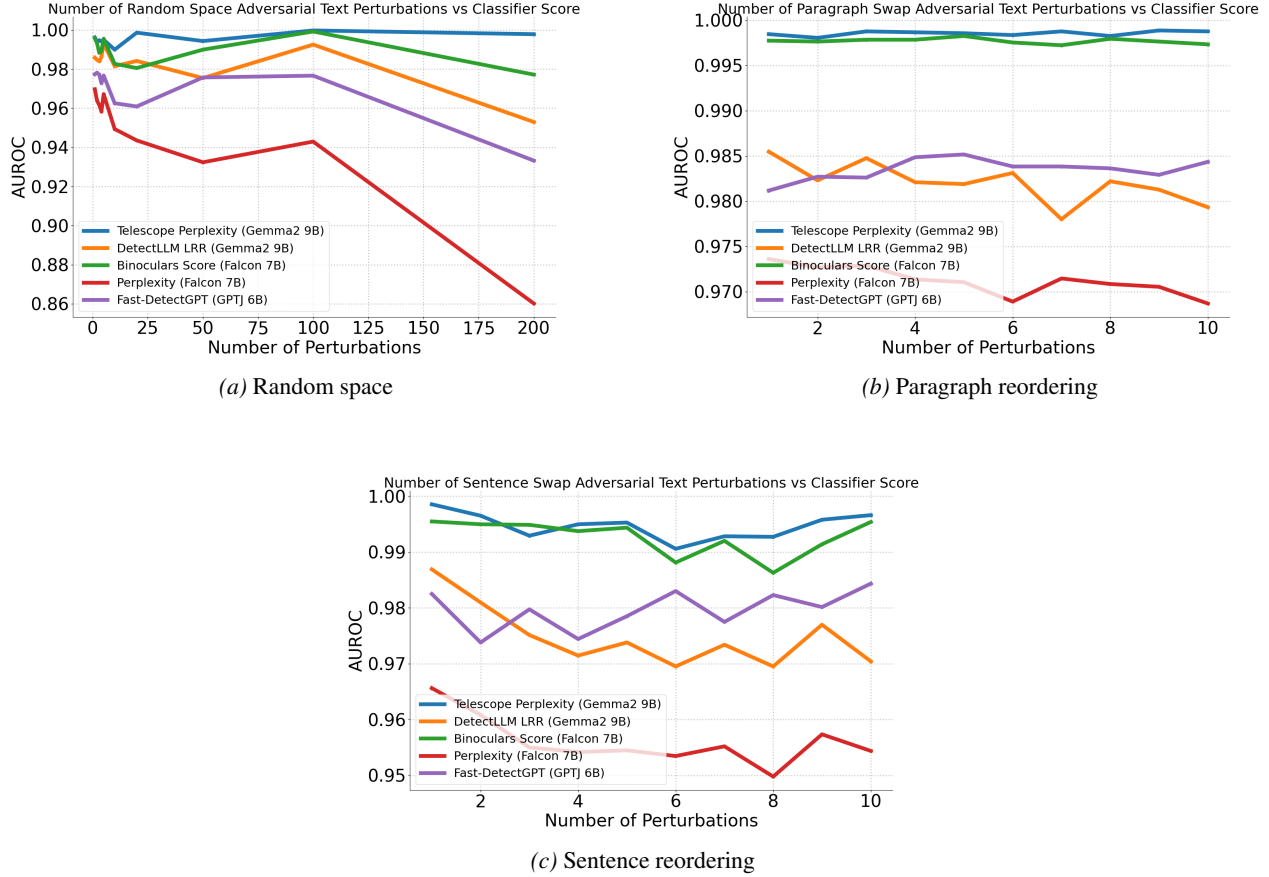


Figure 20. Impact of random space, paragraph reordering, and sentence reordering perturbations on the AUROC of each detector.

8.20. Raw Experimental Results

Table 10. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the AUROC of each reference model on each dataset and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	AUROC				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
Detect LLM Text	Gemma2 2B	0.99639	0.93494	0.90786	0.94416	0.91596
	Gemma2 9B	0.99858	0.95208	0.85133	0.97735	0.93209
	Llama3 8B	0.99296	0.83669	0.93477	0.88760	0.76157
	Falcon 7B	0.98158	0.84827	0.88921	0.91568	0.85181
	SmolLM 135M	0.99547	0.66195	0.90323	0.96461	0.56232
	SmolLM 360M	0.99618	0.65566	0.92177	0.96672	0.52354
	SmolLM 1.7B	0.99727	0.66346	0.89288	0.97815	0.55803
	SmolLM2 135M	0.99138	0.70938	0.86770	0.91307	0.63049
	SmolLM2 360M	0.99316	0.72655	0.88511	0.91758	0.61758
	SmolLM2 1.7B	0.99244	0.77125	0.92608	0.89780	0.62326
	GPT Neo 2.7B	0.98512	0.71435	0.86825	0.89332	0.71570
	GPTJ 6B	0.98570	0.71594	0.86863	0.90173	0.71785
AI vs Human	Gemma2 2B	0.96212	0.89961	0.92421	0.91899	0.87668
	Gemma2 9B	0.98664	0.92074	0.91030	0.96030	0.87927
	Llama3 8B	0.96199	0.87308	0.93684	0.91048	0.70980
	Falcon 7B	0.92515	0.94468	0.91994	0.90292	0.90367
	SmolLM 135M	0.95002	0.79233	0.89341	0.90059	0.59842
	SmolLM 360M	0.95896	0.77711	0.90261	0.91175	0.58062
	SmolLM 1.7B	0.97194	0.75238	0.90740	0.92639	0.53139
	SmolLM2 135M	0.94240	0.88825	0.88505	0.87073	0.80354
	SmolLM2 360M	0.95290	0.89248	0.90376	0.88936	0.75648
	SmolLM2 1.7B	0.96956	0.87438	0.91992	0.90208	0.68395
	GPT Neo 2.7B	0.92374	0.86087	0.88701	0.86402	0.86568
	GPTJ 6B	0.91179	0.87970	0.89867	0.88034	0.88351
HC3	Gemma2 2B	0.99031	0.99596	0.99876	0.99488	0.99403
	Gemma2 9B	0.99751	0.99401	0.99770	0.99123	0.98854
	Llama3 8B	0.99328	0.98834	0.99595	0.98257	0.92389
	Falcon 7B	0.99816	0.99903	0.99854	0.99290	0.99460
	SmolLM 135M	0.99685	0.98999	0.99858	0.99480	0.87981
	SmolLM 360M	0.99873	0.99053	0.99861	0.99415	0.87349
	SmolLM 1.7B	0.99636	0.99060	0.99877	0.99117	0.87977
	SmolLM2 135M	0.98890	0.99395	0.95844	0.89142	0.99546
	SmolLM2 360M	0.99703	0.99894	0.99819	0.99469	0.99154
	SmolLM2 1.7B	0.99503	0.99495	0.99753	0.99029	0.95253
	GPT Neo 2.7B	0.97589	0.99895	0.99788	0.99701	0.99880
	GPTJ 6B	0.97057	0.99772	0.99751	0.99721	0.99756
HC3 Plus	Gemma2 2B	0.98585	0.88405	0.92220	0.91283	0.88782
	Gemma2 9B	0.98446	0.88357	0.91180	0.88890	0.88837
	Llama3 8B	0.98610	0.90728	0.90945	0.88127	0.86948
	Falcon 7B	0.97602	0.87147	0.88716	0.87069	0.84049
	SmolLM 135M	0.99092	0.87831	0.89526	0.86674	0.77864
	SmolLM 360M	0.99046	0.86239	0.89705	0.86342	0.71806
	SmolLM 1.7B	0.98805	0.85848	0.89863	0.86726	0.71921
	SmolLM2 135M	0.99167	0.90861	0.90685	0.85748	0.88758
	SmolLM2 360M	0.99095	0.89558	0.90796	0.86146	0.86440
	SmolLM2 1.7B	0.98718	0.89391	0.91730	0.87319	0.79400
	GPT Neo 2.7B	0.97205	0.89428	0.93543	0.89346	0.89547
	GPTJ 6B	0.97038	0.88330	0.93075	0.89422	0.88545
ESL GPT4o Mini	Gemma2 2B	0.99994	0.85084	0.85037	0.63529	0.79330
	Gemma2 9B	0.99990	0.86277	0.69415	0.82009	0.80347
	Llama3 8B	0.99992	0.76661	0.82423	0.58940	0.60317
	Falcon 7B	0.99946	0.93775	0.80207	0.58803	0.81207
	SmolLM 135M	0.99988	0.66521	0.87070	0.84206	0.31612
	SmolLM 360M	0.99997	0.60768	0.84358	0.84526	0.27595
	SmolLM 1.7B	0.99985	0.64997	0.84296	0.82979	0.36094
	SmolLM2 135M	0.99981	0.88958	0.87037	0.70646	0.60396
	SmolLM2 360M	0.99987	0.85908	0.87069	0.68898	0.54882
	SmolLM2 1.7B	0.99996	0.83895	0.86639	0.63514	0.55588
	GPT Neo 2.7B	0.99950	0.79548	0.78935	0.56247	0.77656
	GPTJ 6B	0.99990	0.83254	0.77787	0.54312	0.82215

Table 11. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the AUROC of each reference model on each dataset and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	AUROC				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Essay ChatGPT	Gemma2 2B	0.99987	0.99845	0.99979	0.99868	0.98578
	Gemma2 9B	1.00000	0.99796	0.99922	0.99956	0.99347
	Llama3 8B	0.99999	0.90958	0.99953	0.99878	0.62873
	Falcon 7B	0.99846	0.99991	0.99932	0.99698	0.97445
	SmolLM 135M	0.99994	0.80145	0.99924	0.99726	0.21200
	SmolLM 360M	1.00000	0.62785	0.99938	0.99898	0.07174
	SmolLM 1.7B	0.99999	0.52616	0.99957	0.99907	0.02705
	SmolLM2 135M	1.00000	0.98192	0.99874	0.99498	0.43133
	SmolLM2 360M	1.00000	0.96320	0.99947	0.99781	0.21501
	SmolLM2 1.7B	1.00000	0.80636	0.99965	0.99889	0.13744
	GPT Neo 2.7B	0.90356	0.99963	0.99118	0.99181	0.99899
	GPTJ 6B	0.93354	0.99965	0.99208	0.99479	0.99893
GB News ChatGPT	Gemma2 2B	0.97920	0.99800	0.99064	0.98780	0.98848
	Gemma2 9B	0.99462	0.99547	0.97712	0.98331	0.98277
	Llama3 8B	0.97055	0.98634	0.98952	0.98963	0.88491
	Falcon 7B	0.85994	0.99975	0.99074	0.99158	0.99754
	SmolLM 135M	0.99511	0.97570	0.99275	0.99237	0.84309
	SmolLM 360M	0.99639	0.96440	0.99214	0.99365	0.78470
	SmolLM 1.7B	0.99594	0.94155	0.99171	0.99405	0.62111
	SmolLM2 135M	0.98123	0.99926	0.99167	0.99233	0.99623
	SmolLM2 360M	0.97502	0.99963	0.99052	0.99242	0.99305
	SmolLM2 1.7B	0.98596	0.99537	0.98714	0.98889	0.94381
	GPT Neo 2.7B	0.52830	0.99893	0.98490	0.99053	0.99881
	GPTJ 6B	0.59538	0.99834	0.97922	0.98940	0.99825
GB Creative ChatGPT	Gemma2 2B	0.99604	0.97944	0.96948	0.94345	0.91437
	Gemma2 9B	0.99895	0.97285	0.96116	0.96909	0.92390
	Llama3 8B	0.99836	0.92792	0.97838	0.94031	0.50947
	Falcon 7B	0.99634	0.99557	0.94199	0.88229	0.97339
	SmolLM 135M	0.99764	0.91721	0.96192	0.96186	0.20938
	SmolLM 360M	0.99881	0.81492	0.96362	0.97309	0.08571
	SmolLM 1.7B	0.99826	0.66920	0.96321	0.97972	0.03862
	SmolLM2 135M	0.99791	0.97418	0.94415	0.90293	0.40744
	SmolLM2 360M	0.99901	0.94520	0.95500	0.91862	0.19595
	SmolLM2 1.7B	0.99972	0.89043	0.96107	0.93553	0.09403
	GPT Neo 2.7B	0.97754	0.96010	0.89463	0.79011	0.95570
	GPTJ 6B	0.96913	0.97450	0.90413	0.82528	0.97233
GB Essay GPT4o Mini	Gemma2 2B	0.99993	0.99683	0.99885	0.99814	0.85312
	Gemma2 9B	1.00000	0.99578	0.99696	0.99957	0.90371
	Llama3 8B	1.00000	0.98817	0.99836	0.99744	0.43800
	Falcon 7B	0.99475	0.99960	0.99571	0.98937	0.99044
	SmolLM 135M	0.99955	0.70467	0.99018	0.99261	0.13627
	SmolLM 360M	0.99990	0.53669	0.99420	0.99786	0.11993
	SmolLM 1.7B	0.99985	0.30103	0.99416	0.99863	0.09223
	SmolLM2 135M	0.99984	0.98026	0.99174	0.98529	0.45491
	SmolLM2 360M	0.99997	0.94472	0.99642	0.99451	0.11646
	SmolLM2 1.7B	0.99999	0.82658	0.99824	0.99733	0.08496
	GPT Neo 2.7B	0.87843	0.99108	0.98302	0.96778	0.99154
	GPTJ 6B	0.90412	0.99515	0.98595	0.98101	0.99564
GB Creative GPT4o Mini	Gemma2 2B	0.99343	0.98986	0.97319	0.92509	0.78200
	Gemma2 9B	0.99865	0.97816	0.96189	0.96531	0.78665
	Llama3 8B	0.99901	0.96322	0.97554	0.93708	0.58860
	Falcon 7B	0.99046	0.99425	0.87453	0.79556	0.99496
	SmolLM 135M	0.99457	0.89682	0.91568	0.90061	0.33064
	SmolLM 360M	0.99852	0.78049	0.93359	0.93911	0.19132
	SmolLM 1.7B	0.99905	0.52650	0.92108	0.97226	0.10307
	SmolLM2 135M	0.99630	0.98213	0.89567	0.80402	0.83478
	SmolLM2 360M	0.99862	0.98275	0.92131	0.85511	0.70817
	SmolLM2 1.7B	0.99978	0.93026	0.94156	0.90110	0.37861
	GPT Neo 2.7B	0.97333	0.95459	0.87377	0.71989	0.97198
	GPTJ 6B	0.97082	0.97404	0.88857	0.76980	0.98679

Table 12. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the AUROC of each reference model on each dataset and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	AUROC				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB News Claude	Gemma2 2B	0.92326	0.94816	0.92999	0.92280	0.80865
	Gemma2 9B	0.98566	0.98377	0.90450	0.97768	0.85818
	Llama3 8B	0.94244	0.89252	0.96908	0.92132	0.69805
	Falcon 7B	0.69352	0.96214	0.92459	0.89289	0.92435
	SmolLM 135M	0.86539	0.83901	0.79028	0.78966	0.69045
	SmolLM 360M	0.93531	0.81669	0.82855	0.85974	0.65129
	SmolLM 1.7B	0.96109	0.84114	0.85459	0.88730	0.66020
	SmolLM2 135M	0.78553	0.90445	0.79296	0.76103	0.78154
	SmolLM2 360M	0.85195	0.93053	0.84060	0.80185	0.82256
	SmolLM2 1.7B	0.90307	0.94830	0.88596	0.83669	0.83607
	GPT Neo 2.7B	0.61459	0.84221	0.87027	0.85370	0.82409
	GPTJ 6B	0.67877	0.80263	0.87400	0.85333	0.77896
GB Creative Claude	Gemma2 2B	0.96752	0.95569	0.93534	0.92088	0.82197
	Gemma2 9B	0.99637	0.97876	0.94173	0.97766	0.87951
	Llama3 8B	0.99272	0.81250	0.97210	0.94611	0.42541
	Falcon 7B	0.97301	0.91781	0.89573	0.85946	0.84564
	SmolLM 135M	0.95966	0.79714	0.82859	0.84268	0.38952
	SmolLM 360M	0.98579	0.64890	0.86804	0.89808	0.26002
	SmolLM 1.7B	0.98967	0.64154	0.88263	0.93873	0.26338
	SmolLM2 135M	0.95666	0.87882	0.82102	0.75189	0.62580
	SmolLM2 360M	0.97669	0.88031	0.87133	0.81404	0.62142
	SmolLM2 1.7B	0.99317	0.89034	0.91813	0.89562	0.57898
	GPT Neo 2.7B	0.89844	0.75945	0.87518	0.80049	0.74052
	GPTJ 6B	0.90283	0.79026	0.90665	0.84817	0.78099
GB Essay Claude	Gemma2 2B	0.96218	0.93386	0.98014	0.97177	0.84094
	Gemma2 9B	0.99443	0.99001	0.96776	0.99572	0.96143
	Llama3 8B	0.99033	0.82850	0.98649	0.98462	0.52133
	Falcon 7B	0.93462	0.96165	0.96284	0.95913	0.88352
	SmolLM 135M	0.94860	0.41458	0.87175	0.92706	0.23946
	SmolLM 360M	0.98438	0.30006	0.90797	0.97047	0.16140
	SmolLM 1.7B	0.98358	0.48143	0.94139	0.98369	0.28624
	SmolLM2 135M	0.96755	0.90387	0.89353	0.90096	0.65658
	SmolLM2 360M	0.98737	0.89085	0.93802	0.95359	0.60700
	SmolLM2 1.7B	0.99449	0.76260	0.96776	0.97681	0.44343
	GPT Neo 2.7B	0.75228	0.89978	0.93997	0.93490	0.89199
	GPTJ 6B	0.80691	0.90736	0.95958	0.95986	0.90269
GB Essay Deepseek V3	Gemma2 2B	0.99996	0.99986	0.99974	0.99949	0.99695
	Gemma2 9B	1.00000	0.99999	0.99950	0.99973	0.99911
	Llama3 8B	1.00000	0.99889	0.99987	0.99679	0.97004
	Falcon 7B	0.99801	0.99997	0.99915	0.99632	0.99424
	SmolLM 135M	0.99976	0.98964	0.99961	0.99508	0.68435
	SmolLM 360M	0.99992	0.97943	0.99982	0.99845	0.63840
	SmolLM 1.7B	0.99999	0.94667	0.99988	0.99845	0.61856
	SmolLM2 135M	0.99992	0.99961	0.99937	0.99393	0.86547
	SmolLM2 360M	0.99997	0.99946	0.99979	0.99804	0.58716
	SmolLM2 1.7B	1.00000	0.99665	0.99993	0.99890	0.58233
	GPT Neo 2.7B	0.89769	0.99797	0.99404	0.99136	0.99729
	GPTJ 6B	0.92291	0.99887	0.99505	0.99506	0.99767
GB Creative Deepseek V3	Gemma2 2B	0.99598	0.99987	0.99806	0.98993	0.99373
	Gemma2 9B	0.99973	0.99993	0.99879	0.99006	0.99678
	Llama3 8B	0.99975	0.99980	0.99889	0.99099	0.91018
	Falcon 7B	0.98690	0.99986	0.98623	0.96526	0.99281
	SmolLM 135M	0.98743	0.98947	0.98617	0.94281	0.81815
	SmolLM 360M	0.99701	0.98976	0.99177	0.96678	0.77653
	SmolLM 1.7B	0.99885	0.97721	0.99343	0.97774	0.71417
	SmolLM2 135M	0.99637	0.99890	0.98290	0.94078	0.94352
	SmolLM2 360M	0.99946	0.99957	0.98931	0.96173	0.89036
	SmolLM2 1.7B	0.99996	0.99952	0.99577	0.98406	0.83254
	GPT Neo 2.7B	0.91809	0.99543	0.96677	0.91081	0.98747
	GPTJ 6B	0.90433	0.99893	0.97411	0.94598	0.99642

Table 13. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across a variety of perturbation schemes and reference models. We report the AUROC of each reference model on each perturbation scheme and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	AUROC				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Spelling Error Perturbation	Gemma2 2B	0.93457	0.95360	0.91127	0.91503	0.95203
	Gemma2 9B	0.98599	0.96196	0.91074	0.95631	0.94367
	Llama3 8B	0.95360	0.83518	0.93290	0.95245	0.52696
	Falcon 7B	0.81250	0.97345	0.89360	0.88848	0.95255
	SmoLLM 135M	0.94586	0.44095	0.85347	0.87552	0.22272
	SmoLLM 360M	0.95307	0.42454	0.87364	0.90855	0.25867
	SmoLLM 1.7B	0.96122	0.38336	0.88712	0.92799	0.20809
	SmoLLM2 135M	0.91127	0.87458	0.83288	0.85546	0.65531
	SmoLLM2 360M	0.94158	0.84772	0.86173	0.88838	0.57912
	SmoLLM2 1.7B	0.96812	0.73986	0.89360	0.91994	0.40040
	GPT Neo 2.7B	0.73871	0.90447	0.82912	0.83340	0.91451
	GPTJ 6B	0.74425	0.92694	0.85117	0.86528	0.93207
GB Character Capitalization Perturbation	Gemma2 2B	0.99683	0.97054	0.93095	0.93750	0.91980
	Gemma2 9B	1.00000	0.98097	0.93678	0.96911	0.92666
	Llama3 8B	0.99795	0.81086	0.95571	0.94640	0.41520
	Falcon 7B	0.92359	0.98609	0.91837	0.90824	0.95264
	SmoLLM 135M	0.95857	0.57334	0.87531	0.88912	0.23977
	SmoLLM 360M	0.96993	0.47412	0.89259	0.92226	0.22085
	SmoLLM 1.7B	0.97259	0.41520	0.90763	0.92881	0.19118
	SmoLLM2 135M	0.91694	0.92349	0.85679	0.85976	0.68586
	SmoLLM2 360M	0.96072	0.86446	0.88830	0.90651	0.49386
	SmoLLM2 1.7B	0.98885	0.76330	0.91714	0.93157	0.39413
	GPT Neo 2.7B	0.78867	0.91408	0.85516	0.85342	0.91960
	GPTJ 6B	0.84820	0.94640	0.87889	0.88216	0.94589
GB Space Insertion Perturbation	Gemma2 2B	0.96750	0.97847	0.94419	0.95234	0.96237
	Gemma2 9B	0.99436	0.99049	0.94367	0.97596	0.96102
	Llama3 8B	0.97366	0.87030	0.96102	0.96875	0.56062
	Falcon 7B	0.87322	0.98997	0.93238	0.92830	0.96917
	SmoLLM 135M	0.94774	0.54933	0.90008	0.91587	0.23913
	SmoLLM 360M	0.95778	0.59751	0.91367	0.93813	0.32212
	SmoLLM 1.7B	0.96332	0.49258	0.92067	0.95182	0.26108
	SmoLLM2 135M	0.91733	0.93792	0.88169	0.89632	0.71091
	SmoLLM2 360M	0.93886	0.90102	0.90792	0.91952	0.60044
	SmoLLM2 1.7B	0.96081	0.84344	0.93384	0.94346	0.49885
	GPT Neo 2.7B	0.73443	0.95098	0.88336	0.88827	0.95401
	GPTJ 6B	0.74812	0.97533	0.89726	0.91879	0.97575
GB Swap Adjacent Paragraphs Perturbation	Gemma2 2B	0.99274	0.99396	0.97883	0.96082	0.92748
	Gemma2 9B	0.99857	0.98558	0.96420	0.98149	0.92277
	Llama3 8B	0.99192	0.89178	0.98128	0.96307	0.57590
	Falcon 7B	0.94415	0.99816	0.97105	0.95090	0.98067
	SmoLLM 135M	0.99632	0.81833	0.96819	0.97893	0.33224
	SmoLLM 360M	0.99632	0.73292	0.96737	0.98609	0.25491
	SmoLLM 1.7B	0.99427	0.61191	0.96553	0.98507	0.17799
	SmoLLM2 135M	0.99478	0.98261	0.96645	0.95816	0.55718
	SmoLLM2 360M	0.99284	0.95591	0.96911	0.97136	0.42400
	SmoLLM2 1.7B	0.99458	0.85475	0.97494	0.97791	0.36538
	GPT Neo 2.7B	0.76892	0.98987	0.93320	0.91459	0.98312
	GPTJ 6B	0.82570	0.98957	0.93372	0.93249	0.98517
GB Paraphrase Paragraphs Perturbation	Gemma2 2B	0.97570	0.98027	0.96133	0.94317	0.90136
	Gemma2 9B	0.99621	0.98576	0.93102	0.97139	0.90280
	Llama3 8B	0.96799	0.84962	0.96224	0.97060	0.54965
	Falcon 7B	0.90645	0.99895	0.95270	0.94735	0.96290
	SmoLLM 135M	0.99660	0.75464	0.96642	0.98145	0.34897
	SmoLLM 360M	0.98498	0.67154	0.95741	0.99007	0.29814
	SmoLLM 1.7B	0.99059	0.59172	0.95532	0.99334	0.25712
	SmoLLM2 135M	0.97949	0.97126	0.96015	0.95244	0.65756
	SmoLLM2 360M	0.98615	0.92527	0.95924	0.96525	0.53123
	SmoLLM2 1.7B	0.98027	0.81617	0.95845	0.97779	0.45479
	GPT Neo 2.7B	0.72002	0.98824	0.91769	0.91756	0.98484
	GPTJ 6B	0.73034	0.98981	0.91220	0.93703	0.98419

Table 14. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across a variety of perturbation schemes and reference models. We report the AUROC of each reference model on each perturbation scheme and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	AUROC				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Swap Adjacent Sentences Perturbation	Gemma2 2B	0.98824	0.98476	0.97432	0.96471	0.93054
	Gemma2 9B	0.99529	0.99131	0.96082	0.97279	0.94006
	Llama3 8B	0.97954	0.90671	0.97054	0.95714	0.63144
	Falcon 7B	0.93351	0.99448	0.95448	0.93781	0.97647
	SmolLM 135M	0.99325	0.83449	0.96072	0.97719	0.46819
	SmolLM 360M	0.99223	0.80953	0.96195	0.98425	0.46389
	SmolLM 1.7B	0.98527	0.73813	0.96471	0.97668	0.43903
	SmolLM2 135M	0.97719	0.99028	0.96358	0.95029	0.73537
	SmolLM2 360M	0.98793	0.97698	0.96461	0.96215	0.51780
	SmolLM2 1.7B	0.99069	0.87817	0.96993	0.96870	0.47289
	GPT Neo 2.7B	0.73271	0.97842	0.93556	0.91244	0.97811
	GPTJ 6B	0.75951	0.98128	0.93474	0.92921	0.97852
GB Paraphrase Sentences Perturbation	Gemma2 2B	0.98750	0.99284	0.98665	0.97340	0.92425
	Gemma2 9B	0.99882	0.99391	0.97511	0.98654	0.94038
	Llama3 8B	0.98814	0.87660	0.98600	0.97938	0.61239
	Falcon 7B	0.94231	0.99840	0.97265	0.94957	0.98451
	SmolLM 135M	0.98835	0.82596	0.97393	0.98365	0.44444
	SmolLM 360M	0.98996	0.74637	0.97479	0.98590	0.41368
	SmolLM 1.7B	0.98344	0.71645	0.97660	0.98675	0.41528
	SmolLM2 135M	0.96966	0.99038	0.97521	0.96058	0.75299
	SmolLM2 360M	0.97831	0.95855	0.97553	0.97372	0.50801
	SmolLM2 1.7B	0.99220	0.83921	0.98013	0.98419	0.42885
	GPT Neo 2.7B	0.76165	0.98910	0.95673	0.93803	0.98921
	GPTJ 6B	0.78921	0.98996	0.95919	0.95363	0.98974
GB Swap Adjacent Words Perturbation	Gemma2 2B	0.72934	0.95029	0.89556	0.90119	0.95765
	Gemma2 9B	0.88513	0.95468	0.90088	0.94374	0.94292
	Llama3 8B	0.73363	0.87029	0.91172	0.91991	0.62050
	Falcon 7B	0.46440	0.97115	0.88277	0.88349	0.95479
	SmolLM 135M	0.85137	0.59278	0.84155	0.87725	0.34012
	SmolLM 360M	0.86753	0.50020	0.84840	0.90538	0.30646
	SmolLM 1.7B	0.88134	0.31659	0.85025	0.91459	0.20448
	SmolLM2 135M	0.73588	0.89351	0.81772	0.84247	0.71266
	SmolLM2 360M	0.68934	0.85454	0.83674	0.85638	0.66448
	SmolLM2 1.7B	0.84032	0.67052	0.86283	0.90313	0.45908
	GPT Neo 2.7B	0.59881	0.81935	0.81219	0.81669	0.85342
	GPTJ 6B	0.53243	0.88063	0.82334	0.83705	0.91039
GB Swap Word with Synonyms Perturbation	Gemma2 2B	0.91039	0.94251	0.92205	0.91090	0.93331
	Gemma2 9B	0.97934	0.92584	0.91049	0.94149	0.91295
	Llama3 8B	0.88482	0.84503	0.93259	0.92962	0.55360
	Falcon 7B	0.71379	0.97105	0.90794	0.89341	0.93668
	SmolLM 135M	0.95172	0.71809	0.89689	0.91858	0.33480
	SmolLM 360M	0.94988	0.60843	0.89945	0.93126	0.27424
	SmolLM 1.7B	0.96410	0.40221	0.90722	0.93402	0.16581
	SmolLM2 135M	0.88963	0.94466	0.88697	0.90528	0.70970
	SmolLM2 360M	0.87991	0.90395	0.89566	0.90978	0.60106
	SmolLM2 1.7B	0.94415	0.75583	0.91377	0.92297	0.44108
	GPT Neo 2.7B	0.70806	0.92911	0.86375	0.87326	0.94169
	GPTJ 6B	0.67359	0.94108	0.87050	0.88482	0.94865

Table 15. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the confidence interval of the AUROC of each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	Telescope	Binoculars	AUROC Perplexity	DetectLLM LRR	Fast-DetectGPT
Detect LLM Text	Gemma2 2B	(0.99549, 0.99729)	(0.93036, 0.93952)	(0.90215, 0.91356)	(0.93980, 0.94852)	(0.91052, 0.92141)
	Gemma2 9B	(0.99800, 0.99916)	(0.94830, 0.95585)	(0.84376, 0.85890)	(0.97478, 0.97991)	(0.92731, 0.93687)
	Llama3 8B	(0.99181, 0.99412)	(0.82836, 0.84502)	(0.93030, 0.93924)	(0.88108, 0.89412)	(0.75156, 0.77157)
	Falcon 7B	(0.97936, 0.98379)	(0.83910, 0.85744)	(0.88255, 0.89586)	(0.91000, 0.92137)	(0.84356, 0.86006)
	SmolLM 135M	(0.99439, 0.99656)	(0.64912, 0.67478)	(0.89727, 0.90920)	(0.96130, 0.96793)	(0.54974, 0.57489)
	SmolLM 360M	(0.99521, 0.99715)	(0.64323, 0.66810)	(0.91661, 0.92692)	(0.96356, 0.96988)	(0.51093, 0.53615)
	SmolLM 1.7B	(0.99644, 0.99811)	(0.65142, 0.67551)	(0.88667, 0.89909)	(0.97572, 0.98059)	(0.54548, 0.57058)
	SmolLM2 135M	(0.98995, 0.99281)	(0.69579, 0.72298)	(0.86034, 0.87506)	(0.90743, 0.91870)	(0.61667, 0.64430)
	SmolLM2 360M	(0.99197, 0.99435)	(0.71372, 0.73939)	(0.87845, 0.89178)	(0.91217, 0.92300)	(0.60398, 0.63117)
	SmolLM2 1.7B	(0.99115, 0.99374)	(0.76072, 0.78178)	(0.92120, 0.93096)	(0.89169, 0.90390)	(0.61089, 0.63563)
	GPT Neo 2.7B	(0.98310, 0.98713)	(0.70075, 0.72794)	(0.86105, 0.87545)	(0.88698, 0.89965)	(0.70226, 0.72915)
	GPTJ 6B	(0.98373, 0.98767)	(0.70237, 0.72951)	(0.86144, 0.87581)	(0.89566, 0.90779)	(0.70441, 0.73128)
AI vs Human	Gemma2 2B	(0.95826, 0.96597)	(0.89267, 0.90655)	(0.91811, 0.93030)	(0.91284, 0.92515)	(0.86937, 0.88398)
	Gemma2 9B	(0.98444, 0.98883)	(0.91476, 0.92672)	(0.90374, 0.91687)	(0.95604, 0.96456)	(0.87225, 0.88629)
	Llama3 8B	(0.95828, 0.96571)	(0.86476, 0.88140)	(0.93144, 0.94224)	(0.90405, 0.91692)	(0.69861, 0.72100)
	Falcon 7B	(0.91852, 0.93177)	(0.93778, 0.95159)	(0.91249, 0.92740)	(0.89473, 0.91112)	(0.89503, 0.91232)
	SmolLM 135M	(0.94541, 0.95462)	(0.78142, 0.80325)	(0.88589, 0.90093)	(0.89351, 0.90768)	(0.58577, 0.61107)
	SmolLM 360M	(0.95492, 0.96300)	(0.76581, 0.78842)	(0.89549, 0.90974)	(0.90517, 0.91833)	(0.56777, 0.59347)
	SmolLM 1.7B	(0.96872, 0.97515)	(0.74072, 0.76405)	(0.90053, 0.91427)	(0.92044, 0.93233)	(0.51839, 0.54440)
	SmolLM2 135M	(0.93743, 0.94737)	(0.87967, 0.89683)	(0.87730, 0.89279)	(0.86264, 0.87882)	(0.79279, 0.81430)
	SmolLM2 360M	(0.94848, 0.95733)	(0.88433, 0.90064)	(0.89674, 0.91079)	(0.88188, 0.89684)	(0.74510, 0.76785)
	SmolLM2 1.7B	(0.96612, 0.97301)	(0.86601, 0.88276)	(0.91357, 0.92627)	(0.89521, 0.90896)	(0.67175, 0.69615)
	GPT Neo 2.7B	(0.91837, 0.92912)	(0.85106, 0.87068)	(0.87952, 0.89450)	(0.85578, 0.87225)	(0.85623, 0.87513)
	GPTJ 6B	(0.90591, 0.91767)	(0.87044, 0.88896)	(0.89163, 0.90570)	(0.87261, 0.88807)	(0.87456, 0.89246)
HC3	Gemma2 2B	(0.98866, 0.99196)	(0.99480, 0.99713)	(0.99816, 0.99937)	(0.99371, 0.99606)	(0.99275, 0.99531)
	Gemma2 9B	(0.99682, 0.99821)	(0.99267, 0.99536)	(0.99690, 0.99850)	(0.98975, 0.99271)	(0.98666, 0.99042)
	Llama3 8B	(0.99159, 0.99496)	(0.98649, 0.99020)	(0.99459, 0.99732)	(0.98024, 0.98489)	(0.91882, 0.92897)
	Falcon 7B	(0.99765, 0.99866)	(0.99825, 0.99981)	(0.99784, 0.99924)	(0.99155, 0.99425)	(0.99341, 0.99579)
	SmolLM 135M	(0.99608, 0.99761)	(0.98837, 0.99161)	(0.99795, 0.99920)	(0.99359, 0.99602)	(0.87330, 0.88632)
	SmolLM 360M	(0.99832, 0.99915)	(0.98899, 0.99207)	(0.99800, 0.99921)	(0.99288, 0.99542)	(0.86679, 0.88020)
	SmolLM 1.7B	(0.99537, 0.99735)	(0.98908, 0.99211)	(0.99822, 0.99932)	(0.98958, 0.99276)	(0.87323, 0.88632)
	SmolLM2 135M	(0.98717, 0.99062)	(0.99266, 0.99525)	(0.95400, 0.96289)	(0.88438, 0.89846)	(0.99444, 0.99648)
	SmolLM2 360M	(0.99628, 0.99779)	(0.99818, 0.99969)	(0.99747, 0.99892)	(0.99346, 0.99592)	(0.99005, 0.99304)
	SmolLM2 1.7B	(0.99356, 0.99649)	(0.99375, 0.99615)	(0.99664, 0.99842)	(0.98841, 0.99217)	(0.94859, 0.95646)
	GPT Neo 2.7B	(0.97271, 0.97907)	(0.99817, 0.99972)	(0.99699, 0.99877)	(0.99598, 0.99804)	(0.99801, 0.99958)
	GPTJ 6B	(0.96679, 0.97434)	(0.99653, 0.99891)	(0.99657, 0.99845)	(0.99620, 0.99823)	(0.99637, 0.99876)
HC3 Plus	Gemma2 2B	(0.98389, 0.98781)	(0.87721, 0.89088)	(0.91703, 0.92737)	(0.90723, 0.91842)	(0.88142, 0.89421)
	Gemma2 9B	(0.98257, 0.98635)	(0.87669, 0.89045)	(0.90617, 0.91743)	(0.88245, 0.89536)	(0.88198, 0.89475)
	Llama3 8B	(0.98374, 0.98846)	(0.90127, 0.91329)	(0.90372, 0.91517)	(0.87452, 0.88802)	(0.86265, 0.87631)
	Falcon 7B	(0.97358, 0.97846)	(0.86392, 0.87903)	(0.88053, 0.89379)	(0.86344, 0.87794)	(0.83248, 0.84850)
	SmolLM 135M	(0.98963, 0.99222)	(0.87132, 0.88530)	(0.88890, 0.90161)	(0.85934, 0.87413)	(0.76969, 0.78760)
	SmolLM 360M	(0.98914, 0.99177)	(0.85505, 0.86972)	(0.89080, 0.90330)	(0.85593, 0.87091)	(0.70817, 0.72795)
	SmolLM 1.7B	(0.98642, 0.98967)	(0.85109, 0.86586)	(0.89247, 0.90479)	(0.85993, 0.87459)	(0.70933, 0.72910)
	SmolLM2 135M	(0.99047, 0.99287)	(0.90253, 0.91469)	(0.90102, 0.91267)	(0.84983, 0.86514)	(0.88100, 0.89416)
	SmolLM2 360M	(0.98967, 0.99224)	(0.88900, 0.90215)	(0.90220, 0.91372)	(0.85396, 0.86896)	(0.85726, 0.87154)
	SmolLM2 1.7B	(0.98526, 0.98909)	(0.88747, 0.90035)	(0.91192, 0.92268)	(0.86609, 0.88029)	(0.78536, 0.80264)
	GPT Neo 2.7B	(0.96844, 0.97565)	(0.88757, 0.90098)	(0.93085, 0.94001)	(0.88705, 0.89987)	(0.88905, 0.90190)
	GPTJ 6B	(0.96648, 0.97428)	(0.87617, 0.89043)	(0.92595, 0.93555)	(0.88780, 0.90064)	(0.87861, 0.89229)
ESL GPT4o Mini	Gemma2 2B	(0.99990, 0.99999)	(0.84127, 0.86040)	(0.84044, 0.86031)	(0.62123, 0.64935)	(0.78182, 0.80478)
	Gemma2 9B	(0.99983, 0.99996)	(0.85366, 0.87187)	(0.68066, 0.70763)	(0.80956, 0.83061)	(0.79229, 0.81465)
	Llama3 8B	(0.99985, 0.99999)	(0.75473, 0.77850)	(0.81350, 0.83495)	(0.57492, 0.60388)	(0.58883, 0.61751)
	Falcon 7B	(0.99906, 0.99985)	(0.93171, 0.94379)	(0.79041, 0.81373)	(0.57316, 0.60291)	(0.80100, 0.82314)
	SmolLM 135M	(0.99981, 0.99995)	(0.65148, 0.67894)	(0.86127, 0.88013)	(0.83210, 0.85201)	(0.30271, 0.32953)
	SmolLM 360M	(0.99995, 0.99999)	(0.59337, 0.62200)	(0.83326, 0.85389)	(0.83545, 0.85506)	(0.26320, 0.28871)
	SmolLM 1.7B	(0.99975, 0.99995)	(0.63611, 0.66383)	(0.83267, 0.85325)	(0.81953, 0.84004)	(0.34698, 0.37490)
	SmolLM2 135M	(0.99968, 0.99994)	(0.88147, 0.89769)	(0.86092, 0.87982)	(0.69323, 0.71969)	(0.58963, 0.61829)
	SmolLM2 360M	(0.99968, 1.00005)	(0.84974, 0.86842)	(0.86129, 0.88009)	(0.67555, 0.70240)	(0.53416, 0.56348)
	SmolLM2 1.7B	(0.99993, 0.99999)	(0.82903, 0.84888)	(0.85697, 0.87582)	(0.62107, 0.64922)	(0.54124, 0.57051)
	GPT Neo 2.7B	(0.99912, 0.99988)	(0.78425, 0.80672)	(0.77719, 0.80150)	(0.54775, 0.57720)	(0.76489, 0.78823)
	GPTJ 6B	(0.99980, 1.00000)	(0.82228, 0.84281)	(0.76552, 0.79022)	(0.52833, 0.55791)	(0.81165, 0.83264)

Table 16. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the confidence interval of the AUROC of each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	Telescope	Binoculars	AUROC Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Essay ChatGPT	Gemma2 2B	(0.99974, 1.00001)	(0.99765, 0.99924)	(0.99939, 1.00019)	(0.99766, 0.99970)	(0.98195, 0.98962)
	Gemma2 9B	(0.99999, 1.00000)	(0.99694, 0.99898)	(0.99839, 1.00006)	(0.99923, 0.99989)	(0.99103, 0.99591)
	Llama3 8B	(0.99998, 1.00001)	(0.89672, 0.92244)	(0.99892, 1.00014)	(0.99761, 0.99995)	(0.60431, 0.65316)
	Falcon 7B	(0.99746, 0.99945)	(0.99982, 1.00000)	(0.99847, 1.00018)	(0.99541, 0.99856)	(0.96788, 0.98103)
	SmolLM 135M	(0.99988, 1.00000)	(0.78211, 0.82079)	(0.99865, 0.99983)	(0.99589, 0.99863)	(0.19217, 0.23184)
	SmolLM 360M	(0.99999, 1.00000)	(0.60300, 0.65271)	(0.99885, 0.99992)	(0.99816, 0.99980)	(0.06055, 0.08293)
	SmolLM 1.7B	(0.99997, 1.00001)	(0.50024, 0.55209)	(0.99913, 1.00002)	(0.99819, 0.99994)	(0.02039, 0.03370)
	SmolLM2 135M	(0.99999, 1.00000)	(0.97729, 0.98655)	(0.99782, 0.99966)	(0.99272, 0.99723)	(0.40573, 0.45693)
	SmolLM2 360M	(0.99999, 1.00000)	(0.95590, 0.97049)	(0.99891, 1.00003)	(0.99653, 0.99910)	(0.19445, 0.23557)
	SmolLM2 1.7B	(0.99999, 1.00000)	(0.78690, 0.82582)	(0.99926, 1.00004)	(0.99780, 0.99997)	(0.12090, 0.15398)
GB News ChatGPT	GPT Neo 2.7B	(0.88995, 0.91717)	(0.99930, 0.99995)	(0.98649, 0.99587)	(0.98869, 0.99492)	(0.99749, 1.00050)
	GPTJ 6B	(0.92232, 0.94476)	(0.99921, 1.00009)	(0.98751, 0.99664)	(0.99232, 0.99727)	(0.99711, 1.00076)
	Gemma2 2B	(0.97425, 0.98415)	(0.99678, 0.99921)	(0.98680, 0.99447)	(0.98354, 0.99205)	(0.98470, 0.99227)
	Gemma2 9B	(0.99248, 0.99675)	(0.99358, 0.99736)	(0.97056, 0.98369)	(0.97795, 0.98867)	(0.97787, 0.98767)
	Llama3 8B	(0.96311, 0.97800)	(0.98246, 0.99022)	(0.98532, 0.99372)	(0.98595, 0.99332)	(0.87061, 0.89921)
	Falcon 7B	(0.84393, 0.87595)	(0.99935, 1.00015)	(0.98695, 0.99452)	(0.98813, 0.99504)	(0.99603, 0.99905)
	SmolLM 135M	(0.99316, 0.99705)	(0.96923, 0.98217)	(0.98940, 0.99609)	(0.98897, 0.99577)	(0.82578, 0.86041)
	SmolLM 360M	(0.99469, 0.99808)	(0.95610, 0.97271)	(0.98871, 0.99557)	(0.99046, 0.99684)	(0.76440, 0.80500)
	SmolLM 1.7B	(0.99442, 0.99745)	(0.93106, 0.95204)	(0.98814, 0.99527)	(0.99114, 0.99695)	(0.59634, 0.64587)
	SmolLM2 135M	(0.97652, 0.98594)	(0.99822, 1.00030)	(0.98803, 0.99531)	(0.98883, 0.99583)	(0.99388, 0.99858)
GB Creative ChatGPT	SmolLM2 360M	(0.96907, 0.98098)	(0.99932, 0.99993)	(0.98664, 0.99440)	(0.98900, 0.99584)	(0.99008, 0.99602)
	SmolLM2 1.7B	(0.98177, 0.99014)	(0.99290, 0.99784)	(0.98253, 0.99174)	(0.98488, 0.99290)	(0.93364, 0.95397)
	GPT Neo 2.7B	(0.50294, 0.55366)	(0.99822, 0.99964)	(0.97988, 0.98991)	(0.98670, 0.99435)	(0.99807, 0.99955)
	GPTJ 6B	(0.57059, 0.62016)	(0.99718, 0.99951)	(0.97312, 0.98531)	(0.98539, 0.99341)	(0.99711, 0.99939)
	Gemma2 2B	(0.99341, 0.99867)	(0.97317, 0.98572)	(0.96159, 0.97737)	(0.93308, 0.95382)	(0.90201, 0.92674)
	Gemma2 9B	(0.99743, 1.00047)	(0.96613, 0.97957)	(0.95222, 0.97011)	(0.96161, 0.97657)	(0.91221, 0.93559)
	Llama3 8B	(0.99749, 0.99923)	(0.91644, 0.93939)	(0.97225, 0.98451)	(0.92904, 0.95158)	(0.48396, 0.53499)
	Falcon 7B	(0.99418, 0.99851)	(0.99282, 0.99832)	(0.93103, 0.95295)	(0.86684, 0.89773)	(0.96644, 0.98034)
	SmolLM 135M	(0.99554, 0.99974)	(0.90447, 0.92995)	(0.95292, 0.97092)	(0.95358, 0.97014)	(0.18967, 0.22908)
	SmolLM 360M	(0.99781, 0.99980)	(0.79584, 0.83400)	(0.95489, 0.97235)	(0.96604, 0.98014)	(0.07300, 0.09841)
GB Essay GPT4o Mini	SmolLM 1.7B	(0.99683, 0.99968)	(0.64516, 0.69325)	(0.95445, 0.97196)	(0.97354, 0.98591)	(0.03067, 0.04658)
	SmolLM2 135M	(0.99657, 0.99925)	(0.96751, 0.98085)	(0.93320, 0.95509)	(0.88893, 0.91693)	(0.38233, 0.43254)
	SmolLM2 360M	(0.99828, 0.99975)	(0.93514, 0.95526)	(0.94526, 0.96474)	(0.90583, 0.93141)	(0.17637, 0.21553)
	SmolLM2 1.7B	(0.99948, 0.99995)	(0.87609, 0.90477)	(0.95213, 0.97002)	(0.92415, 0.94691)	(0.08020, 0.10787)
	GPT Neo 2.7B	(0.97154, 0.98354)	(0.95055, 0.96964)	(0.87979, 0.90946)	(0.77009, 0.81013)	(0.94638, 0.96501)
	GPTJ 6B	(0.96206, 0.97620)	(0.96681, 0.98219)	(0.88996, 0.91831)	(0.80672, 0.84384)	(0.96518, 0.97948)
	Gemma2 2B	(0.99986, 1.00000)	(0.99538, 0.99829)	(0.99763, 1.00007)	(0.99695, 0.99933)	(0.83652, 0.86973)
	Gemma2 9B	(0.99999, 1.00000)	(0.99414, 0.99743)	(0.99478, 0.99914)	(0.99924, 0.99990)	(0.89042, 0.91699)
	Llama3 8B	(0.99999, 1.00000)	(0.98438, 0.99195)	(0.99693, 0.99980)	(0.99552, 0.99937)	(0.41274, 0.46326)
	Falcon 7B	(0.99236, 0.99714)	(0.99910, 1.00009)	(0.99349, 0.99793)	(0.98588, 0.99285)	(0.98645, 0.99444)
GB Creative GPT4o Mini	SmolLM 135M	(0.99923, 0.99988)	(0.68187, 0.72747)	(0.98691, 0.99345)	(0.98997, 0.99526)	(0.12027, 0.15228)
	SmolLM 360M	(0.99980, 1.00000)	(0.51098, 0.56240)	(0.99184, 0.99655)	(0.99651, 0.99920)	(0.10530, 0.13457)
	SmolLM 1.7B	(0.99966, 1.00004)	(0.27789, 0.32418)	(0.99161, 0.99671)	(0.99752, 0.99973)	(0.07936, 0.10511)
	SmolLM2 135M	(0.99970, 0.99997)	(0.97508, 0.98543)	(0.98876, 0.99472)	(0.98106, 0.98953)	(0.42931, 0.48051)
	SmolLM2 360M	(0.99994, 1.00001)	(0.93511, 0.95434)	(0.99461, 0.99823)	(0.99224, 0.99678)	(0.10123, 0.13169)
	SmolLM2 1.7B	(0.99998, 1.00001)	(0.80854, 0.84463)	(0.99698, 0.99951)	(0.99557, 0.99908)	(0.07271, 0.09721)
	GPT Neo 2.7B	(0.86300, 0.89386)	(0.98794, 0.99422)	(0.97719, 0.98885)	(0.96116, 0.97441)	(0.98831, 0.99476)
	GPTJ 6B	(0.89051, 0.91774)	(0.99267, 0.99763)	(0.98040, 0.99149)	(0.97614, 0.98588)	(0.99310, 0.99818)
	Gemma2 2B	(0.99009, 0.99677)	(0.98626, 0.99346)	(0.96718, 0.97920)	(0.91398, 0.93620)	(0.76197, 0.80204)
	Gemma2 9B	(0.99680, 1.00051)	(0.97285, 0.98348)	(0.95443, 0.96936)	(0.95835, 0.97226)	(0.76679, 0.80650)
GB News Claude	Llama3 8B	(0.99814, 0.99989)	(0.95547, 0.97097)	(0.96964, 0.98145)	(0.92730, 0.94686)	(0.56357, 0.61362)
	Falcon 7B	(0.98707, 0.99386)	(0.99121, 0.99729)	(0.85911, 0.88994)	(0.77604, 0.81509)	(0.99270, 0.99722)
	SmolLM 135M	(0.99226, 0.99687)	(0.88301, 0.91063)	(0.90317, 0.92819)	(0.88749, 0.91373)	(0.30696, 0.35432)
	SmolLM 360M	(0.99707, 0.99998)	(0.76032, 0.80065)	(0.92262, 0.94455)	(0.92923, 0.94899)	(0.17234, 0.21030)
	SmolLM 1.7B	(0.99785, 1.00026)	(0.50101, 0.55198)	(0.90911, 0.93305)	(0.96620, 0.97832)	(0.08934, 0.11680)
	SmolLM2 135M	(0.99450, 0.99810)	(0.97695, 0.98731)	(0.88171, 0.90963)	(0.78510, 0.82295)	(0.81690, 0.85267)
	SmolLM2 360M	(0.99763, 0.99960)	(0.97775, 0.98775)	(0.90934, 0.93328)	(0.83885, 0.87137)	(0.68458, 0.73176)
	SmolLM2 1.7B	(0.99950, 1.00006)	(0.91918, 0.94134)	(0.93159, 0.95153)	(0.88811, 0.91408)	(0.35351, 0.40370)
	GPT Neo 2.7B	(0.96592, 0.98073)	(0.94593, 0.96326)	(0.85856, 0.88898)	(0.69776, 0.74202)	(0.96576, 0.97819)
	GPTJ 6B	(0.96347, 0.97816)	(0.96762, 0.98046)	(0.87441, 0.90274)	(0.74947, 0.79013)	(0.98299, 0.99060)
GB Essay GPT4o Mini	Gemma2 2B	(0.91202, 0.93450)	(0.93879, 0.95752)	(0.91911, 0.94087)	(0.91150, 0.93410)	(0.79008, 0.82723)
	Gemma2 9B	(0.98163, 0.98970)	(0.97896, 0.98859)	(0.89145, 0.91756)	(0.97242, 0.98293)	(0.84216, 0.87420)
	Llama3 8B	(0.93283, 0.95204)	(0.87719, 0.90785)	(0.96244, 0.97572)	(0.90999, 0.93266)	(0.67430, 0.72180)
	Falcon 7B	(0.66973, 0.71731)	(0.95452, 0.96977)	(0.91298, 0.93621)	(0.87885, 0.90693)	(0.91295, 0.93576)
	SmolLM 135M	(0.84970, 0.88108)	(0.82162, 0.85640)	(0.77059, 0.80997)	(0.77006, 0.80925)	(0.66744, 0.71345)
	SmolLM 360M	(0.92497, 0.94565)	(0.79771, 0.83566)	(0.81075, 0.84634)	(0.84386, 0.87562)	(0.62727, 0.67532)
	SmolLM 1.7B	(0.95367, 0.96852)	(0.82335, 0.85893)	(0.83828, 0.87091)	(0.87304, 0.90157)	(0.63631, 0.68410)
	SmolLM2 135M	(0.76592, 0.80514)	(0.89150, 0.91741)	(0.77348, 0.81245)	(0.74038, 0.78168)	(0.76155, 0.80153)
	SmolLM2 360M	(0.83582, 0.86809)	(0.91988, 0.94118)	(0.82345, 0.85774)	(0.78289, 0.82081)	(0.80453, 0.84059)
	SmolLM2 1.7B	(0.89044, 0.91570)	(0.93924, 0.95737)	(0.87165, 0.90026)	(0.81944, 0.85394)	(0.81861, 0.85353)
GB News Claude	GPT Neo 2.7B	(0.59010, 0.63907)	(0.82514, 0.85928)	(0.85503, 0.88551)	(0.83742, 0.86997)	(0.80614, 0.84205)
	GPTJ 6B	(0.65561, 0.70192)	(0.78355, 0.82171)	(0.85892, 0.88908)	(0.83715, 0.86951)	(0.75888, 0.79903)

Table 17. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the confidence interval of the AUROC of each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	Telescope	Binoculars	AUROC Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Creative Claude	Gemma2 2B	(0.96049, 0.97455)	(0.94717, 0.96422)	(0.92490, 0.94578)	(0.90933, 0.93242)	(0.80374, 0.84019)
	Gemma2 9B	(0.99398, 0.99876)	(0.97323, 0.98428)	(0.93178, 0.95168)	(0.97222, 0.98310)	(0.86452, 0.89450)
	Llama3 8B	(0.98991, 0.99553)	(0.79302, 0.83197)	(0.96541, 0.97879)	(0.93697, 0.95524)	(0.39983, 0.45100)
	Falcon 7B	(0.96698, 0.97903)	(0.90546, 0.93016)	(0.88176, 0.90971)	(0.84342, 0.87551)	(0.82852, 0.86276)
	SmolLM 135M	(0.95176, 0.96755)	(0.77757, 0.81671)	(0.81058, 0.84661)	(0.82561, 0.85976)	(0.36485, 0.41419)
	SmolLM 360M	(0.98166, 0.98992)	(0.62437, 0.67343)	(0.85219, 0.88389)	(0.88462, 0.91154)	(0.23835, 0.28169)
	SmolLM 1.7B	(0.98578, 0.99356)	(0.61655, 0.66652)	(0.86775, 0.89751)	(0.92864, 0.94882)	(0.24152, 0.28524)
	SmolLM2 135M	(0.94863, 0.96468)	(0.86355, 0.89409)	(0.80265, 0.83940)	(0.73081, 0.77297)	(0.60135, 0.65025)
	SmolLM2 360M	(0.97125, 0.98213)	(0.86516, 0.89547)	(0.85574, 0.88693)	(0.79547, 0.83261)	(0.59684, 0.64600)
	SmolLM2 1.7B	(0.99071, 0.99563)	(0.87514, 0.90554)	(0.90593, 0.93033)	(0.88198, 0.90927)	(0.55359, 0.60438)
GB Essay Claude	GPT Neo 2.7B	(0.88467, 0.91222)	(0.73851, 0.78040)	(0.86002, 0.89034)	(0.78148, 0.81950)	(0.71903, 0.76201)
	GPTJ 6B	(0.88917, 0.91650)	(0.77053, 0.80998)	(0.89372, 0.91958)	(0.83165, 0.86469)	(0.76103, 0.80094)
	Gemma2 2B	(0.95442, 0.96995)	(0.92362, 0.94409)	(0.97523, 0.98506)	(0.96568, 0.97786)	(0.82398, 0.85791)
	Gemma2 9B	(0.99156, 0.99730)	(0.98683, 0.99318)	(0.96092, 0.97461)	(0.99397, 0.99748)	(0.95390, 0.96896)
	Llama3 8B	(0.98705, 0.99360)	(0.80982, 0.84719)	(0.98218, 0.99080)	(0.98034, 0.98891)	(0.49545, 0.54720)
	Falcon 7B	(0.92394, 0.94530)	(0.95359, 0.96972)	(0.95535, 0.97033)	(0.95146, 0.96680)	(0.86863, 0.89841)
	SmolLM 135M	(0.93941, 0.95779)	(0.38879, 0.44037)	(0.85647, 0.88703)	(0.91594, 0.93818)	(0.21815, 0.26076)
	SmolLM 360M	(0.97962, 0.98914)	(0.27582, 0.32429)	(0.89536, 0.92059)	(0.96394, 0.97699)	(0.14342, 0.17938)
	SmolLM 1.7B	(0.97895, 0.98822)	(0.45532, 0.50754)	(0.93159, 0.95119)	(0.97938, 0.98799)	(0.26351, 0.30898)
	SmolLM2 135M	(0.96070, 0.97440)	(0.89070, 0.91704)	(0.87975, 0.90730)	(0.88764, 0.91427)	(0.63268, 0.68048)
GB Essay Deepseek V3	SmolLM2 360M	(0.98358, 0.99116)	(0.87552, 0.90618)	(0.92801, 0.94804)	(0.94524, 0.96194)	(0.58189, 0.63210)
	SmolLM2 1.7B	(0.99209, 0.99688)	(0.74031, 0.78490)	(0.96091, 0.97461)	(0.97145, 0.98217)	(0.41730, 0.46956)
	GPT Neo 2.7B	(0.73116, 0.77340)	(0.88661, 0.91294)	(0.92952, 0.95042)	(0.92452, 0.94529)	(0.87837, 0.90561)
	GPTJ 6B	(0.78802, 0.82579)	(0.89459, 0.92012)	(0.95097, 0.96819)	(0.95220, 0.96752)	(0.88971, 0.91567)
	Gemma2 2B	(0.99987, 1.00004)	(0.99961, 1.00012)	(0.99936, 1.00011)	(0.99903, 0.99995)	(0.99556, 0.99833)
	Gemma2 9B	(1.00000, 1.00000)	(0.99998, 1.00001)	(0.99877, 1.00023)	(0.99937, 1.00008)	(0.99844, 0.99978)
	Llama3 8B	(0.99999, 1.00000)	(0.99703, 1.00075)	(0.99973, 1.00001)	(0.99425, 0.99933)	(0.96327, 0.97680)
	Falcon 7B	(0.99695, 0.99907)	(0.99993, 1.00001)	(0.99824, 1.00007)	(0.99391, 0.99873)	(0.99113, 0.99735)
	SmolLM 135M	(0.99948, 1.00003)	(0.98591, 0.99337)	(0.99918, 1.00005)	(0.99307, 0.99709)	(0.66061, 0.70808)
	SmolLM 360M	(0.99981, 1.00002)	(0.97395, 0.98491)	(0.99961, 1.00004)	(0.99736, 0.99954)	(0.61354, 0.66327)
GB Creative Deepseek V3	SmolLM 1.7B	(0.99996, 1.00001)	(0.93695, 0.95640)	(0.99975, 1.00001)	(0.99721, 0.99969)	(0.59327, 0.64384)
	SmolLM2 135M	(0.99980, 1.00003)	(0.99932, 0.99991)	(0.99873, 1.00000)	(0.99135, 0.99651)	(0.84910, 0.88184)
	SmolLM2 360M	(0.99993, 1.00001)	(0.99908, 0.99985)	(0.99953, 1.00006)	(0.99678, 0.99929)	(0.56134, 0.61297)
	SmolLM2 1.7B	(0.99999, 1.00000)	(0.99392, 0.99937)	(0.99986, 1.00000)	(0.99791, 0.99989)	(0.55654, 0.60812)
	GPT Neo 2.7B	(0.88319, 0.91219)	(0.99569, 1.00026)	(0.99020, 0.99788)	(0.98757, 0.99515)	(0.99491, 0.99966)
	GPTJ 6B	(0.91044, 0.93537)	(0.99756, 1.00019)	(0.99172, 0.99838)	(0.99201, 0.99810)	(0.99543, 0.99991)
	Gemma2 2B	(0.99406, 0.99789)	(0.99972, 1.00001)	(0.99677, 0.99936)	(0.98584, 0.99403)	(0.99085, 0.99661)
	Gemma2 9B	(0.99928, 1.00019)	(0.99984, 1.00002)	(0.99795, 0.99964)	(0.98563, 0.99450)	(0.99500, 0.99856)
	Llama3 8B	(0.99953, 0.99998)	(0.99958, 1.00002)	(0.99794, 0.99984)	(0.98691, 0.99507)	(0.89512, 0.92525)
	Falcon 7B	(0.98277, 0.99104)	(0.99973, 1.00000)	(0.98165, 0.99080)	(0.95679, 0.97373)	(0.98965, 0.99598)
GB Essay Claude	SmolLM 135M	(0.98310, 0.99176)	(0.98542, 0.99352)	(0.98117, 0.99117)	(0.93113, 0.95449)	(0.79717, 0.83914)
	SmolLM 360M	(0.99566, 0.99837)	(0.98570, 0.99382)	(0.98809, 0.99546)	(0.95788, 0.97568)	(0.75319, 0.79987)
	SmolLM 1.7B	(0.99820, 0.99951)	(0.97014, 0.98429)	(0.99047, 0.99639)	(0.97099, 0.98448)	(0.68834, 0.74000)
	SmolLM2 135M	(0.99451, 0.99824)	(0.99808, 0.99973)	(0.97697, 0.98882)	(0.92878, 0.95278)	(0.93219, 0.95484)
	SmolLM2 360M	(0.99902, 0.99990)	(0.99913, 1.00002)	(0.98493, 0.99369)	(0.95230, 0.97117)	(0.87348, 0.90723)
	SmolLM2 1.7B	(0.99990, 1.00002)	(0.99901, 1.00003)	(0.99369, 0.99785)	(0.97843, 0.98969)	(0.81193, 0.85316)
	GPT Neo 2.7B	(0.90370, 0.93248)	(0.99325, 0.99762)	(0.95864, 0.97490)	(0.89619, 0.92542)	(0.98364, 0.99131)
	GPTJ 6B	(0.88896, 0.91970)	(0.99815, 0.99971)	(0.96717, 0.98105)	(0.93483, 0.95713)	(0.99489, 0.99794)

Table 18. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across a variety of perturbation schemes and reference models. We report the confidence interval of the AUROC of each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	Telescope	Binoculars	AUROC Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Spelling Error Perturbation	Gemma2 2B	(0.89989, 0.96925)	(0.92012, 0.98707)	(0.86572, 0.95682)	(0.87086, 0.95920)	(0.92540, 0.97865)
	Gemma2 9B	(0.97207, 0.99992)	(0.93764, 0.98627)	(0.86572, 0.95577)	(0.92041, 0.99221)	(0.91320, 0.97413)
	Llama3 8B	(0.92188, 0.98531)	(0.77753, 0.89283)	(0.89162, 0.97418)	(0.91567, 0.98922)	(0.44545, 0.60848)
	Falcon 7B	(0.75326, 0.87174)	(0.94422, 1.00269)	(0.84473, 0.94247)	(0.83952, 0.93745)	(0.92151, 0.98359)
	SmolLM 135M	(0.91310, 0.97862)	(0.35586, 0.52604)	(0.79624, 0.91070)	(0.82338, 0.92767)	(0.14875, 0.29669)
	SmolLM 360M	(0.92134, 0.98480)	(0.33736, 0.51172)	(0.81973, 0.92756)	(0.86239, 0.95471)	(0.17887, 0.33848)
	SmolLM 1.7B	(0.93498, 0.98747)	(0.29785, 0.46888)	(0.83546, 0.93879)	(0.88659, 0.96939)	(0.13934, 0.27684)
	SmolLM2 135M	(0.86924, 0.95329)	(0.82073, 0.92843)	(0.77280, 0.89296)	(0.79979, 0.91112)	(0.57240, 0.73822)
	SmolLM2 360M	(0.90919, 0.97396)	(0.78797, 0.90748)	(0.80614, 0.91731)	(0.83812, 0.93864)	(0.49205, 0.66618)
	SmolLM2 1.7B	(0.94436, 0.99189)	(0.66873, 0.81099)	(0.84429, 0.94292)	(0.87594, 0.96394)	(0.31601, 0.48478)
	GPT Neo 2.7B	(0.67024, 0.80718)	(0.85988, 0.94907)	(0.76927, 0.88897)	(0.77388, 0.89293)	(0.87411, 0.95490)
	GPTJ 6B	(0.67617, 0.81233)	(0.88309, 0.97080)	(0.79407, 0.90827)	(0.81018, 0.92038)	(0.89182, 0.97231)
GB Character Capitalization Perturbation	Gemma2 2B	(0.99293, 1.00073)	(0.94845, 0.99263)	(0.88807, 0.97384)	(0.90054, 0.97446)	(0.88289, 0.95672)
	Gemma2 9B	(1.00000, 1.00000)	(0.96527, 0.99668)	(0.90124, 0.97233)	(0.94608, 0.99214)	(0.89269, 0.96062)
	Llama3 8B	(0.99537, 1.00053)	(0.74916, 0.87257)	(0.92362, 0.98780)	(0.90897, 0.98383)	(0.33534, 0.49506)
	Falcon 7B	(0.88866, 0.95852)	(0.96982, 1.00236)	(0.87180, 0.96494)	(0.86075, 0.95574)	(0.92288, 0.98240)
	SmolLM 135M	(0.92536, 0.99179)	(0.48816, 0.65853)	(0.81949, 0.93112)	(0.83655, 0.94169)	(0.16396, 0.31558)
	SmolLM 360M	(0.94528, 0.99457)	(0.38881, 0.55944)	(0.83942, 0.94577)	(0.87393, 0.97059)	(0.15020, 0.29149)
	SmolLM 1.7B	(0.95249, 0.99268)	(0.32968, 0.50072)	(0.85691, 0.95835)	(0.88268, 0.97494)	(0.12382, 0.25854)
	SmolLM2 135M	(0.87258, 0.96130)	(0.88288, 0.96409)	(0.79837, 0.91521)	(0.80220, 0.91731)	(0.60536, 0.76636)
	SmolLM2 360M	(0.93154, 0.98990)	(0.80820, 0.92073)	(0.83419, 0.94241)	(0.85631, 0.95670)	(0.40338, 0.58434)
	SmolLM2 1.7B	(0.97361, 1.00409)	(0.69194, 0.83466)	(0.86898, 0.96531)	(0.88842, 0.97472)	(0.30939, 0.47887)
	GPT Neo 2.7B	(0.72674, 0.85060)	(0.86751, 0.96064)	(0.79639, 0.91392)	(0.79469, 0.91214)	(0.87619, 0.96301)
	GPTJ 6B	(0.79666, 0.89973)	(0.90884, 0.98395)	(0.82439, 0.93339)	(0.82840, 0.93592)	(0.90982, 0.98195)
GB Space Insertion Perturbation	Gemma2 2B	(0.94603, 0.98896)	(0.95956, 0.99738)	(0.90587, 0.98250)	(0.91857, 0.98611)	(0.93999, 0.98476)
	Gemma2 9B	(0.98709, 1.00163)	(0.97974, 1.00124)	(0.90822, 0.97911)	(0.95259, 0.99933)	(0.93875, 0.98328)
	Llama3 8B	(0.95138, 0.99594)	(0.81983, 0.92077)	(0.92996, 0.99207)	(0.94140, 0.99610)	(0.47975, 0.64149)
	Falcon 7B	(0.82319, 0.92325)	(0.97149, 1.00844)	(0.89055, 0.97421)	(0.88650, 0.97010)	(0.94505, 0.99328)
	SmolLM 135M	(0.91218, 0.98330)	(0.46379, 0.63487)	(0.85159, 0.94857)	(0.87019, 0.96154)	(0.16336, 0.31490)
	SmolLM 360M	(0.92729, 0.98826)	(0.51066, 0.68437)	(0.86850, 0.95884)	(0.89578, 0.98047)	(0.23658, 0.40765)
	SmolLM 1.7B	(0.93658, 0.99006)	(0.40491, 0.58025)	(0.87648, 0.96487)	(0.91438, 0.98926)	(0.18544, 0.33671)
	SmolLM2 135M	(0.87416, 0.96050)	(0.90146, 0.97438)	(0.82978, 0.93359)	(0.84406, 0.94858)	(0.63083, 0.79100)
	SmolLM2 360M	(0.90310, 0.97462)	(0.85731, 0.94474)	(0.86087, 0.95498)	(0.87324, 0.96581)	(0.51152, 0.68936)
	SmolLM2 1.7B	(0.93050, 0.99111)	(0.78468, 0.90219)	(0.89316, 0.97452)	(0.90328, 0.98364)	(0.41290, 0.58480)
	GPT Neo 2.7B	(0.66539, 0.80347)	(0.91810, 0.98386)	(0.83174, 0.93498)	(0.83624, 0.94030)	(0.92461, 0.98341)
	GPTJ 6B	(0.68071, 0.81552)	(0.95168, 0.99898)	(0.84854, 0.94599)	(0.87431, 0.96328)	(0.95412, 0.99739)
GB Swap Adjacent Paragraphs Perturbation	Gemma2 2B	(0.98553, 0.99994)	(0.98816, 0.99977)	(0.95950, 0.99815)	(0.93483, 0.98682)	(0.89124, 0.96371)
	Gemma2 9B	(0.99615, 1.00099)	(0.97442, 0.99673)	(0.94125, 0.98714)	(0.96707, 0.99590)	(0.88624, 0.95930)
	Llama3 8B	(0.98395, 0.99988)	(0.84380, 0.93976)	(0.96451, 0.99805)	(0.93268, 0.99346)	(0.49595, 0.65585)
	Falcon 7B	(0.91202, 0.97628)	(0.99496, 1.00135)	(0.94789, 0.99421)	(0.91910, 0.98270)	(0.96274, 0.99859)
	SmolLM 135M	(0.99221, 1.00043)	(0.75739, 0.87927)	(0.94398, 0.99239)	(0.96241, 0.99544)	(0.25256, 0.41192)
	SmolLM 360M	(0.99214, 1.00050)	(0.65963, 0.80621)	(0.94252, 0.99222)	(0.97214, 1.00004)	(0.17865, 0.33117)
	SmolLM 1.7B	(0.98837, 1.00018)	(0.52944, 0.69438)	(0.94011, 0.99095)	(0.96743, 1.00270)	(0.11279, 0.24318)
	SmolLM2 135M	(0.98903, 1.00054)	(0.96771, 0.99751)	(0.94109, 0.99180)	(0.92868, 0.98764)	(0.47274, 0.64162)
	SmolLM2 360M	(0.98500, 1.00068)	(0.92975, 0.98208)	(0.94362, 0.99460)	(0.94667, 0.99604)	(0.33599, 0.51201)
	SmolLM2 1.7B	(0.98688, 1.00227)	(0.79801, 0.91148)	(0.95272, 0.99715)	(0.95548, 1.00033)	(0.27751, 0.45326)
	GPT Neo 2.7B	(0.70466, 0.83318)	(0.97757, 1.00218)	(0.89804, 0.96837)	(0.87028, 0.95889)	(0.96658, 0.99967)
	GPTJ 6B	(0.76890, 0.88249)	(0.97566, 1.00347)	(0.89861, 0.96882)	(0.89300, 0.97197)	(0.97036, 0.99998)
GB Paraphrase Paragraphs Perturbation	Gemma2 2B	(0.95453, 0.99686)	(0.96301, 0.99753)	(0.93071, 0.99195)	(0.91226, 0.97407)	(0.85709, 0.94562)
	Gemma2 9B	(0.99128, 1.00114)	(0.97300, 0.99852)	(0.89301, 0.96903)	(0.95171, 0.99106)	(0.85825, 0.94734)
	Llama3 8B	(0.94312, 0.99287)	(0.79178, 0.90746)	(0.93161, 0.99287)	(0.94703, 0.99418)	(0.46387, 0.63542)
	Falcon 7B	(0.86140, 0.95150)	(0.99712, 1.00079)	(0.91982, 0.98558)	(0.91636, 0.97834)	(0.93658, 0.98921)
	SmolLM 135M	(0.99175, 1.00145)	(0.68071, 0.82857)	(0.93797, 0.99487)	(0.96697, 0.99592)	(0.26288, 0.43505)
	SmolLM 360M	(0.96494, 1.00501)	(0.58988, 0.75321)	(0.92413, 0.99069)	(0.98057, 0.99957)	(0.21362, 0.38267)
	SmolLM 1.7B	(0.97921, 1.00198)	(0.50487, 0.67857)	(0.92226, 0.98837)	(0.98581, 1.00087)	(0.17857, 0.33567)
	SmolLM2 135M	(0.96003, 0.99894)	(0.95182, 0.99069)	(0.92866, 0.99164)	(0.92286, 0.98203)	(0.57263, 0.74250)
	SmolLM2 360M	(0.97310, 0.99920)	(0.88689, 0.96364)	(0.92784, 0.99063)	(0.94028, 0.99022)	(0.43800, 0.62445)
	SmolLM2 1.7B	(0.96123, 0.99931)	(0.75219, 0.88016)	(0.92710, 0.98981)	(0.95937, 0.99621)	(0.36177, 0.54782)
	GPT Neo 2.7B	(0.64509, 0.79494)	(0.97717, 0.99932)	(0.87634, 0.95904)	(0.87622, 0.95890)	(0.97033, 0.99936)
	GPTJ 6B	(0.65679, 0.80389)	(0.97795, 1.00166)	(0.86927, 0.95514)	(0.90145, 0.97261)	(0.96928, 0.99910)

Table 19. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across a variety of perturbation schemes and reference models. We report the confidence interval of the AUROC of each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	Telescope	Binoculars	AUROC Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Swap Adjacent Sentences Perturbation	Gemma2 2B	(0.97790, 0.99857)	(0.96978, 0.99973)	(0.94978, 0.99887)	(0.93792, 0.99149)	(0.89712, 0.96397)
	Gemma2 9B	(0.98880, 1.00179)	(0.98373, 0.99888)	(0.93414, 0.98750)	(0.95097, 0.99461)	(0.90950, 0.97061)
	Llama3 8B	(0.96177, 0.99732)	(0.86639, 0.94703)	(0.94562, 0.99546)	(0.92639, 0.98789)	(0.55384, 0.70905)
	Falcon 7B	(0.89741, 0.96961)	(0.98723, 1.00172)	(0.92433, 0.98463)	(0.90332, 0.97230)	(0.95585, 0.99709)
	SmolLM 135M	(0.98690, 0.99960)	(0.77561, 0.89338)	(0.93156, 0.98988)	(0.95636, 0.99802)	(0.38334, 0.55303)
	SmolLM 360M	(0.98429, 1.00016)	(0.74477, 0.87430)	(0.93231, 0.99159)	(0.96746, 1.00104)	(0.37919, 0.54859)
	SmolLM 1.7B	(0.97331, 0.99723)	(0.66462, 0.81165)	(0.93559, 0.99383)	(0.95188, 1.00148)	(0.35367, 0.52440)
	SmolLM2 135M	(0.95655, 0.99783)	(0.98022, 1.00035)	(0.93511, 0.99205)	(0.91801, 0.98257)	(0.66106, 0.80969)
	SmolLM2 360M	(0.97503, 1.00083)	(0.95798, 0.99599)	(0.93554, 0.99367)	(0.93210, 0.99221)	(0.43038, 0.60521)
	SmolLM2 1.7B	(0.98065, 1.00073)	(0.82663, 0.92971)	(0.94380, 0.99605)	(0.93986, 0.99754)	(0.38338, 0.56241)
GB Paraphrase Sentences Perturbation	GPT Neo 2.7B	(0.66408, 0.80135)	(0.95796, 0.99887)	(0.89911, 0.97200)	(0.86806, 0.95682)	(0.95834, 0.99788)
	GPTJ 6B	(0.69378, 0.82524)	(0.96317, 0.99940)	(0.89724, 0.97224)	(0.88847, 0.96996)	(0.96070, 0.99633)
	Gemma2 2B	(0.97380, 1.00120)	(0.98534, 1.00035)	(0.96882, 1.00447)	(0.95086, 0.99594)	(0.88820, 0.96030)
	Gemma2 9B	(0.99702, 1.00063)	(0.98789, 0.99993)	(0.95505, 0.99517)	(0.96941, 1.00367)	(0.90880, 0.97197)
	Llama3 8B	(0.97725, 0.99903)	(0.82810, 0.92510)	(0.96990, 1.00211)	(0.95663, 1.00213)	(0.53240, 0.69238)
	Falcon 7B	(0.90946, 0.97515)	(0.99517, 1.00163)	(0.94844, 0.99686)	(0.91837, 0.98077)	(0.96592, 1.00309)
	SmolLM 135M	(0.97589, 1.00082)	(0.76550, 0.88642)	(0.95061, 0.99725)	(0.96795, 0.99936)	(0.35831, 0.53057)
	SmolLM 360M	(0.97848, 1.00143)	(0.67214, 0.82059)	(0.95027, 0.99930)	(0.96806, 1.00373)	(0.32883, 0.49852)
	SmolLM 1.7B	(0.96678, 1.00010)	(0.64000, 0.79291)	(0.95292, 1.00028)	(0.96542, 1.00808)	(0.33036, 0.50019)
	SmolLM2 135M	(0.94615, 0.99317)	(0.98173, 0.99903)	(0.95110, 0.99933)	(0.93490, 0.98625)	(0.67782, 0.82816)
GB Swap Adjacent Words Perturbation	SmolLM2 360M	(0.95713, 0.99949)	(0.93101, 0.98608)	(0.95124, 0.99983)	(0.94921, 0.99822)	(0.41715, 0.59887)
	SmolLM2 1.7B	(0.98111, 1.00329)	(0.77896, 0.89946)	(0.95749, 1.00276)	(0.96187, 1.00651)	(0.33919, 0.51851)
	GPT Neo 2.7B	(0.69541, 0.82788)	(0.97408, 1.00413)	(0.92852, 0.98494)	(0.90337, 0.97270)	(0.97577, 1.00265)
	GPTJ 6B	(0.72588, 0.85254)	(0.97549, 1.00442)	(0.93168, 0.98669)	(0.92309, 0.98418)	(0.97636, 1.00313)
	Gemma2 2B	(0.65897, 0.79970)	(0.91337, 0.98720)	(0.84459, 0.94653)	(0.84976, 0.95261)	(0.93098, 0.98433)
	Gemma2 9B	(0.83745, 0.93280)	(0.92101, 0.98836)	(0.85152, 0.95024)	(0.90587, 0.98161)	(0.91131, 0.97453)
	Llama3 8B	(0.66428, 0.80299)	(0.81849, 0.92210)	(0.86333, 0.96012)	(0.87248, 0.96733)	(0.54215, 0.69885)
	Falcon 7B	(0.38339, 0.54542)	(0.94161, 1.00070)	(0.82908, 0.93647)	(0.83019, 0.93679)	(0.92366, 0.98592)
	SmolLM 135M	(0.79631, 0.90643)	(0.50968, 0.67588)	(0.78142, 0.90168)	(0.82283, 0.93167)	(0.25886, 0.42138)
	SmolLM 360M	(0.81503, 0.92003)	(0.41417, 0.58624)	(0.78944, 0.90736)	(0.85480, 0.95596)	(0.22802, 0.38491)
GB Swap Word with Synonyms Perturbation	SmolLM 1.7B	(0.83285, 0.92984)	(0.23830, 0.39488)	(0.79142, 0.90907)	(0.86563, 0.96355)	(0.13987, 0.26909)
	SmolLM2 135M	(0.66547, 0.80630)	(0.84506, 0.94197)	(0.75491, 0.88053)	(0.78297, 0.90197)	(0.63945, 0.78588)
	SmolLM2 360M	(0.61557, 0.76311)	(0.80004, 0.90904)	(0.77636, 0.89713)	(0.79868, 0.91408)	(0.58474, 0.74423)
	SmolLM2 1.7B	(0.78403, 0.89662)	(0.59300, 0.74804)	(0.80640, 0.91925)	(0.85209, 0.95417)	(0.37251, 0.54566)
	GPT Neo 2.7B	(0.51989, 0.67774)	(0.75907, 0.87963)	(0.74922, 0.87517)	(0.75368, 0.87971)	(0.80142, 0.90541)
	GPTJ 6B	(0.45176, 0.61309)	(0.83045, 0.93080)	(0.76158, 0.88510)	(0.77696, 0.89714)	(0.87054, 0.95025)
	Gemma2 2B	(0.86835, 0.95243)	(0.90295, 0.98207)	(0.87625, 0.96786)	(0.86256, 0.95924)	(0.89967, 0.96695)
	Gemma2 9B	(0.95780, 1.00088)	(0.88291, 0.96877)	(0.86382, 0.95717)	(0.89853, 0.98445)	(0.87284, 0.95306)
	Llama3 8B	(0.83842, 0.93122)	(0.78974, 0.90032)	(0.88840, 0.97678)	(0.88392, 0.97533)	(0.47277, 0.63443)
	Falcon 7B	(0.64254, 0.78504)	(0.94000, 1.00210)	(0.85967, 0.95621)	(0.84336, 0.94346)	(0.89827, 0.97509)
GB Swap Word with Synonyms Perturbation	SmolLM 135M	(0.92043, 0.98301)	(0.64166, 0.79451)	(0.84608, 0.94770)	(0.87171, 0.96544)	(0.25397, 0.41563)
	SmolLM 360M	(0.91857, 0.98118)	(0.52475, 0.69210)	(0.84872, 0.95018)	(0.88552, 0.97700)	(0.19749, 0.35100)
	SmolLM 1.7B	(0.94084, 0.98735)	(0.31916, 0.48526)	(0.85815, 0.95630)	(0.88863, 0.97941)	(0.10818, 0.22345)
	SmolLM2 135M	(0.84108, 0.93818)	(0.90813, 0.98119)	(0.83470, 0.93924)	(0.85638, 0.95417)	(0.63160, 0.78780)
	SmolLM2 360M	(0.82932, 0.93050)	(0.85710, 0.95080)	(0.84457, 0.94676)	(0.86117, 0.95838)	(0.51315, 0.68898)
	SmolLM2 1.7B	(0.91185, 0.97645)	(0.68557, 0.82610)	(0.86613, 0.96141)	(0.87643, 0.96952)	(0.35553, 0.52663)
	GPT Neo 2.7B	(0.63674, 0.77938)	(0.88809, 0.97014)	(0.80799, 0.91951)	(0.81899, 0.92753)	(0.90779, 0.97560)
	GPTJ 6B	(0.59932, 0.74786)	(0.90461, 0.97755)	(0.81580, 0.92520)	(0.83198, 0.93766)	(0.91683, 0.98047)

Table 20. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT when transferring a classification threshold from all of the other Ghostbusters datasets to this dataset. We report the F1-Score of each reference model on each test dataset.

Test Dataset	Reference Model	F1-Score				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Essay ChatGPT	Gemma2 2B	0.98331	0.97774	0.98128	0.97623	0.90996
	Gemma2 9B	0.97724	0.97521	0.98331	0.98887	0.93627
	Llama3 8B	0.99140	0.83308	0.96965	0.97673	0.59737
	Falcon 7B	0.92757	0.97775	0.96586	0.96793	0.91878
	SmolLM 135M	0.93374	0.72534	0.85230	0.86191	0.70157
	SmolLM 360M	0.96105	0.60445	0.87759	0.90794	0.78705
	SmolLM 1.7B	0.94031	0.57258	0.91705	0.92059	0.81285
	SmolLM2 135M	0.98128	0.93424	0.88518	0.89226	0.50278
	SmolLM2 360M	0.99039	0.89833	0.92362	0.93627	0.37835
	SmolLM2 1.7B	0.99140	0.73596	0.96206	0.96763	0.74659
	GPT Neo 2.7B	0.80071	0.95397	0.95296	0.94992	0.95549
	GPTJ 6B	0.83966	0.96257	0.96358	0.96358	0.95852
GB News ChatGPT	Gemma2 2B	0.93105	0.93558	0.80121	0.79970	0.72874
	Gemma2 9B	0.96880	0.94514	0.71666	0.85757	0.71364
	Llama3 8B	0.90840	0.94011	0.83694	0.84852	0.78812
	Falcon 7B	0.76614	0.99339	0.81240	0.81800	0.95933
	SmolLM 135M	0.94967	0.91897	0.88978	0.95622	0.34273
	SmolLM 360M	0.96376	0.88123	0.85606	0.95169	0.38802
	SmolLM 1.7B	0.95471	0.84801	0.82285	0.96074	0.46905
	SmolLM2 135M	0.93608	0.98440	0.87670	0.92099	0.86563
	SmolLM2 360M	0.92652	0.97735	0.84449	0.90488	0.17765
	SmolLM2 1.7B	0.95219	0.95169	0.81681	0.87821	0.29391
	GPT Neo 2.7B	0.51384	0.95068	0.75440	0.81379	0.94363
	GPTJ 6B	0.56266	0.92501	0.73427	0.80473	0.91495
GB Creative ChatGPT	Gemma2 2B	0.95335	0.94270	0.93611	0.87069	0.83621
	Gemma2 9B	0.99087	0.92799	0.91531	0.90720	0.82150
	Llama3 8B	0.98225	0.85903	0.93103	0.88742	0.51318
	Falcon 7B	0.97146	0.96316	0.89206	0.81370	0.92060
	SmolLM 135M	0.96298	0.81694	0.91430	0.90467	0.68560
	SmolLM 360M	0.97819	0.71805	0.91227	0.92951	0.80020
	SmolLM 1.7B	0.98986	0.61105	0.91075	0.94625	0.84736
	SmolLM2 135M	0.92901	0.91481	0.89402	0.83773	0.46450
	SmolLM2 360M	0.96298	0.88793	0.91379	0.85700	0.33215
	SmolLM2 1.7B	0.97870	0.81440	0.91988	0.87880	0.64503
	GPT Neo 2.7B	0.93458	0.90822	0.81085	0.72160	0.89148
	GPTJ 6B	0.91633	0.93864	0.82556	0.75761	0.91988
GB Essay GPT4o	Gemma2 2B	0.98278	0.97214	0.98024	0.97518	0.72948
	Gemma2 9B	0.97720	0.96251	0.97974	0.98987	0.78622
	Llama3 8B	0.99139	0.93617	0.96910	0.97467	0.47264
	Falcon 7B	0.92562	0.97695	0.96386	0.94762	0.94081
	SmolLM 135M	0.93364	0.64539	0.85258	0.86322	0.75228
	SmolLM 360M	0.96099	0.55623	0.88095	0.90831	0.77052
	SmolLM 1.7B	0.94073	0.46454	0.91895	0.91895	0.79686
	SmolLM2 135M	0.98126	0.93262	0.88652	0.89108	0.50152
	SmolLM2 360M	0.99037	0.87285	0.92351	0.93617	0.34904
	SmolLM2 1.7B	0.99139	0.72442	0.96099	0.96657	0.76089
	GPT Neo 2.7B	0.78267	0.94732	0.94478	0.90122	0.94580
	GPTJ 6B	0.81256	0.95947	0.95795	0.92806	0.95491
GB Creative GPT4o	Gemma2 2B	0.95388	0.95438	0.85555	0.77547	0.64622
	Gemma2 9B	0.99088	0.91992	0.80335	0.82311	0.62646
	Llama3 8B	0.98834	0.91029	0.78409	0.79524	0.57121
	Falcon 7B	0.95424	0.96672	0.68903	0.65887	0.94852
	SmolLM 135M	0.96097	0.80791	0.66700	0.73695	0.60922
	SmolLM 360M	0.97821	0.70299	0.67816	0.78459	0.73543
	SmolLM 1.7B	0.99189	0.52560	0.66244	0.86113	0.81247
	SmolLM2 135M	0.93208	0.92600	0.69133	0.67613	0.73441
	SmolLM2 360M	0.96401	0.92296	0.72631	0.71515	0.61125
	SmolLM2 1.7B	0.97973	0.86265	0.77496	0.77851	0.59301
	GPT Neo 2.7B	0.93006	0.89103	0.70958	0.62392	0.90623
	GPTJ 6B	0.92803	0.92397	0.71870	0.65231	0.93411

Table 21. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT when transferring a classification threshold from all of the other Ghostbusters datasets to this dataset. We report the F1-Score of each reference model on each test dataset.

Test Dataset	Reference Model	F1-Score				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB News Claude	Gemma2 2B	0.78034	0.88465	0.80441	0.79238	0.67503
	Gemma2 9B	0.89819	0.93330	0.72217	0.86108	0.71063
	Llama3 8B	0.86409	0.85005	0.84554	0.82297	0.66851
	Falcon 7B	0.61840	0.81818	0.79314	0.77518	0.80348
	SmolLM 135M	0.57472	0.73521	0.69960	0.63842	0.38415
	SmolLM 360M	0.70712	0.75677	0.74975	0.72518	0.42126
	SmolLM 1.7B	0.73019	0.77332	0.76530	0.75125	0.43831
	SmolLM2 135M	0.57523	0.72317	0.70411	0.64744	0.69860
	SmolLM2 360M	0.66901	0.79238	0.76229	0.70913	0.74824
	SmolLM2 1.7B	0.72869	0.87813	0.79539	0.75125	0.30241
	GPT Neo 2.7B	0.57372	0.72969	0.74824	0.75827	0.68455
	GPTJ 6B	0.61184	0.70512	0.73119	0.75577	0.66800
GB Creative Claude	Gemma2 2B	0.91006	0.88479	0.74583	0.77362	0.68570
	Gemma2 9B	0.96614	0.93128	0.76705	0.89944	0.73219
	Llama3 8B	0.95200	0.75644	0.79636	0.83274	0.48711
	Falcon 7B	0.90312	0.81042	0.75365	0.75052	0.70937
	SmolLM 135M	0.87367	0.72713	0.58211	0.67408	0.56746
	SmolLM 360M	0.93684	0.62052	0.59626	0.73977	0.67964
	SmolLM 1.7B	0.92218	0.60081	0.62203	0.78373	0.67964
	SmolLM2 135M	0.88833	0.79939	0.60940	0.63416	0.58919
	SmolLM2 360M	0.92370	0.80798	0.64629	0.67711	0.58615
	SmolLM2 1.7B	0.96160	0.83274	0.72764	0.78929	0.45983
	GPT Neo 2.7B	0.71299	0.67105	0.72865	0.70743	0.64275
	GPTJ 6B	0.73269	0.67761	0.76908	0.74836	0.65538
GB Essay Claude	Gemma2 2B	0.86343	0.79059	0.90794	0.89277	0.71523
	Gemma2 9B	0.97016	0.93930	0.85787	0.97218	0.89934
	Llama3 8B	0.93677	0.77137	0.94841	0.93121	0.54476
	Falcon 7B	0.86465	0.89797	0.87611	0.87246	0.77928
	SmolLM 135M	0.88467	0.52706	0.78199	0.82752	0.67729
	SmolLM 360M	0.94689	0.49722	0.82752	0.89074	0.72939
	SmolLM 1.7B	0.93020	0.55387	0.87253	0.91098	0.66211
	SmolLM2 135M	0.84977	0.80880	0.81740	0.81184	0.61204
	SmolLM2 360M	0.92312	0.83915	0.86444	0.88316	0.59181
	SmolLM2 1.7B	0.96611	0.74153	0.89782	0.91452	0.54578
	GPT Neo 2.7B	0.68639	0.78907	0.86343	0.86444	0.77086
	GPTJ 6B	0.73596	0.79110	0.88670	0.88771	0.77339
GB Essay Deepseek V3	Gemma2 2B	0.98144	0.98250	0.98144	0.97720	0.91622
	Gemma2 9B	0.97720	0.99099	0.98303	0.99205	0.93849
	Llama3 8B	0.99099	0.94804	0.97031	0.97455	0.78314
	Falcon 7B	0.92659	0.97934	0.96520	0.96629	0.94399
	SmolLM 135M	0.93584	0.91835	0.85472	0.86850	0.40933
	SmolLM 360M	0.96076	0.89502	0.88123	0.91092	0.44062
	SmolLM 1.7B	0.93955	0.86373	0.91888	0.91888	0.46501
	SmolLM2 135M	0.98091	0.96501	0.88918	0.89661	0.77359
	SmolLM2 360M	0.98993	0.96341	0.92418	0.93796	0.56681
	SmolLM2 1.7B	0.99099	0.93902	0.96288	0.96554	0.46660
	GPT Neo 2.7B	0.78314	0.95440	0.95122	0.94804	0.95387
	GPTJ 6B	0.82344	0.96288	0.96394	0.96554	0.95652
GB Creative Deepseek V3	Gemma2 2B	0.94631	0.96846	0.97651	0.95503	0.91477
	Gemma2 9B	0.99396	0.97383	0.98322	0.96242	0.93893
	Llama3 8B	0.98926	0.93557	0.98792	0.95638	0.75570
	Falcon 7B	0.93776	0.96854	0.93981	0.89603	0.94391
	SmolLM 135M	0.94094	0.82685	0.94631	0.87181	0.32886
	SmolLM 360M	0.96913	0.77450	0.96174	0.91074	0.40403
	SmolLM 1.7B	0.98188	0.73557	0.96913	0.92617	0.44966
	SmolLM2 135M	0.92483	0.92752	0.94430	0.87450	0.77114
	SmolLM2 360M	0.95839	0.92483	0.95570	0.89799	0.66309
	SmolLM2 1.7B	0.97785	0.90134	0.96577	0.93826	0.35503
	GPT Neo 2.7B	0.81812	0.93356	0.90604	0.83289	0.91611
	GPTJ 6B	0.80805	0.95839	0.91946	0.87987	0.94430

Table 22. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT when transferring a classification threshold from all of the other Ghostbusters datasets to this dataset. We report the Confidence Interval of the F1-Score of each reference model on each test dataset.

Test Dataset	Reference Model	Telescope	Binoculars	F1-Score Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Essay ChatGPT	Gemma2 2B	(0.97766, 0.98896)	(0.97124, 0.98425)	(0.97531, 0.98726)	(0.96951, 0.98294)	(0.89735, 0.92258)
	Gemma2 9B	(0.97066, 0.98381)	(0.96836, 0.98207)	(0.97766, 0.98896)	(0.98425, 0.99350)	(0.92550, 0.94703)
	Llama3 8B	(0.98733, 0.99547)	(0.81664, 0.84952)	(0.96209, 0.97721)	(0.97009, 0.98338)	(0.57575, 0.61899)
	Falcon 7B	(0.91602, 0.93913)	(0.97118, 0.98433)	(0.95776, 0.97395)	(0.96007, 0.97578)	(0.90660, 0.93096)
	SmolLM 135M	(0.92277, 0.94470)	(0.70567, 0.74502)	(0.83666, 0.86794)	(0.84670, 0.87712)	(0.68140, 0.72174)
	SmolLM 360M	(0.95252, 0.96958)	(0.58290, 0.62601)	(0.86314, 0.89204)	(0.89520, 0.92069)	(0.76900, 0.80510)
	SmolLM 1.7B	(0.92987, 0.95076)	(0.55078, 0.59439)	(0.90489, 0.92920)	(0.90867, 0.93251)	(0.79565, 0.83004)
	SmolLM2 135M	(0.97531, 0.98726)	(0.92332, 0.94517)	(0.87113, 0.89923)	(0.87859, 0.90593)	(0.48074, 0.52482)
	SmolLM2 360M	(0.98609, 0.99469)	(0.88501, 0.91165)	(0.91191, 0.93533)	(0.92550, 0.94703)	(0.35697, 0.39973)
	SmolLM2 1.7B	(0.98733, 0.99547)	(0.71653, 0.75540)	(0.95364, 0.97048)	(0.95983, 0.97543)	(0.72741, 0.76576)
	GPT Neo 2.7B	(0.78310, 0.81832)	(0.94473, 0.96321)	(0.94363, 0.96229)	(0.94031, 0.95954)	(0.94640, 0.96458)
GB News ChatGPT	GPTJ 6B	(0.82348, 0.85583)	(0.95420, 0.97094)	(0.95532, 0.97184)	(0.95532, 0.97184)	(0.94973, 0.96731)
	Gemma2 2B	(0.91991, 0.94219)	(0.92479, 0.94638)	(0.78366, 0.81876)	(0.78210, 0.81730)	(0.70919, 0.74829)
	Gemma2 9B	(0.96115, 0.97644)	(0.93513, 0.95516)	(0.69684, 0.73647)	(0.84221, 0.87294)	(0.69376, 0.73352)
	Llama3 8B	(0.89572, 0.92109)	(0.92968, 0.95054)	(0.82070, 0.85318)	(0.83275, 0.86428)	(0.77016, 0.80609)
	Falcon 7B	(0.74744, 0.78485)	(0.98981, 0.99697)	(0.79515, 0.82966)	(0.80095, 0.83505)	(0.95060, 0.96806)
	SmolLM 135M	(0.94006, 0.95929)	(0.90698, 0.93097)	(0.87601, 0.90355)	(0.94722, 0.96521)	(0.32186, 0.36360)
	SmolLM 360M	(0.95555, 0.97198)	(0.86700, 0.89545)	(0.84063, 0.87150)	(0.94226, 0.96111)	(0.36660, 0.40945)
	SmolLM 1.7B	(0.94556, 0.96385)	(0.83223, 0.86380)	(0.80606, 0.83964)	(0.95221, 0.96928)	(0.44711, 0.49099)
	SmolLM2 135M	(0.92533, 0.94684)	(0.97895, 0.98985)	(0.86224, 0.89115)	(0.90913, 0.93285)	(0.85063, 0.88062)
	SmolLM2 360M	(0.91505, 0.93799)	(0.97081, 0.98389)	(0.82856, 0.86042)	(0.89198, 0.91778)	(0.16085, 0.19446)
	SmolLM2 1.7B	(0.94281, 0.96157)	(0.94226, 0.96111)	(0.79980, 0.83382)	(0.86383, 0.89259)	(0.27388, 0.31394)
GB Creative ChatGPT	GPT Neo 2.7B	(0.49186, 0.53582)	(0.94116, 0.96020)	(0.73548, 0.77333)	(0.79667, 0.83091)	(0.93349, 0.95377)
	GPTJ 6B	(0.54085, 0.58447)	(0.91343, 0.93659)	(0.71485, 0.75369)	(0.78730, 0.82216)	(0.90268, 0.92721)
	Gemma2 2B	(0.94404, 0.96265)	(0.93244, 0.95296)	(0.92531, 0.94690)	(0.85588, 0.88550)	(0.81987, 0.85254)
	Gemma2 9B	(0.98667, 0.99507)	(0.91658, 0.93940)	(0.90303, 0.92760)	(0.89439, 0.92001)	(0.80460, 0.83840)
	Llama3 8B	(0.97642, 0.98808)	(0.84367, 0.87439)	(0.91985, 0.94222)	(0.87347, 0.90137)	(0.49112, 0.53524)
	Falcon 7B	(0.96402, 0.97889)	(0.95474, 0.97157)	(0.87821, 0.90591)	(0.79632, 0.83108)	(0.90853, 0.93267)
	SmolLM 135M	(0.95465, 0.97131)	(0.79987, 0.83401)	(0.90195, 0.92665)	(0.89170, 0.91763)	(0.66511, 0.70609)
	SmolLM 360M	(0.97175, 0.98464)	(0.69819, 0.73791)	(0.89979, 0.92476)	(0.91822, 0.94081)	(0.78256, 0.81785)
	SmolLM 1.7B	(0.98544, 0.99428)	(0.58954, 0.63257)	(0.89817, 0.92333)	(0.93629, 0.95620)	(0.83149, 0.86324)
	SmolLM2 135M	(0.91767, 0.94034)	(0.90249, 0.92713)	(0.88043, 0.90760)	(0.82146, 0.85400)	(0.44249, 0.48652)
	SmolLM2 360M	(0.95465, 0.97131)	(0.87401, 0.90185)	(0.90141, 0.92618)	(0.84155, 0.87245)	(0.31136, 0.35294)
GB Essay GPT4o	SmolLM2 1.7B	(0.97233, 0.98507)	(0.79724, 0.83156)	(0.90790, 0.93186)	(0.86440, 0.89321)	(0.62391, 0.66615)
	GPT Neo 2.7B	(0.92367, 0.94550)	(0.89547, 0.92096)	(0.79357, 0.82814)	(0.70182, 0.74138)	(0.87775, 0.90521)
	GPTJ 6B	(0.90411, 0.92855)	(0.92805, 0.94923)	(0.80881, 0.84231)	(0.73869, 0.77652)	(0.90790, 0.93186)
	Gemma2 2B	(0.97704, 0.98852)	(0.96488, 0.97940)	(0.97410, 0.98638)	(0.96831, 0.98204)	(0.70989, 0.74908)
	Gemma2 9B	(0.97062, 0.98379)	(0.95413, 0.97089)	(0.97352, 0.98595)	(0.98545, 0.99429)	(0.76814, 0.80431)
	Llama3 8B	(0.98731, 0.99546)	(0.92539, 0.94695)	(0.96146, 0.97673)	(0.96774, 0.98160)	(0.45062, 0.49467)
	Falcon 7B	(0.91384, 0.93739)	(0.97022, 0.98368)	(0.95548, 0.97223)	(0.93762, 0.95761)	(0.93022, 0.95139)
	SmolLM 135M	(0.92266, 0.94462)	(0.62429, 0.66649)	(0.83694, 0.86822)	(0.84806, 0.87838)	(0.73324, 0.77132)
	SmolLM 360M	(0.95245, 0.96953)	(0.53431, 0.57815)	(0.86667, 0.89524)	(0.89558, 0.92104)	(0.75197, 0.78907)
	SmolLM 1.7B	(0.93031, 0.95115)	(0.44254, 0.48654)	(0.90691, 0.93099)	(0.90691, 0.93099)	(0.77911, 0.81461)
	SmolLM2 135M	(0.97527, 0.98724)	(0.92157, 0.94368)	(0.87253, 0.90052)	(0.87734, 0.90483)	(0.47946, 0.52358)
GB Essay GPT4o	SmolLM2 360M	(0.98607, 0.99468)	(0.85815, 0.88754)	(0.91178, 0.93523)	(0.92539, 0.94695)	(0.32801, 0.37007)
	SmolLM2 1.7B	(0.98731, 0.99546)	(0.70471, 0.74413)	(0.95245, 0.96953)	(0.95864, 0.97450)	(0.74208, 0.77971)
	GPT Neo 2.7B	(0.76448, 0.80087)	(0.93746, 0.95717)	(0.93471, 0.95486)	(0.88805, 0.91438)	(0.93581, 0.95578)
	GPTJ 6B	(0.79535, 0.82978)	(0.95077, 0.96817)	(0.94910, 0.96681)	(0.91667, 0.93946)	(0.94576, 0.96407)

Table 23. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT when transferring a classification threshold from all of the other Ghostbusters datasets to this dataset. We report the Confidence Interval of the F1-Score of each reference model on each test dataset.

Test Dataset	Reference Model	Telescope	Binoculars	F1-Score Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Creative GPT4o	Gemma2 2B	(0.94462, 0.96313)	(0.94518, 0.96359)	(0.84004, 0.87106)	(0.75706, 0.79388)	(0.62513, 0.66732)
	Gemma2 9B	(0.98668, 0.99507)	(0.90794, 0.93190)	(0.78581, 0.82088)	(0.80628, 0.83995)	(0.60511, 0.64780)
	Llama3 8B	(0.98361, 0.99308)	(0.89768, 0.92290)	(0.76593, 0.80224)	(0.77743, 0.81304)	(0.54937, 0.59305)
	Falcon 7B	(0.94490, 0.96358)	(0.95870, 0.97474)	(0.66834, 0.70972)	(0.63768, 0.68006)	(0.93864, 0.95839)
	SmolLM 135M	(0.95243, 0.96952)	(0.79052, 0.82529)	(0.64621, 0.68780)	(0.71752, 0.75638)	(0.58769, 0.63075)
	SmolLM 360M	(0.97176, 0.98465)	(0.68283, 0.72315)	(0.65754, 0.69877)	(0.76645, 0.80273)	(0.71596, 0.75489)
	SmolLM 1.7B	(0.98793, 0.99585)	(0.50356, 0.54763)	(0.64158, 0.68331)	(0.84587, 0.87638)	(0.79524, 0.82969)
	SmolLM2 135M	(0.92098, 0.94319)	(0.91445, 0.93755)	(0.67095, 0.71172)	(0.65548, 0.69678)	(0.71493, 0.75390)
	SmolLM2 360M	(0.95580, 0.97223)	(0.91119, 0.93473)	(0.70663, 0.74598)	(0.69524, 0.73507)	(0.58974, 0.63276)
	SmolLM2 1.7B	(0.97351, 0.98595)	(0.84746, 0.87783)	(0.75654, 0.79339)	(0.76019, 0.79683)	(0.57133, 0.61468)
	GPT Neo 2.7B	(0.91880, 0.94131)	(0.87728, 0.90478)	(0.68955, 0.72961)	(0.60255, 0.64530)	(0.89337, 0.91910)
	GPTJ 6B	(0.91662, 0.93943)	(0.91228, 0.93567)	(0.69886, 0.73854)	(0.63129, 0.67332)	(0.92316, 0.94506)
GB News Claude	Gemma2 2B	(0.76217, 0.79851)	(0.87063, 0.89867)	(0.78700, 0.82182)	(0.77457, 0.81018)	(0.65447, 0.69558)
	Gemma2 9B	(0.88492, 0.91147)	(0.92235, 0.94425)	(0.70251, 0.74183)	(0.84590, 0.87626)	(0.69073, 0.73054)
	Llama3 8B	(0.84905, 0.87913)	(0.83438, 0.86572)	(0.82967, 0.86140)	(0.80622, 0.83972)	(0.64784, 0.68917)
	Falcon 7B	(0.59619, 0.64061)	(0.80054, 0.83582)	(0.77462, 0.81166)	(0.75609, 0.79427)	(0.78531, 0.82166)
	SmolLM 135M	(0.55302, 0.59642)	(0.71584, 0.75457)	(0.67948, 0.71972)	(0.61733, 0.65950)	(0.36280, 0.40550)
	SmolLM 360M	(0.68715, 0.72710)	(0.73794, 0.77560)	(0.73074, 0.76876)	(0.70558, 0.74477)	(0.39959, 0.44294)
	SmolLM 1.7B	(0.71071, 0.74967)	(0.75494, 0.79170)	(0.74669, 0.78390)	(0.73228, 0.77023)	(0.41654, 0.46009)
	SmolLM2 135M	(0.55353, 0.59692)	(0.70353, 0.74281)	(0.68408, 0.72415)	(0.62647, 0.66841)	(0.67846, 0.71874)
	SmolLM2 360M	(0.64835, 0.68966)	(0.77457, 0.81018)	(0.74360, 0.78097)	(0.68919, 0.72906)	(0.72919, 0.76729)
	SmolLM2 1.7B	(0.70917, 0.74820)	(0.86378, 0.89249)	(0.77768, 0.81309)	(0.73228, 0.77023)	(0.28225, 0.32257)
	GPT Neo 2.7B	(0.55202, 0.59543)	(0.71020, 0.74918)	(0.72919, 0.76729)	(0.73948, 0.77707)	(0.66416, 0.70495)
	GPTJ 6B	(0.59045, 0.63323)	(0.68510, 0.72513)	(0.71173, 0.75065)	(0.73691, 0.77462)	(0.64733, 0.68867)
GB Creative Claude	Gemma2 2B	(0.89745, 0.92266)	(0.87072, 0.89886)	(0.72665, 0.76501)	(0.75519, 0.79206)	(0.66525, 0.70615)
	Gemma2 9B	(0.95818, 0.97411)	(0.92013, 0.94242)	(0.74843, 0.78568)	(0.88619, 0.91269)	(0.71268, 0.75170)
	Llama3 8B	(0.94258, 0.96141)	(0.73753, 0.77535)	(0.77862, 0.81410)	(0.81630, 0.84919)	(0.46509, 0.50914)
	Falcon 7B	(0.88989, 0.91636)	(0.79288, 0.82795)	(0.73437, 0.77292)	(0.73117, 0.76988)	(0.68907, 0.72968)
	SmolLM 135M	(0.85904, 0.88831)	(0.70751, 0.74676)	(0.56038, 0.60384)	(0.65343, 0.69473)	(0.54563, 0.58929)
	SmolLM 360M	(0.92612, 0.94755)	(0.59914, 0.64189)	(0.57464, 0.61788)	(0.72044, 0.75910)	(0.65908, 0.70019)
	SmolLM 1.7B	(0.91038, 0.93399)	(0.57923, 0.62239)	(0.60067, 0.64339)	(0.76559, 0.80187)	(0.65908, 0.70019)
	SmolLM2 135M	(0.87445, 0.90220)	(0.78175, 0.81704)	(0.58790, 0.63089)	(0.61294, 0.65538)	(0.56751, 0.61086)
	SmolLM2 360M	(0.91200, 0.93540)	(0.79063, 0.82534)	(0.62522, 0.66735)	(0.65651, 0.69771)	(0.56446, 0.60785)
	SmolLM2 1.7B	(0.95313, 0.97006)	(0.81630, 0.84919)	(0.70803, 0.74725)	(0.77132, 0.80726)	(0.43787, 0.48179)
	GPT Neo 2.7B	(0.69306, 0.73292)	(0.65035, 0.69175)	(0.70906, 0.74824)	(0.68738, 0.72747)	(0.62164, 0.66386)
	GPTJ 6B	(0.71320, 0.75219)	(0.65702, 0.69821)	(0.75051, 0.78764)	(0.72924, 0.76748)	(0.63444, 0.67632)
GB Essay Claude	Gemma2 2B	(0.84829, 0.87857)	(0.77266, 0.80853)	(0.89520, 0.92069)	(0.87913, 0.90641)	(0.69533, 0.73512)
	Gemma2 9B	(0.96266, 0.97766)	(0.92878, 0.94983)	(0.84247, 0.87326)	(0.96493, 0.97943)	(0.88608, 0.91261)
	Llama3 8B	(0.92605, 0.94750)	(0.75286, 0.78988)	(0.93866, 0.95816)	(0.92005, 0.94237)	(0.52281, 0.56672)
	Falcon 7B	(0.84936, 0.87995)	(0.88443, 0.91151)	(0.86137, 0.89084)	(0.85755, 0.88738)	(0.76074, 0.79783)
	SmolLM 135M	(0.87059, 0.89875)	(0.50505, 0.54907)	(0.76379, 0.80019)	(0.81086, 0.84417)	(0.65668, 0.69790)
	SmolLM 360M	(0.93700, 0.95677)	(0.47518, 0.51926)	(0.81086, 0.84417)	(0.87699, 0.90449)	(0.70980, 0.74897)
	SmolLM 1.7B	(0.91896, 0.94143)	(0.53196, 0.57578)	(0.85783, 0.88723)	(0.89842, 0.92353)	(0.64126, 0.68296)
	SmolLM2 135M	(0.83402, 0.86552)	(0.79147, 0.82614)	(0.80037, 0.83443)	(0.79461, 0.82906)	(0.59056, 0.63352)
	SmolLM2 360M	(0.91137, 0.93486)	(0.82296, 0.85534)	(0.84935, 0.87953)	(0.86900, 0.89732)	(0.57014, 0.61347)
	SmolLM2 1.7B	(0.95813, 0.97409)	(0.72223, 0.76083)	(0.88447, 0.91118)	(0.90219, 0.92684)	(0.52383, 0.56772)
	GPT Neo 2.7B	(0.66594, 0.70684)	(0.77109, 0.80706)	(0.84829, 0.87857)	(0.84935, 0.87953)	(0.75234, 0.78939)
	GPTJ 6B	(0.71653, 0.75540)	(0.77318, 0.80902)	(0.87273, 0.90067)	(0.87379, 0.90163)	(0.75494, 0.79185)
GB Essay Deepseek V3	Gemma2 2B	(0.97535, 0.98753)	(0.97659, 0.98842)	(0.97535, 0.98753)	(0.97046, 0.98394)	(0.90372, 0.92873)
	Gemma2 9B	(0.97046, 0.98394)	(0.98672, 0.99525)	(0.97720, 0.98886)	(0.98804, 0.99606)	(0.92765, 0.94934)
	Llama3 8B	(0.98672, 0.99525)	(0.93802, 0.95806)	(0.96265, 0.97797)	(0.96744, 0.98166)	(0.76454, 0.80174)
	Falcon 7B	(0.91467, 0.93851)	(0.97283, 0.98584)	(0.95682, 0.97358)	(0.95804, 0.97454)	(0.93348, 0.95450)
	SmolLM 135M	(0.92478, 0.94690)	(0.90599, 0.93070)	(0.83882, 0.87062)	(0.85325, 0.88376)	(0.38714, 0.43152)
	SmolLM 360M	(0.95200, 0.96953)	(0.88118, 0.90885)	(0.86663, 0.89583)	(0.89807, 0.92378)	(0.41821, 0.46302)
	SmolLM 1.7B	(0.92880, 0.95031)	(0.84825, 0.87922)	(0.90655, 0.93120)	(0.90655, 0.93120)	(0.44250, 0.48752)
	SmolLM2 135M	(0.97474, 0.98709)	(0.95671, 0.97330)	(0.87502, 0.90335)	(0.88287, 0.91035)	(0.75471, 0.79248)
	SmolLM2 360M	(0.98542, 0.99443)	(0.95494, 0.97189)	(0.91223, 0.93612)	(0.92708, 0.94885)	(0.54444, 0.58917)
	SmolLM2 1.7B	(0.98672, 0.99525)	(0.92823, 0.94982)	(0.95435, 0.97142)	(0.95730, 0.97377)	(0.44408, 0.48911)
	GPT Neo 2.7B	(0.76454, 0.80174)	(0.94499, 0.96382)	(0.94150, 0.96094)	(0.93802, 0.95806)	(0.94440, 0.96334)
	GPTJ 6B	(0.80623, 0.84064)	(0.95435, 0.97142)	(0.95553, 0.97236)	(0.95730, 0.97377)	(0.94732, 0.96573)
GB Creative Deepseek V3	Gemma2 2B	(0.93486, 0.95775)	(0.95958, 0.97733)	(0.96882, 0.98420)	(0.94451, 0.96556)	(0.90059, 0.92894)
	Gemma2 9B	(0.99003, 0.99789)	(0.96572, 0.98193)	(0.97670, 0.98974)	(0.95276, 0.97207)	(0.92677, 0.95109)
	Llama3 8B	(0.98403, 0.99450)	(0.92310, 0.94804)	(0.98237, 0.99347)	(0.94600, 0.96675)	(0.73389, 0.77752)
	Falcon 7B	(0.92537, 0.95014)	(0.95959, 0.97748)	(0.92762, 0.95200)	(0.88039, 0.91168)	(0.93212, 0.95571)
	SmolLM 135M	(0.92897, 0.95291)	(0.80763, 0.84606)	(0.93486, 0.95775)	(0.85484, 0.88879)	(0.30500, 0.35271)
	SmolLM 360M	(0.96034, 0.97791)	(0.75328, 0.79572)	(0.95201, 0.97148)	(0.89626, 0.92522)	(0.37911, 0.42894)
	SmolLM 1.7B	(0.97511, 0.98865)	(0.71318, 0.75796)	(0.96034, 0.97791)	(0.91290, 0.93945)	(0.42441, 0.47492)
	SmolLM2 135M	(0.91144, 0.93822)	(0.91435, 0.94068)	(0.93265, 0.95594)	(0.85768, 0.89132)	(0.74981, 0.79247)
	SmolLM2 360M	(0.94825, 0.96853)	(0.91144, 0.93822)	(0.94526, 0.96615)	(0.88262, 0.91335)	(0.63909, 0.68709)
	SmolLM2 1.7B	(0.97038, 0.98532)	(0.88620, 0.91648)	(0.95654, 0.97500)	(0.92603, 0.95048)	(0.33074, 0.37933)
	GPT Neo 2.7B	(0.79853, 0.83771)	(0.92091, 0.94620)	(0.89123, 0.92086)	(0.81394, 0.85183)	(0.90203, 0.93018)
	GPTJ 6B	(0.78806, 0.82805)	(0.94825, 0.96853)	(0.90565, 0.93328)	(0.86336, 0.89637)	(0.93265, 0.95594)

Table 24. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the true positive rate at a false positive rate of 5% of each reference model on each dataset and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
Detect LLM Text	Gemma2 2B	0.98689	0.70103	0.65742	0.76557	0.62869
	Gemma2 9B	0.99622	0.75296	0.55029	0.89463	0.65516
	Llama3 8B	0.96723	0.53390	0.65793	0.57373	0.39551
	Falcon 7B	0.91009	0.72005	0.65798	0.64725	0.62759
	SmolLM 135M	0.98412	0.49508	0.65163	0.84825	0.26368
	SmolLM 360M	0.98538	0.43257	0.66473	0.84321	0.21553
	SmolLM 1.7B	0.98866	0.40837	0.62365	0.89009	0.21124
	SmolLM2 135M	0.95841	0.65263	0.63675	0.63701	0.48072
	SmolLM2 360M	0.96975	0.63625	0.64280	0.64608	0.44643
	SmolLM2 1.7B	0.96622	0.55382	0.66398	0.59743	0.35316
	GPT Neo 2.7B	0.94379	0.67154	0.62087	0.56920	0.66196
	GPTJ 6B	0.94580	0.67356	0.61608	0.60676	0.66675
AI vs Human	Gemma2 2B	0.85605	0.71685	0.78579	0.75119	0.61992
	Gemma2 9B	0.94585	0.74670	0.74881	0.86978	0.57079
	Llama3 8B	0.84997	0.70734	0.80032	0.71712	0.33281
	Falcon 7B	0.75804	0.88194	0.78253	0.73355	0.77156
	SmolLM 135M	0.83862	0.61912	0.74142	0.72610	0.28209
	SmolLM 360M	0.85209	0.59403	0.75753	0.74274	0.26889
	SmolLM 1.7B	0.89223	0.55256	0.75647	0.77126	0.20866
	SmolLM2 135M	0.81749	0.79424	0.72293	0.66165	0.62335
	SmolLM2 360M	0.84337	0.78500	0.75383	0.70761	0.52324
	SmolLM2 1.7B	0.88933	0.72187	0.78156	0.72187	0.39884
	GPT Neo 2.7B	0.71210	0.77734	0.70602	0.65003	0.77021
	GPTJ 6B	0.68251	0.80851	0.73323	0.68278	0.80164
HC3	Gemma2 2B	0.96359	0.98780	0.99906	0.98405	0.98236
	Gemma2 9B	0.99775	0.97823	0.99794	0.96359	0.96002
	Llama3 8B	0.99775	0.95721	0.99850	0.93318	0.66141
	Falcon 7B	0.99699	0.99887	0.99812	0.97107	0.97953
	SmolLM 135M	0.99381	0.96396	0.99756	0.98423	0.53360
	SmolLM 360M	0.99906	0.96471	0.99869	0.97785	0.54542
	SmolLM 1.7B	0.99850	0.96453	0.99869	0.96978	0.54580
	SmolLM2 135M	0.95627	0.97954	0.91047	0.77947	0.98292
	SmolLM2 360M	0.99474	0.99850	0.99812	0.98330	0.96697
	SmolLM2 1.7B	0.99906	0.98386	0.99887	0.97185	0.80724
	GPT Neo 2.7B	0.88232	0.99831	0.99869	0.99099	0.99756
	GPTJ 6B	0.86111	0.99643	0.99794	0.99249	0.99531
HC3 Plus	Gemma2 2B	0.93207	0.76587	0.78432	0.77510	0.69706
	Gemma2 9B	0.92089	0.76516	0.77634	0.74335	0.70380
	Llama3 8B	0.95548	0.77350	0.77527	0.73909	0.54824
	Falcon 7B	0.89110	0.78521	0.77120	0.75842	0.68375
	SmolLM 135M	0.95424	0.73838	0.77616	0.75523	0.37336
	SmolLM 360M	0.94129	0.67400	0.77403	0.75044	0.26694
	SmolLM 1.7B	0.94094	0.65662	0.77368	0.74548	0.25949
	SmolLM2 135M	0.95779	0.80259	0.77616	0.74956	0.74672
	SmolLM2 360M	0.95459	0.78556	0.77687	0.74973	0.67559
	SmolLM2 1.7B	0.93828	0.74016	0.77989	0.75115	0.43455
	GPT Neo 2.7B	0.84941	0.79656	0.79124	0.77102	0.78024
	GPTJ 6B	0.86839	0.78875	0.78716	0.77510	0.77403
ESL GPT4o Mini	Gemma2 2B	1.00000	0.34781	0.23232	0.06263	0.13973
	Gemma2 9B	1.00000	0.42559	0.05892	0.30774	0.15522
	Llama3 8B	1.00000	0.24983	0.17609	0.04276	0.05387
	Falcon 7B	1.00000	0.71780	0.14356	0.04493	0.28817
	SmolLM 135M	1.00000	0.22896	0.22525	0.28350	0.02088
	SmolLM 360M	1.00000	0.15926	0.16768	0.32458	0.01717
	SmolLM 1.7B	1.00000	0.19226	0.18182	0.29832	0.02424
	SmolLM2 135M	0.99966	0.56094	0.22256	0.08013	0.09596
	SmolLM2 360M	1.00000	0.48047	0.24983	0.09024	0.08956
	SmolLM2 1.7B	1.00000	0.41044	0.26768	0.06296	0.08451
	GPT Neo 2.7B	1.00000	0.26869	0.02020	0.02525	0.22088
	GPTJ 6B	1.00000	0.33300	0.02054	0.01953	0.29933

Table 25. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the true positive rate at a false positive rate of 5% of each reference model on each dataset and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Essay ChatGPT	Gemma2 2B	1.00000	0.99200	1.00000	0.99800	0.92700
	Gemma2 9B	1.00000	0.99600	1.00000	0.99700	0.97700
	Llama3 8B	1.00000	0.66800	1.00000	0.99800	0.15400
	Falcon 7B	0.99899	1.00000	0.99899	0.99194	0.90927
	SmolLM 135M	1.00000	0.42800	0.99900	0.99000	0.00800
	SmolLM 360M	1.00000	0.24200	0.99900	0.99500	0.00000
	SmolLM 1.7B	1.00000	0.17300	1.00000	0.99700	0.00100
	SmolLM2 135M	1.00000	0.92600	0.99700	0.98500	0.10600
	SmolLM2 360M	1.00000	0.84200	0.99900	0.99300	0.04400
	SmolLM2 1.7B	1.00000	0.47400	1.00000	0.99600	0.01500
	GPT Neo 2.7B	0.46900	0.99900	0.99100	0.96000	0.99900
	GPTJ 6B	0.57900	1.00000	0.99500	0.98200	0.99900
GB News ChatGPT	Gemma2 2B	0.91440	0.99194	0.97080	0.95368	0.95267
	Gemma2 9B	0.96979	0.97482	0.92649	0.94663	0.93152
	Llama3 8B	0.91541	0.93152	0.96375	0.96073	0.54884
	Falcon 7B	0.53877	0.99899	0.96677	0.96777	0.98792
	SmolLM 135M	0.97784	0.91944	0.97281	0.96576	0.56093
	SmolLM 360M	0.97986	0.88520	0.97180	0.96878	0.45720
	SmolLM 1.7B	0.96979	0.80060	0.97180	0.97583	0.22054
	SmolLM2 135M	0.92246	0.99799	0.97080	0.97080	0.98389
	SmolLM2 360M	0.91339	0.99597	0.96878	0.97180	0.97281
	SmolLM2 1.7B	0.94159	0.98489	0.96073	0.96073	0.81672
	GPT Neo 2.7B	0.04632	0.99295	0.95368	0.96878	0.98993
	GPTJ 6B	0.10373	0.99295	0.93555	0.96375	0.99094
GB Creative ChatGPT	Gemma2 2B	0.99492	0.93401	0.92386	0.80406	0.52284
	Gemma2 9B	0.99898	0.90558	0.89239	0.89848	0.56041
	Llama3 8B	0.98985	0.72081	0.93706	0.82335	0.07208
	Falcon 7B	0.98769	0.98462	0.81641	0.62462	0.90769
	SmolLM 135M	0.99086	0.67411	0.90355	0.87208	0.00711
	SmolLM 360M	0.99492	0.43350	0.89848	0.91269	0.00203
	SmolLM 1.7B	0.99289	0.23858	0.90457	0.94315	0.00000
	SmolLM2 135M	0.98985	0.90863	0.83756	0.69340	0.06497
	SmolLM2 360M	0.99492	0.76244	0.87208	0.74518	0.02234
	SmolLM2 1.7B	0.99797	0.54721	0.89746	0.80102	0.00711
	GPT Neo 2.7B	0.91878	0.90964	0.65076	0.38376	0.87716
	GPTJ 6B	0.87513	0.93909	0.68223	0.47614	0.91777
GB Essay GPT4o	Gemma2 2B	1.00000	0.98796	0.99799	0.99498	0.33601
	Gemma2 9B	1.00000	0.98195	0.99498	0.99799	0.47141
	Llama3 8B	1.00000	0.94885	0.99799	0.99198	0.04614
	Falcon 7B	0.98760	0.99897	0.98864	0.94215	0.96591
	SmolLM 135M	0.99900	0.25978	0.95687	0.96991	0.00301
	SmolLM 360M	1.00000	0.14243	0.97994	0.99298	0.00301
	SmolLM 1.7B	0.99900	0.03310	0.98094	0.99498	0.00000
	SmolLM2 135M	1.00000	0.92076	0.96790	0.93882	0.09829
	SmolLM2 360M	1.00000	0.76028	0.99198	0.97693	0.01404
	SmolLM2 1.7B	1.00000	0.43430	0.99699	0.98897	0.00100
	GPT Neo 2.7B	0.44032	0.96590	0.94183	0.82648	0.96389
	GPTJ 6B	0.51655	0.98395	0.96389	0.90572	0.98195
GB Creative GPT4o	Gemma2 2B	0.98580	0.95538	0.86613	0.64300	0.21095
	Gemma2 9B	1.00000	0.89858	0.78702	0.82759	0.22211
	Llama3 8B	1.00000	0.81846	0.88032	0.70183	0.13793
	Falcon 7B	0.96189	0.98867	0.39959	0.29145	0.97528
	SmolLM 135M	0.98073	0.51927	0.55274	0.56795	0.02434
	SmolLM 360M	0.99696	0.28093	0.64909	0.70284	0.01217
	SmolLM 1.7B	0.99899	0.07505	0.58012	0.86410	0.00203
	SmolLM2 135M	0.98580	0.92292	0.47363	0.32759	0.49290
	SmolLM2 360M	0.99594	0.90568	0.55578	0.43306	0.40264
	SmolLM2 1.7B	1.00000	0.63895	0.67647	0.57404	0.08621
	GPT Neo 2.7B	0.91075	0.78803	0.42596	0.20690	0.86815
	GPTJ 6B	0.88235	0.86815	0.47465	0.27181	0.93205

Table 26. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the true positive rate at a false positive rate of 5% of each reference model on each dataset and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB News Claude	Gemma2 2B	0.65700	0.81400	0.65100	0.65500	0.40500
	Gemma2 9B	0.93700	0.93400	0.55600	0.90500	0.41500
	Llama3 8B	0.73300	0.73900	0.83200	0.63200	0.37800
	Falcon 7B	0.20278	0.82966	0.63615	0.56779	0.63499
	SmolLM 135M	0.52100	0.46900	0.27700	0.34900	0.17600
	SmolLM 360M	0.74700	0.45100	0.35500	0.49400	0.17900
	SmolLM 1.7B	0.82300	0.55800	0.40900	0.58600	0.18200
	SmolLM2 135M	0.30300	0.64200	0.26900	0.26500	0.26900
	SmolLM2 360M	0.44100	0.67400	0.35900	0.31200	0.34700
	SmolLM2 1.7B	0.61300	0.79300	0.47100	0.38900	0.41800
	GPT Neo 2.7B	0.12000	0.45300	0.46300	0.42900	0.33700
	GPTJ 6B	0.19400	0.36500	0.45600	0.41900	0.25900
GB Creative Claude	Gemma2 2B	0.82863	0.81048	0.69758	0.62198	0.28024
	Gemma2 9B	0.99194	0.91633	0.73387	0.89718	0.39012
	Llama3 8B	0.97581	0.52520	0.89819	0.76411	0.11089
	Falcon 7B	0.87293	0.68079	0.53099	0.46591	0.39876
	SmolLM 135M	0.78931	0.34778	0.33770	0.44657	0.02520
	SmolLM 360M	0.94153	0.23387	0.44556	0.58669	0.01613
	SmolLM 1.7B	0.96371	0.25504	0.48286	0.73992	0.01613
	SmolLM2 135M	0.79234	0.53125	0.30645	0.27923	0.11492
	SmolLM2 360M	0.86089	0.53327	0.41431	0.36794	0.13911
	SmolLM2 1.7B	0.96976	0.62702	0.60484	0.59980	0.14113
	GPT Neo 2.7B	0.51008	0.33972	0.47379	0.36290	0.25605
	GPTJ 6B	0.47681	0.38105	0.56351	0.45161	0.32661
GB Essay Claude	Gemma2 2B	0.83000	0.70800	0.90100	0.86200	0.34800
	Gemma2 9B	0.98400	0.95300	0.82900	0.98200	0.79700
	Llama3 8B	0.95600	0.56800	0.94800	0.92000	0.15800
	Falcon 7B	0.70816	0.85612	0.79082	0.79286	0.55714
	SmolLM 135M	0.76500	0.10900	0.48000	0.69200	0.02600
	SmolLM 360M	0.94600	0.10000	0.55600	0.87400	0.01600
	SmolLM 1.7B	0.91900	0.15200	0.68500	0.92700	0.04200
	SmolLM2 135M	0.85000	0.67500	0.54000	0.63200	0.18200
	SmolLM2 360M	0.95000	0.70800	0.67300	0.78600	0.17500
	SmolLM2 1.7B	0.97800	0.51500	0.81700	0.87600	0.14300
	GPT Neo 2.7B	0.25100	0.62300	0.68600	0.72900	0.57400
	GPTJ 6B	0.31800	0.62800	0.79500	0.82700	0.56100
GB Essay Deepseek V3	Gemma2 2B	1.00000	0.99895	0.99895	0.99685	0.98846
	Gemma2 9B	1.00000	1.00000	0.99895	0.99895	0.99685
	Llama3 8B	1.00000	0.99790	0.99895	0.99265	0.86464
	Falcon 7B	0.99362	1.00000	0.99787	0.98830	0.97872
	SmolLM 135M	0.99895	0.95593	1.00000	0.97901	0.21931
	SmolLM 360M	1.00000	0.92445	1.00000	0.99475	0.18363
	SmolLM 1.7B	1.00000	0.77335	1.00000	0.99475	0.20462
	SmolLM2 135M	1.00000	0.99895	0.99895	0.98006	0.57712
	SmolLM2 360M	1.00000	1.00000	1.00000	0.99370	0.20357
	SmolLM2 1.7B	1.00000	0.99370	1.00000	0.99475	0.16055
	GPT Neo 2.7B	0.58132	0.99580	0.99580	0.96642	0.99475
	GPTJ 6B	0.64113	0.99685	0.99580	0.98636	0.99475
GB Creative Deepseek V3	Gemma2 2B	0.97718	1.00000	0.99144	0.96006	0.98003
	Gemma2 9B	0.99857	1.00000	0.99429	0.97004	0.98716
	Llama3 8B	1.00000	0.99857	0.99715	0.95720	0.73466
	Falcon 7B	0.92571	1.00000	0.93429	0.86000	0.96000
	SmolLM 135M	0.93723	0.95578	0.94437	0.79030	0.39943
	SmolLM 360M	0.98003	0.96006	0.97147	0.87019	0.37375
	SmolLM 1.7B	0.99572	0.91583	0.97718	0.91869	0.27960
	SmolLM2 135M	0.98288	0.99572	0.94151	0.78887	0.77746
	SmolLM2 360M	0.99572	0.99857	0.95720	0.84879	0.67047
	SmolLM2 1.7B	1.00000	1.00000	0.97575	0.93153	0.47932
	GPT Neo 2.7B	0.65906	0.98431	0.86305	0.65335	0.93295
	GPTJ 6B	0.62482	1.00000	0.88160	0.78745	0.97575

Table 27. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across a variety of perturbation schemes and reference models. We report the true positive rate at a false positive rate of 5% of each reference model on each dataset and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Spelling Error Perturbation	Gemma2 2B	0.77174	0.90217	0.77174	0.75000	0.76087
	Gemma2 9B	0.94565	0.86957	0.73913	0.92391	0.79348
	Llama3 8B	0.89130	0.52174	0.82609	0.89130	0.10870
	Falcon 7B	0.51087	0.95652	0.72826	0.65217	0.88043
	SmolLM 135M	0.84783	0.18478	0.64130	0.70652	0.09783
	SmolLM 360M	0.85870	0.23913	0.70652	0.80435	0.14130
	SmolLM 1.7B	0.85870	0.17391	0.70652	0.81522	0.03261
	SmolLM2 135M	0.71739	0.72826	0.59783	0.63043	0.34783
	SmolLM2 360M	0.79348	0.70652	0.65217	0.71739	0.34783
	SmolLM2 1.7B	0.90217	0.41304	0.72826	0.76087	0.11957
	GPT Neo 2.7B	0.27174	0.73913	0.46739	0.54348	0.73913
	GPTJ 6B	0.31522	0.81522	0.58696	0.66304	0.81522
GB Character Capitalization Perturbation	Gemma2 2B	0.98936	0.90426	0.85106	0.85106	0.73404
	Gemma2 9B	1.00000	0.93617	0.85106	0.91489	0.70213
	Llama3 8B	1.00000	0.50000	0.88298	0.89362	0.05319
	Falcon 7B	0.67021	0.96809	0.79787	0.78723	0.82979
	SmolLM 135M	0.87234	0.23404	0.75532	0.71277	0.10638
	SmolLM 360M	0.92553	0.18085	0.78723	0.87234	0.07447
	SmolLM 1.7B	0.91489	0.17021	0.79787	0.89362	0.05319
	SmolLM2 135M	0.80851	0.79787	0.71277	0.67021	0.36170
	SmolLM2 360M	0.88298	0.69149	0.76596	0.78723	0.34043
	SmolLM2 1.7B	0.96809	0.50000	0.80851	0.85106	0.19149
	GPT Neo 2.7B	0.36170	0.79787	0.57447	0.64894	0.79787
	GPTJ 6B	0.41489	0.88298	0.67021	0.73404	0.87234
GB Space Insertion Perturbation	Gemma2 2B	0.81522	0.93478	0.85870	0.88043	0.83696
	Gemma2 9B	0.97826	0.94565	0.85870	0.92391	0.81522
	Llama3 8B	0.92391	0.59783	0.89130	0.93478	0.13043
	Falcon 7B	0.60870	0.98913	0.83696	0.79348	0.91304
	SmolLM 135M	0.85870	0.25000	0.75000	0.79348	0.09783
	SmolLM 360M	0.86957	0.33696	0.76087	0.86957	0.17391
	SmolLM 1.7B	0.90217	0.25000	0.80435	0.92391	0.07609
	SmolLM2 135M	0.72826	0.85870	0.70652	0.79348	0.53261
	SmolLM2 360M	0.82609	0.70652	0.77174	0.83696	0.40217
	SmolLM2 1.7B	0.91304	0.63043	0.79348	0.90217	0.27174
	GPT Neo 2.7B	0.23913	0.84783	0.64130	0.68478	0.81522
	GPTJ 6B	0.27174	0.94565	0.70652	0.77174	0.92391
GB Swap Adjacent Paragraphs Perturbation	Gemma2 2B	0.96809	0.94681	0.92553	0.84043	0.75532
	Gemma2 9B	0.98936	0.90426	0.89362	0.92553	0.74468
	Llama3 8B	0.94681	0.63830	0.92553	0.92553	0.17021
	Falcon 7B	0.82979	0.98936	0.87234	0.86170	0.93617
	SmolLM 135M	0.97872	0.45745	0.89362	0.91489	0.12766
	SmolLM 360M	0.96809	0.38298	0.89362	0.93617	0.09574
	SmolLM 1.7B	0.95745	0.28723	0.89362	0.95745	0.02128
	SmolLM2 135M	0.96809	0.91489	0.88298	0.89362	0.29787
	SmolLM2 360M	0.94681	0.78723	0.90426	0.90426	0.28723
	SmolLM2 1.7B	0.97872	0.67021	0.90426	0.92553	0.23404
	GPT Neo 2.7B	0.28723	0.97872	0.73404	0.76596	0.94681
	GPTJ 6B	0.45745	0.96809	0.75532	0.82979	0.94681
GB Paraphrase Paragraphs Perturbation	Gemma2 2B	0.92135	0.93258	0.85393	0.74157	0.66292
	Gemma2 9B	0.96629	0.92135	0.77528	0.86517	0.58427
	Llama3 8B	0.86517	0.64045	0.88764	0.87640	0.14607
	Falcon 7B	0.73034	0.98876	0.87640	0.76404	0.87640
	SmolLM 135M	0.97753	0.41573	0.89888	0.91011	0.15730
	SmolLM 360M	0.96629	0.28090	0.87640	0.94382	0.15730
	SmolLM 1.7B	0.97753	0.22472	0.87640	0.96629	0.06742
	SmolLM2 135M	0.92135	0.82022	0.89888	0.78652	0.46067
	SmolLM2 360M	0.93258	0.70787	0.87640	0.86517	0.38202
	SmolLM2 1.7B	0.94382	0.51685	0.85393	0.92135	0.31461
	GPT Neo 2.7B	0.26966	0.92135	0.66292	0.71910	0.95506
	GPTJ 6B	0.33708	0.96629	0.67416	0.77528	0.93258

Table 28. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across a variety of perturbation schemes and reference models. We report the true positive rate at a false positive rate of 5% of each reference model on each dataset and detection technique. The best performance on a dataset is bolded.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Swap Adjacent Sentences Perturbation	Gemma2 2B	0.93617	0.95745	0.92553	0.85106	0.70213
	Gemma2 9B	0.97872	0.94681	0.86170	0.90426	0.68085
	Llama3 8B	0.94681	0.68085	0.91489	0.87234	0.21277
	Falcon 7B	0.80851	0.97872	0.85106	0.77660	0.92553
	SmolLM 135M	0.94681	0.55319	0.89362	0.90426	0.18085
	SmolLM 360M	0.95745	0.57447	0.90426	0.94681	0.19149
	SmolLM 1.7B	0.91489	0.41489	0.92553	0.95745	0.17021
	SmolLM2 135M	0.94681	0.96809	0.88298	0.86170	0.48936
	SmolLM2 360M	0.95745	0.91489	0.91489	0.91489	0.30851
	SmolLM2 1.7B	0.95745	0.71277	0.90426	0.92553	0.27660
	GPT Neo 2.7B	0.26596	0.94681	0.78723	0.75532	0.94681
	GPTJ 6B	0.28723	0.94681	0.78723	0.78723	0.91489
GB Paraphrase Sentences Perturbation	Gemma2 2B	0.95556	0.94444	0.95556	0.88889	0.71111
	Gemma2 9B	0.98889	0.97778	0.91111	0.95556	0.82222
	Llama3 8B	0.96667	0.65556	0.95556	0.93333	0.25556
	Falcon 7B	0.81111	0.98889	0.88889	0.77778	0.95556
	SmolLM 135M	0.96667	0.53333	0.92222	0.94444	0.18889
	SmolLM 360M	0.94444	0.52222	0.92222	0.95556	0.15556
	SmolLM 1.7B	0.95556	0.37778	0.94444	0.97778	0.11111
	SmolLM2 135M	0.88889	0.93333	0.93333	0.78889	0.56667
	SmolLM2 360M	0.95556	0.86667	0.95556	0.93333	0.33333
	SmolLM2 1.7B	0.97778	0.65556	0.93333	0.96667	0.25556
	GPT Neo 2.7B	0.26667	0.97778	0.81111	0.77778	0.96667
	GPTJ 6B	0.32222	0.96667	0.80000	0.82222	0.96667
GB Swap Adjacent Words Perturbation	Gemma2 2B	0.26596	0.91489	0.76596	0.84043	0.82979
	Gemma2 9B	0.64894	0.89362	0.77660	0.89362	0.78723
	Llama3 8B	0.22340	0.63830	0.80851	0.86170	0.09574
	Falcon 7B	0.02128	0.95745	0.74468	0.72340	0.85106
	SmolLM 135M	0.62766	0.27660	0.60638	0.68085	0.08511
	SmolLM 360M	0.64894	0.26596	0.59574	0.78723	0.09574
	SmolLM 1.7B	0.65957	0.11702	0.64894	0.82979	0.04255
	SmolLM2 135M	0.38298	0.75532	0.50000	0.61702	0.35106
	SmolLM2 360M	0.27660	0.62766	0.57447	0.61702	0.36170
	SmolLM2 1.7B	0.59574	0.36170	0.69149	0.76596	0.19149
	GPT Neo 2.7B	0.11702	0.42553	0.46809	0.52128	0.47872
	GPTJ 6B	0.07447	0.68085	0.46809	0.59574	0.65957
GB Swap Word with Synonyms Perturbation	Gemma2 2B	0.70213	0.89362	0.84043	0.81915	0.74468
	Gemma2 9B	0.93617	0.84043	0.78723	0.89362	0.62766
	Llama3 8B	0.58511	0.52128	0.86170	0.86170	0.13830
	Falcon 7B	0.32979	0.95745	0.76596	0.70213	0.84043
	SmolLM 135M	0.86170	0.46809	0.79787	0.77660	0.09574
	SmolLM 360M	0.85106	0.32979	0.80851	0.86170	0.06383
	SmolLM 1.7B	0.86170	0.13830	0.79787	0.90426	0.02128
	SmolLM2 135M	0.67021	0.87234	0.74468	0.73404	0.47872
	SmolLM2 360M	0.65957	0.78723	0.78723	0.79787	0.40426
	SmolLM2 1.7B	0.85106	0.47872	0.77660	0.82979	0.17021
	GPT Neo 2.7B	0.30851	0.84043	0.58511	0.67021	0.84043
	GPTJ 6B	0.24468	0.84043	0.59574	0.72340	0.81915

Table 29. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the confidence interval of the true positive rate at a false positive rate of 5% for each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
Detect LLM Text	Gemma2 2B	(0.98336, 0.99143)	(0.67960, 0.71894)	(0.64254, 0.67357)	(0.75145, 0.78171)	(0.60599, 0.65592)
	Gemma2 9B	(0.99395, 0.99798)	(0.73305, 0.77136)	(0.53617, 0.57020)	(0.88177, 0.90774)	(0.63119, 0.68164)
	Llama3 8B	(0.96017, 0.97480)	(0.51549, 0.55357)	(0.63473, 0.67835)	(0.55709, 0.59692)	(0.37887, 0.41671)
	Falcon 7B	(0.89423, 0.92210)	(0.70625, 0.73615)	(0.64291, 0.67357)	(0.62119, 0.67280)	(0.61149, 0.64343)
	SmolLM 135M	(0.97933, 0.98840)	(0.47820, 0.51248)	(0.63599, 0.66726)	(0.83160, 0.86413)	(0.24679, 0.28058)
	SmolLM 360M	(0.98135, 0.98941)	(0.41567, 0.44996)	(0.64532, 0.68137)	(0.82455, 0.85909)	(0.19965, 0.23015)
	SmolLM 1.7B	(0.98538, 0.99193)	(0.39349, 0.42576)	(0.60474, 0.64180)	(0.87396, 0.90119)	(0.19435, 0.22889)
	SmolLM2 135M	(0.95110, 0.96547)	(0.63700, 0.66853)	(0.62086, 0.65265)	(0.61608, 0.66045)	(0.46458, 0.49811)
	SmolLM2 360M	(0.96395, 0.97605)	(0.62112, 0.65214)	(0.62540, 0.65818)	(0.62818, 0.66625)	(0.42929, 0.46284)
	SmolLM2 1.7B	(0.95942, 0.97227)	(0.53643, 0.57248)	(0.64178, 0.68238)	(0.57751, 0.61937)	(0.33652, 0.36956)
	GPT Neo 2.7B	(0.93496, 0.95210)	(0.65616, 0.68591)	(0.60120, 0.63752)	(0.54626, 0.58786)	(0.64860, 0.67684)
	GPTJ 6B	(0.93622, 0.95337)	(0.65868, 0.68868)	(0.59843, 0.63324)	(0.58101, 0.62768)	(0.65212, 0.68112)
AI vs Human	Gemma2 2B	(0.84152, 0.86714)	(0.69941, 0.73323)	(0.76994, 0.80005)	(0.73533, 0.77073)	(0.59693, 0.64078)
	Gemma2 9B	(0.93661, 0.95298)	(0.73164, 0.76335)	(0.73217, 0.76413)	(0.85790, 0.88193)	(0.54807, 0.59748)
	Llama3 8B	(0.83677, 0.86371)	(0.69017, 0.72319)	(0.78764, 0.81616)	(0.70018, 0.73376)	(0.31220, 0.35579)
	Falcon 7B	(0.73977, 0.77778)	(0.86988, 0.89474)	(0.76388, 0.79971)	(0.71454, 0.75073)	(0.75328, 0.78947)
	SmolLM 135M	(0.82620, 0.85103)	(0.60274, 0.63603)	(0.72820, 0.75938)	(0.70971, 0.74379)	(0.26148, 0.29822)
	SmolLM 360M	(0.83835, 0.86504)	(0.57527, 0.61146)	(0.74274, 0.77259)	(0.72847, 0.76175)	(0.25277, 0.28605)
	SmolLM 1.7B	(0.88193, 0.90386)	(0.53486, 0.57053)	(0.74035, 0.77179)	(0.75621, 0.78553)	(0.18991, 0.22689)
	SmolLM2 135M	(0.80269, 0.83043)	(0.78156, 0.80771)	(0.70734, 0.74010)	(0.64342, 0.68279)	(0.60697, 0.64210)
	SmolLM2 360M	(0.83148, 0.85817)	(0.77021, 0.79952)	(0.73718, 0.76943)	(0.68516, 0.72399)	(0.50686, 0.54226)
	SmolLM2 1.7B	(0.87876, 0.89911)	(0.70575, 0.73772)	(0.76545, 0.79662)	(0.70443, 0.73877)	(0.38061, 0.41839)
	GPT Neo 2.7B	(0.69202, 0.73111)	(0.76413, 0.79239)	(0.68436, 0.72319)	(0.63232, 0.66614)	(0.75674, 0.78288)
	GPTJ 6B	(0.66376, 0.69942)	(0.79556, 0.82013)	(0.71684, 0.74776)	(0.66560, 0.70074)	(0.78816, 0.81537)
HC3	Gemma2 2B	(0.95138, 0.97091)	(0.98461, 0.99099)	(0.99831, 0.99981)	(0.98010, 0.98818)	(0.97635, 0.98630)
	Gemma2 9B	(0.99606, 0.99944)	(0.97410, 0.98236)	(0.99662, 0.99925)	(0.95664, 0.96922)	(0.95063, 0.96659)
	Llama3 8B	(0.99606, 0.99887)	(0.95064, 0.96359)	(0.99737, 0.99944)	(0.92098, 0.94426)	(0.64451, 0.68844)
	Falcon 7B	(0.99530, 0.99831)	(0.99793, 0.99962)	(0.99681, 0.99906)	(0.96543, 0.97633)	(0.97464, 0.98385)
	SmolLM 135M	(0.99136, 0.99606)	(0.95777, 0.96866)	(0.99625, 0.99887)	(0.97935, 0.98762)	(0.50617, 0.55612)
	SmolLM 360M	(0.99794, 0.99981)	(0.95795, 0.96997)	(0.99737, 0.99944)	(0.97241, 0.98255)	(0.52289, 0.56982)
	SmolLM 1.7B	(0.99700, 0.99962)	(0.95852, 0.97072)	(0.99737, 0.99944)	(0.96396, 0.97485)	(0.51726, 0.57433)
	SmolLM2 135M	(0.94876, 0.96509)	(0.97560, 0.98311)	(0.90109, 0.91929)	(0.76370, 0.79374)	(0.97954, 0.98686)
	SmolLM2 360M	(0.99118, 0.99718)	(0.99756, 0.99944)	(0.99662, 0.99925)	(0.97842, 0.98780)	(0.96152, 0.97260)
	SmolLM2 1.7B	(0.99756, 0.99981)	(0.98011, 0.98742)	(0.99775, 0.99962)	(0.96396, 0.97842)	(0.78848, 0.82940)
	GPT Neo 2.7B	(0.83915, 0.91405)	(0.99718, 0.99944)	(0.99756, 0.99944)	(0.98818, 0.99343)	(0.99625, 0.99987)
	GPTJ 6B	(0.81868, 0.89941)	(0.99456, 0.99794)	(0.99662, 0.99906)	(0.98986, 0.99493)	(0.99343, 0.99700)
HC3 Plus	Gemma2 2B	(0.91450, 0.94768)	(0.75186, 0.77883)	(0.77332, 0.79567)	(0.76250, 0.78699)	(0.67790, 0.71356)
	Gemma2 9B	(0.91184, 0.93456)	(0.75364, 0.77794)	(0.76533, 0.78769)	(0.73182, 0.75878)	(0.68712, 0.71853)
	Llama3 8B	(0.94732, 0.96524)	(0.75967, 0.78787)	(0.76481, 0.78627)	(0.72383, 0.75275)	(0.52908, 0.57415)
	Falcon 7B	(0.87354, 0.90174)	(0.77368, 0.79639)	(0.75824, 0.78272)	(0.74707, 0.77031)	(0.67081, 0.69936)
	SmolLM 135M	(0.94252, 0.96205)	(0.72437, 0.75187)	(0.76570, 0.78680)	(0.74281, 0.76747)	(0.34569, 0.39961)
	SmolLM 360M	(0.93100, 0.95459)	(0.65927, 0.69386)	(0.76339, 0.78521)	(0.73926, 0.76251)	(0.24813, 0.28787)
	SmolLM 1.7B	(0.92391, 0.94999)	(0.64171, 0.67471)	(0.76250, 0.78450)	(0.73448, 0.75789)	(0.23980, 0.27759)
	SmolLM2 135M	(0.94306, 0.96843)	(0.79142, 0.81235)	(0.76445, 0.78699)	(0.73731, 0.76126)	(0.73342, 0.76074)
	SmolLM2 360M	(0.94413, 0.96399)	(0.77492, 0.79692)	(0.76569, 0.78822)	(0.73660, 0.76126)	(0.65945, 0.69085)
	SmolLM2 1.7B	(0.92355, 0.94714)	(0.72685, 0.75417)	(0.76836, 0.79159)	(0.73873, 0.76411)	(0.41167, 0.45548)
	GPT Neo 2.7B	(0.79832, 0.89341)	(0.78538, 0.80862)	(0.77917, 0.80365)	(0.75949, 0.78255)	(0.76889, 0.79106)
	GPTJ 6B	(0.82103, 0.90282)	(0.77811, 0.80028)	(0.77527, 0.79887)	(0.76392, 0.78680)	(0.76215, 0.78450)
ESL GPT4o Mini	Gemma2 2B	(1.00000, 1.00000)	(0.31145, 0.39091)	(0.18551, 0.27745)	(0.05051, 0.07441)	(0.10707, 0.16971)
	Gemma2 9B	(1.00000, 1.00000)	(0.38821, 0.45488)	(0.04007, 0.07643)	(0.26969, 0.33974)	(0.13502, 0.21077)
	Llama3 8B	(1.00000, 1.00000)	(0.21918, 0.28956)	(0.13737, 0.22930)	(0.03333, 0.05051)	(0.03973, 0.06802)
	Falcon 7B	(1.00000, 1.00000)	(0.69287, 0.74939)	(0.11899, 0.18044)	(0.03299, 0.05756)	(0.24956, 0.32819)
	SmolLM 135M	(1.00000, 1.00000)	(0.20034, 0.25286)	(0.18889, 0.28013)	(0.24376, 0.32829)	(0.01481, 0.02660)
	SmolLM 360M	(1.00000, 1.00000)	(0.14141, 0.18013)	(0.13535, 0.22863)	(0.28788, 0.36566)	(0.01246, 0.02290)
	SmolLM 1.7B	(1.00000, 1.00000)	(0.16599, 0.21482)	(0.13906, 0.22156)	(0.27103, 0.32997)	(0.01818, 0.03300)
	SmolLM2 135M	(0.99899, 1.00000)	(0.52054, 0.60034)	(0.17441, 0.29327)	(0.06464, 0.09832)	(0.07946, 0.11246)
	SmolLM2 360M	(1.00000, 1.00000)	(0.44577, 0.51112)	(0.18247, 0.29529)	(0.06934, 0.11077)	(0.07306, 0.10606)
	SmolLM2 1.7B	(1.00000, 1.00000)	(0.37035, 0.44043)	(0.21414, 0.31987)	(0.04983, 0.07811)	(0.07339, 0.10237)
	GPT Neo 2.7B	(0.99999, 1.00000)	(0.23332, 0.29192)	(0.00774, 0.03940)	(0.01717, 0.03266)	(0.19394, 0.24815)
	GPTJ 6B	(1.00000, 1.00000)	(0.30471, 0.36364)	(0.00640, 0.04074)	(0.01347, 0.02626)	(0.25959, 0.33300)

Table 30. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the confidence interval of the true positive rate at a false positive rate of 5% for each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Essay ChatGPT	Gemma2 2B	(1.00000, 1.00000)	(0.98700, 0.99900)	(1.00000, 1.00000)	(0.99100, 1.00000)	(0.88398, 0.95100)
	Gemma2 9B	(1.00000, 1.00000)	(0.98900, 0.99900)	(1.00000, 1.00000)	(0.99400, 1.00000)	(0.96198, 0.98600)
	Llama3 8B	(1.00000, 1.00000)	(0.62100, 0.73400)	(1.00000, 1.00000)	(0.99300, 1.00000)	(0.12400, 0.19000)
	Falcon 7B	(0.99698, 1.00000)	(1.00000, 1.00000)	(0.99698, 1.00000)	(0.98185, 0.99698)	(0.88808, 0.92744)
	SmolLM 135M	(1.00000, 1.00000)	(0.37400, 0.48000)	(0.99600, 1.00000)	(0.98200, 0.99600)	(0.00300, 0.01600)
	SmolLM 360M	(1.00000, 1.00000)	(0.21000, 0.28003)	(0.99700, 1.00000)	(0.99100, 0.99900)	(0.00000, 0.00100)
	SmolLM 1.7B	(1.00000, 1.00000)	(0.14600, 0.20800)	(1.00000, 1.00000)	(0.99300, 1.00000)	(0.00000, 0.00300)
	SmolLM2 135M	(1.00000, 1.00000)	(0.90300, 0.94300)	(0.99200, 1.00000)	(0.97000, 0.99200)	(0.08400, 0.13202)
	SmolLM2 360M	(1.00000, 1.00000)	(0.80900, 0.86902)	(0.99700, 1.00000)	(0.98798, 0.99800)	(0.02900, 0.05800)
	SmolLM2 1.7B	(1.00000, 1.00000)	(0.43200, 0.51602)	(1.00000, 1.00000)	(0.99200, 1.00000)	(0.00800, 0.02300)
	GPT Neo 2.7B	(0.37298, 0.54900)	(0.99700, 1.00000)	(0.98100, 0.99800)	(0.94100, 0.97702)	(0.99600, 1.00000)
	GPTJ 6B	(0.52000, 0.68600)	(0.99800, 1.00000)	(0.98798, 0.99900)	(0.96700, 0.99100)	(0.99700, 1.00000)
GB News ChatGPT	Gemma2 2B	(0.88318, 0.93454)	(0.98590, 0.99799)	(0.95468, 0.98087)	(0.94056, 0.96777)	(0.93454, 0.96878)
	Gemma2 9B	(0.95770, 0.97986)	(0.96274, 0.98993)	(0.90937, 0.94159)	(0.93051, 0.95972)	(0.90937, 0.95065)
	Llama3 8B	(0.89225, 0.93051)	(0.91239, 0.95569)	(0.95065, 0.97784)	(0.94864, 0.97281)	(0.48940, 0.59416)
	Falcon 7B	(0.49345, 0.58613)	(0.99695, 1.00000)	(0.95569, 0.97885)	(0.95667, 0.97784)	(0.97885, 0.99396)
	SmolLM 135M	(0.96777, 0.98691)	(0.89325, 0.93756)	(0.95871, 0.98288)	(0.95365, 0.97986)	(0.50856, 0.60123)
	SmolLM 360M	(0.97180, 0.98993)	(0.85700, 0.91037)	(0.95972, 0.98087)	(0.95770, 0.98187)	(0.40081, 0.49648)
	SmolLM 1.7B	(0.95670, 0.98389)	(0.74418, 0.83890)	(0.96073, 0.98389)	(0.96274, 0.98691)	(0.16213, 0.26387)
	SmolLM2 135M	(0.90332, 0.94159)	(0.99496, 1.00000)	(0.95670, 0.98187)	(0.95972, 0.98087)	(0.97482, 0.99197)
	SmolLM2 360M	(0.89121, 0.93051)	(0.99194, 1.00000)	(0.95368, 0.98087)	(0.96173, 0.98187)	(0.95869, 0.98389)
	SmolLM2 1.7B	(0.92548, 0.95569)	(0.97482, 0.99194)	(0.94562, 0.97281)	(0.94562, 0.97382)	(0.77845, 0.84595)
	GPT Neo 2.7B	(0.03525, 0.06347)	(0.98590, 0.99799)	(0.93756, 0.96677)	(0.95468, 0.97888)	(0.98288, 0.99597)
	GPTJ 6B	(0.07754, 0.13900)	(0.98792, 0.99799)	(0.91742, 0.95267)	(0.95267, 0.97482)	(0.98489, 0.99698)
GB Creative ChatGPT	Gemma2 2B	(0.98782, 0.99898)	(0.91574, 0.95533)	(0.90048, 0.94114)	(0.76751, 0.84264)	(0.46495, 0.60812)
	Gemma2 9B	(0.99594, 1.00000)	(0.88119, 0.93099)	(0.87513, 0.91779)	(0.87510, 0.91777)	(0.46901, 0.62335)
	Llama3 8B	(0.98274, 0.99594)	(0.68426, 0.78683)	(0.91675, 0.95330)	(0.79695, 0.85076)	(0.04162, 0.09543)
	Falcon 7B	(0.97846, 0.99385)	(0.97741, 0.99282)	(0.78049, 0.85744)	(0.57538, 0.68410)	(0.88718, 0.92823)
	SmolLM 135M	(0.98477, 0.99695)	(0.61419, 0.72792)	(0.87310, 0.92693)	(0.83553, 0.89442)	(0.00203, 0.01218)
	SmolLM 360M	(0.98985, 0.99898)	(0.38376, 0.51982)	(0.87716, 0.92386)	(0.89036, 0.93503)	(0.00000, 0.00508)
	SmolLM 1.7B	(0.98782, 0.99797)	(0.20708, 0.29949)	(0.86497, 0.92794)	(0.92792, 0.95838)	(0.00000, 0.00000)
	SmolLM2 135M	(0.98274, 0.99594)	(0.87815, 0.93604)	(0.80102, 0.87310)	(0.65074, 0.72995)	(0.04467, 0.08528)
	SmolLM2 360M	(0.98985, 0.99898)	(0.71368, 0.81830)	(0.84772, 0.89949)	(0.69239, 0.78782)	(0.01421, 0.03350)
	SmolLM2 1.7B	(0.99492, 1.00000)	(0.49137, 0.60005)	(0.86904, 0.91980)	(0.76954, 0.83452)	(0.00203, 0.01320)
	GPT Neo 2.7B	(0.86599, 0.93706)	(0.88528, 0.93096)	(0.56650, 0.69239)	(0.32586, 0.42944)	(0.85178, 0.90457)
	GPTJ 6B	(0.82843, 0.90558)	(0.92284, 0.95838)	(0.63449, 0.73299)	(0.42944, 0.52995)	(0.89442, 0.93604)
GB Essay GPT4o	Gemma2 2B	(1.00000, 1.00000)	(0.97891, 0.99599)	(0.99298, 1.00000)	(0.98897, 0.99900)	(0.24273, 0.38618)
	Gemma2 9B	(1.00000, 1.00000)	(0.96891, 0.98997)	(0.98997, 0.99900)	(0.99398, 1.00000)	(0.38516, 0.56169)
	Llama3 8B	(1.00000, 1.00000)	(0.92879, 0.96790)	(0.99498, 1.00000)	(0.98596, 0.99699)	(0.03408, 0.06219)
	Falcon 7B	(0.97417, 0.99383)	(0.99690, 1.00000)	(0.97521, 0.99587)	(0.92149, 0.96694)	(0.95145, 0.97624)
	SmolLM 135M	(0.99599, 1.00000)	(0.21163, 0.31196)	(0.92678, 0.97392)	(0.95186, 0.98195)	(0.00000, 0.00702)
	SmolLM 360M	(0.99900, 1.00000)	(0.11334, 0.17252)	(0.96790, 0.98997)	(0.98295, 0.99799)	(0.00000, 0.00702)
	SmolLM 1.7B	(0.99699, 1.00000)	(0.02207, 0.04714)	(0.96690, 0.99298)	(0.98997, 0.99900)	(0.00000, 0.00000)
	SmolLM2 135M	(1.00000, 1.00000)	(0.89569, 0.93882)	(0.93681, 0.98395)	(0.90471, 0.95888)	(0.07623, 0.12437)
	SmolLM2 360M	(1.00000, 1.00000)	(0.71314, 0.81547)	(0.98295, 0.99699)	(0.96590, 0.98796)	(0.00702, 0.02307)
	SmolLM2 1.7B	(1.00000, 1.00000)	(0.38816, 0.47643)	(0.99298, 1.00000)	(0.98092, 0.99699)	(0.00000, 0.00301)
	GPT Neo 2.7B	(0.31093, 0.51056)	(0.94280, 0.97994)	(0.90973, 0.96891)	(0.77131, 0.86359)	(0.94885, 0.97593)
	GPTJ 6B	(0.43929, 0.60682)	(0.97292, 0.99097)	(0.94383, 0.98094)	(0.85858, 0.93180)	(0.97091, 0.99298)
GB Creative GPT4o	Gemma2 2B	(0.97566, 0.99493)	(0.93507, 0.97771)	(0.80325, 0.90061)	(0.59229, 0.70487)	(0.17137, 0.27181)
	Gemma2 9B	(0.99899, 1.00000)	(0.86105, 0.92698)	(0.74744, 0.83164)	(0.79615, 0.86311)	(0.15010, 0.27183)
	Llama3 8B	(1.00000, 1.00000)	(0.77987, 0.87525)	(0.82959, 0.91078)	(0.65213, 0.74645)	(0.09026, 0.17546)
	Falcon 7B	(0.94333, 0.97837)	(0.98043, 0.99485)	(0.33780, 0.50260)	(0.23893, 0.35018)	(0.96189, 0.98970)
	SmolLM 135M	(0.96957, 0.98986)	(0.45332, 0.58522)	(0.46651, 0.62680)	(0.51014, 0.61460)	(0.01521, 0.03651)
	SmolLM 360M	(0.99087, 1.00000)	(0.23020, 0.33976)	(0.56691, 0.71095)	(0.65616, 0.76169)	(0.00507, 0.02028)
	SmolLM 1.7B	(0.99696, 1.00000)	(0.05578, 0.10243)	(0.48580, 0.65418)	(0.82556, 0.90368)	(0.00000, 0.00710)
	SmolLM2 135M	(0.97160, 0.99594)	(0.88641, 0.95132)	(0.38943, 0.54871)	(0.27584, 0.38238)	(0.42089, 0.54668)
	SmolLM2 360M	(0.99189, 1.00000)	(0.87323, 0.94526)	(0.50910, 0.63185)	(0.36308, 0.49800)	(0.36613, 0.44222)
	SmolLM2 1.7B	(1.00000, 1.00000)	(0.56995, 0.70385)	(0.60446, 0.75456)	(0.51724, 0.64097)	(0.06288, 0.11460)
	GPT Neo 2.7B	(0.86711, 0.93306)	(0.73223, 0.83471)	(0.36004, 0.48377)	(0.16531, 0.24544)	(0.83466, 0.89858)
	GPTJ 6B	(0.84280, 0.91988)	(0.83063, 0.91278)	(0.41988, 0.53448)	(0.22515, 0.31440)	(0.90667, 0.95132)

Table 31. Two standard deviation confidence interval of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across datasets and reference models. We report the confidence interval of the true positive rate at a false positive rate of 5% for each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB News Claude	Gemma2 2B	(0.57700, 0.69600)	(0.78400, 0.85502)	(0.55200, 0.71602)	(0.58098, 0.71500)	(0.34400, 0.47600)
	Gemma2 9B	(0.91900, 0.95800)	(0.90298, 0.95300)	(0.50595, 0.60702)	(0.87798, 0.92500)	(0.34395, 0.47302)
	Llama3 8B	(0.65098, 0.76800)	(0.69700, 0.77905)	(0.76600, 0.88900)	(0.59700, 0.67700)	(0.33400, 0.41700)
	Falcon 7B	(0.16107, 0.24450)	(0.77520, 0.86327)	(0.57819, 0.69293)	(0.50521, 0.61066)	(0.56663, 0.67442)
	SmolLM 135M	(0.47897, 0.57600)	(0.39500, 0.53102)	(0.22800, 0.32800)	(0.26993, 0.40400)	(0.13500, 0.22100)
	SmolLM 360M	(0.71900, 0.79800)	(0.40698, 0.52502)	(0.30498, 0.39500)	(0.41800, 0.56400)	(0.13898, 0.20802)
	SmolLM 1.7B	(0.79697, 0.86100)	(0.46493, 0.61100)	(0.33600, 0.49000)	(0.52198, 0.63802)	(0.11400, 0.22500)
	SmolLM2 135M	(0.24800, 0.35800)	(0.56200, 0.69900)	(0.22298, 0.33100)	(0.20900, 0.31900)	(0.22500, 0.32302)
	SmolLM2 360M	(0.39400, 0.50205)	(0.63600, 0.73802)	(0.30700, 0.42300)	(0.26198, 0.37602)	(0.27198, 0.39603)
	SmolLM2 1.7B	(0.56398, 0.65502)	(0.74000, 0.82700)	(0.40100, 0.52702)	(0.33697, 0.44900)	(0.35497, 0.47102)
GB Creative Claude	GPT Neo 2.7B	(0.09000, 0.15300)	(0.37598, 0.51802)	(0.38900, 0.50902)	(0.36100, 0.47800)	(0.28400, 0.41000)
	GPTJ 6B	(0.15300, 0.23600)	(0.30600, 0.43003)	(0.38200, 0.52900)	(0.37798, 0.48002)	(0.20200, 0.30600)
	Gemma2 2B	(0.77016, 0.87702)	(0.77518, 0.85181)	(0.64010, 0.75305)	(0.58266, 0.69360)	(0.23992, 0.33269)
	Gemma2 9B	(0.98488, 0.99698)	(0.88710, 0.94052)	(0.69453, 0.78427)	(0.87195, 0.92238)	(0.30645, 0.44156)
	Llama3 8B	(0.95867, 0.98790)	(0.49088, 0.57765)	(0.84980, 0.92440)	(0.71673, 0.80746)	(0.07460, 0.13407)
	Falcon 7B	(0.83365, 0.90599)	(0.61978, 0.72624)	(0.45865, 0.60127)	(0.41733, 0.53208)	(0.34811, 0.46591)
	SmolLM 135M	(0.74997, 0.83871)	(0.28730, 0.41638)	(0.26210, 0.40524)	(0.38808, 0.50504)	(0.01610, 0.04133)
	SmolLM 360M	(0.91935, 0.95766)	(0.19655, 0.29640)	(0.35277, 0.52623)	(0.54534, 0.64617)	(0.00806, 0.02520)
	SmolLM 1.7B	(0.95262, 0.98085)	(0.21976, 0.31250)	(0.38405, 0.57664)	(0.68143, 0.79133)	(0.00706, 0.03128)
	SmolLM2 135M	(0.73788, 0.83972)	(0.44443, 0.59879)	(0.24186, 0.36492)	(0.22984, 0.32661)	(0.07457, 0.15728)
GB Essay Claude	SmolLM2 360M	(0.82356, 0.91532)	(0.47881, 0.60990)	(0.36089, 0.47986)	(0.30945, 0.42442)	(0.11694, 0.17341)
	SmolLM2 1.7B	(0.95363, 0.98185)	(0.57961, 0.68851)	(0.53629, 0.69156)	(0.53931, 0.66230)	(0.10381, 0.17641)
	GPT Neo 2.7B	(0.40723, 0.57460)	(0.28427, 0.39216)	(0.40121, 0.52923)	(0.30746, 0.41031)	(0.20260, 0.31855)
	GPTJ 6B	(0.41026, 0.54743)	(0.33163, 0.43553)	(0.52117, 0.62702)	(0.39514, 0.49496)	(0.27112, 0.38511)
	Gemma2 2B	(0.79300, 0.86705)	(0.66197, 0.77202)	(0.86700, 0.92000)	(0.81700, 0.89100)	(0.27598, 0.39703)
	Gemma2 9B	(0.97400, 0.99100)	(0.93200, 0.96800)	(0.78498, 0.86900)	(0.96900, 0.99000)	(0.72598, 0.85000)
	Llama3 8B	(0.93800, 0.96900)	(0.52000, 0.61900)	(0.91200, 0.96700)	(0.89200, 0.94200)	(0.13500, 0.18900)
	Falcon 7B	(0.64184, 0.75408)	(0.82755, 0.88367)	(0.74898, 0.83776)	(0.75306, 0.84286)	(0.50814, 0.61635)
	SmolLM 135M	(0.71900, 0.81200)	(0.08800, 0.13302)	(0.39200, 0.53400)	(0.65398, 0.74400)	(0.01500, 0.03600)
	SmolLM 360M	(0.92600, 0.96300)	(0.07900, 0.12100)	(0.50200, 0.61200)	(0.83900, 0.90700)	(0.00898, 0.02500)
GB Essay Deepseek V3	SmolLM 1.7B	(0.89600, 0.94600)	(0.12200, 0.18600)	(0.61198, 0.74600)	(0.90600, 0.94300)	(0.02600, 0.05600)
	SmolLM2 135M	(0.77200, 0.88100)	(0.61300, 0.71300)	(0.43897, 0.60302)	(0.57400, 0.68202)	(0.14700, 0.23602)
	SmolLM2 360M	(0.92498, 0.96602)	(0.66197, 0.75602)	(0.59100, 0.73500)	(0.75300, 0.81900)	(0.14000, 0.21200)
	SmolLM2 1.7B	(0.96800, 0.98700)	(0.47897, 0.55200)	(0.78200, 0.86500)	(0.83600, 0.90800)	(0.11500, 0.17103)
	GPT Neo 2.7B	(0.18800, 0.29900)	(0.55900, 0.66100)	(0.59500, 0.77407)	(0.68098, 0.78700)	(0.52198, 0.61105)
	GPTJ 6B	(0.26998, 0.37700)	(0.54900, 0.67900)	(0.72895, 0.84600)	(0.77200, 0.86400)	(0.52798, 0.63600)
	Gemma2 2B	(0.99895, 1.00000)	(0.99580, 1.00000)	(0.99685, 1.00000)	(0.99265, 1.00000)	(0.97796, 0.99475)
	Gemma2 9B	(1.00000, 1.00000)	(1.00000, 1.00000)	(0.99580, 1.00000)	(0.99685, 1.00000)	(0.99265, 1.00000)
	Llama3 8B	(1.00000, 1.00000)	(0.99475, 1.00000)	(0.99685, 1.00000)	(0.98741, 0.99790)	(0.83736, 0.89192)
	Falcon 7B	(0.98830, 0.99894)	(1.00000, 1.00000)	(0.99362, 1.00000)	(0.98191, 0.99574)	(0.97021, 0.98830)
GB Creative Deepseek V3	SmolLM 135M	(0.99580, 1.00000)	(0.93599, 0.96852)	(0.99685, 1.00000)	(0.96747, 0.98846)	(0.17733, 0.26443)
	SmolLM 360M	(0.99895, 1.00000)	(0.89819, 0.94124)	(1.00000, 1.00000)	(0.98846, 0.99895)	(0.14373, 0.21721)
	SmolLM 1.7B	(1.00000, 1.00000)	(0.73242, 0.81742)	(1.00000, 1.00000)	(0.99056, 0.99895)	(0.16472, 0.23610)
	SmolLM2 135M	(0.99895, 1.00000)	(0.99685, 1.00000)	(0.99580, 1.00000)	(0.96222, 0.98951)	(0.53410, 0.63694)
	SmolLM2 360M	(1.00000, 1.00000)	(1.00000, 1.00000)	(0.99790, 1.00000)	(0.98846, 0.99790)	(0.16789, 0.23924)
	SmolLM2 1.7B	(1.00000, 1.00000)	(0.98951, 0.99895)	(1.00000, 1.00000)	(0.99056, 0.99895)	(0.13012, 0.18995)
	GPT Neo 2.7B	(0.45115, 0.64536)	(0.99161, 0.99895)	(0.99161, 0.99895)	(0.94963, 0.98009)	(0.98951, 0.99895)
	GPTJ 6B	(0.59706, 0.71566)	(0.99265, 1.00000)	(0.99056, 0.99895)	(0.97899, 0.99370)	(0.98951, 0.99895)
	Gemma2 2B	(0.96291, 0.99144)	(0.99857, 1.00000)	(0.98146, 0.99715)	(0.94437, 0.97575)	(0.96862, 0.99144)
	Gemma2 9B	(0.99572, 1.00000)	(1.00000, 1.00000)	(0.98716, 0.99857)	(0.95432, 0.98146)	(0.97432, 0.99715)
GB Creative Deepseek V3	Llama3 8B	(0.99711, 1.00000)	(0.99572, 1.00000)	(0.99144, 1.00000)	(0.94290, 0.97432)	(0.67190, 0.77175)
	Falcon 7B	(0.90139, 0.95571)	(1.00000, 1.00000)	(0.91000, 0.95714)	(0.80711, 0.89861)	(0.94143, 0.98143)
	SmolLM 135M	(0.91869, 0.96006)	(0.93438, 0.97290)	(0.92154, 0.96434)	(0.75175, 0.82168)	(0.35521, 0.46933)
	SmolLM 360M	(0.97001, 0.99001)	(0.94294, 0.97432)	(0.95578, 0.98431)	(0.83452, 0.90014)	(0.29672, 0.41797)
	SmolLM 1.7B	(0.98716, 1.00000)	(0.89016, 0.94583)	(0.96002, 0.98716)	(0.88588, 0.93866)	(0.20681, 0.32525)
	SmolLM2 135M	(0.97147, 0.99287)	(0.99144, 1.00000)	(0.91441, 0.96006)	(0.75178, 0.82026)	(0.72892, 0.81455)
	SmolLM2 360M	(0.99001, 1.00000)	(0.99572, 1.00000)	(0.94151, 0.97290)	(0.81455, 0.88302)	(0.63053, 0.70756)
	SmolLM2 1.7B	(1.00000, 1.00000)	(1.00000, 1.00000)	(0.96434, 0.99001)	(0.91155, 0.95435)	(0.43795, 0.54782)
	GPT Neo 2.7B	(0.60339, 0.71897)	(0.97432, 0.99287)	(0.81740, 0.89158)	(0.59201, 0.69900)	(0.88873, 0.95292)
	GPTJ 6B	(0.55777, 0.67618)	(0.99715, 1.00000)	(0.85877, 0.91583)	(0.73174, 0.82603)	(0.96291, 0.98859)

Table 32. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across a variety of perturbation schemes and reference models. We report the confidence interval of the true positive rate at a false positive rate of 5% for each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Spelling Error Perturbation	Gemma2 2B	(0.56522, 0.89130)	(0.78261, 0.96739)	(0.65217, 0.86957)	(0.64130, 0.85870)	(0.67391, 0.90217)
	Gemma2 9B	(0.88043, 0.98913)	(0.77174, 0.93478)	(0.50000, 0.86957)	(0.85870, 0.97826)	(0.54348, 0.88043)
	Llama3 8B	(0.69565, 0.95652)	(0.35870, 0.67391)	(0.73886, 0.91304)	(0.82609, 0.96739)	(0.04348, 0.18505)
	Falcon 7B	(0.38043, 0.63043)	(0.90217, 1.00000)	(0.48913, 0.83696)	(0.54348, 0.79348)	(0.73913, 0.94565)
	SmolLM 135M	(0.71739, 0.91304)	(0.09783, 0.26087)	(0.43478, 0.77174)	(0.51087, 0.80462)	(0.04348, 0.16304)
	SmolLM 360M	(0.75000, 0.94565)	(0.14130, 0.33696)	(0.47799, 0.79348)	(0.64130, 0.88043)	(0.06522, 0.21739)
	SmolLM 1.7B	(0.73913, 0.93478)	(0.09783, 0.26114)	(0.54321, 0.82636)	(0.70652, 0.89130)	(0.00000, 0.10870)
	SmolLM2 135M	(0.55435, 0.83696)	(0.58696, 0.81522)	(0.38043, 0.71739)	(0.43451, 0.73913)	(0.25000, 0.45652)
	SmolLM2 360M	(0.61957, 0.90217)	(0.55435, 0.79348)	(0.48913, 0.76087)	(0.56522, 0.80435)	(0.25000, 0.47826)
	SmolLM2 1.7B	(0.83696, 0.95652)	(0.26087, 0.52174)	(0.54348, 0.82609)	(0.66304, 0.88043)	(0.05435, 0.20652)
	GPT Neo 2.7B	(0.14130, 0.41304)	(0.59755, 0.83696)	(0.35870, 0.64130)	(0.39130, 0.67391)	(0.58696, 0.83723)
	GPTJ 6B	(0.06522, 0.44565)	(0.71739, 0.91304)	(0.40217, 0.68478)	(0.48886, 0.77174)	(0.70652, 0.90217)
GB Character Capitalization Perturbation	Gemma2 2B	(0.94681, 1.00000)	(0.79787, 0.95745)	(0.71277, 0.93617)	(0.73404, 0.91489)	(0.55293, 0.85106)
	Gemma2 9B	(1.00000, 1.00000)	(0.85106, 0.97872)	(0.67021, 0.92553)	(0.84043, 0.96809)	(0.51064, 0.85106)
	Llama3 8B	(0.93617, 1.00000)	(0.27660, 0.69176)	(0.80851, 0.95745)	(0.81915, 0.94681)	(0.00000, 0.11702)
	Falcon 7B	(0.53191, 0.82979)	(0.92553, 1.00000)	(0.71250, 0.90426)	(0.69149, 0.87234)	(0.75532, 0.92553)
	SmolLM 135M	(0.79787, 0.96809)	(0.13830, 0.39362)	(0.61702, 0.84043)	(0.61702, 0.82979)	(0.04255, 0.17021)
	SmolLM 360M	(0.85106, 0.96835)	(0.10638, 0.28723)	(0.68085, 0.87234)	(0.77660, 0.93617)	(0.02128, 0.13856)
	SmolLM 1.7B	(0.80824, 0.96809)	(0.07447, 0.27686)	(0.70186, 0.89362)	(0.80824, 0.94681)	(0.01064, 0.12766)
	SmolLM2 135M	(0.64867, 0.88298)	(0.65957, 0.89362)	(0.55319, 0.80851)	(0.51064, 0.79814)	(0.27660, 0.55319)
	SmolLM2 360M	(0.80851, 0.94681)	(0.59548, 0.79787)	(0.67021, 0.85133)	(0.69149, 0.90452)	(0.25532, 0.43617)
	SmolLM2 1.7B	(0.92553, 1.00000)	(0.37234, 0.64894)	(0.73404, 0.90426)	(0.78723, 0.93617)	(0.11702, 0.29787)
	GPT Neo 2.7B	(0.15957, 0.51064)	(0.70213, 0.89362)	(0.45745, 0.79787)	(0.54255, 0.74468)	(0.70213, 0.89362)
	GPTJ 6B	(0.32979, 0.62793)	(0.79787, 0.94681)	(0.54229, 0.81915)	(0.63830, 0.82979)	(0.79787, 0.93617)
GB Space Insertion Perturbation	Gemma2 2B	(0.67364, 0.96739)	(0.86957, 0.97826)	(0.75000, 0.94565)	(0.81522, 0.94565)	(0.73913, 0.91304)
	Gemma2 9B	(0.94565, 1.00000)	(0.90217, 0.98913)	(0.66304, 0.93478)	(0.84783, 0.97826)	(0.70652, 0.92391)
	Llama3 8B	(0.85870, 0.97826)	(0.46739, 0.71739)	(0.82582, 0.95652)	(0.86957, 0.97826)	(0.06522, 0.23913)
	Falcon 7B	(0.45652, 0.77174)	(0.95652, 1.00000)	(0.72826, 0.92391)	(0.70652, 0.90217)	(0.82582, 0.96739)
	SmolLM 135M	(0.75000, 0.93478)	(0.16304, 0.36957)	(0.59783, 0.84783)	(0.70652, 0.88043)	(0.04348, 0.17391)
	SmolLM 360M	(0.77174, 0.95652)	(0.22826, 0.44565)	(0.67391, 0.86957)	(0.79348, 0.93478)	(0.09783, 0.27174)
	SmolLM 1.7B	(0.81522, 0.95652)	(0.16304, 0.35870)	(0.69565, 0.89130)	(0.85870, 0.96739)	(0.03261, 0.14130)
	SmolLM2 135M	(0.61957, 0.86957)	(0.73913, 0.92391)	(0.58696, 0.81522)	(0.70652, 0.88043)	(0.41304, 0.63071)
	SmolLM2 360M	(0.71739, 0.92391)	(0.59783, 0.80435)	(0.67391, 0.86957)	(0.73913, 0.91304)	(0.31522, 0.52174)
	SmolLM2 1.7B	(0.84783, 0.96739)	(0.50000, 0.71739)	(0.71739, 0.91304)	(0.82609, 0.95652)	(0.15217, 0.35897)
	GPT Neo 2.7B	(0.13043, 0.40217)	(0.72826, 0.93478)	(0.47826, 0.82609)	(0.58696, 0.83696)	(0.72826, 0.93478)
	GPTJ 6B	(0.10870, 0.43478)	(0.89130, 0.97826)	(0.54348, 0.83696)	(0.66304, 0.89130)	(0.86957, 0.97826)
GB Swap Adjacent Paragraphs Perturbation	Gemma2 2B	(0.90426, 1.00000)	(0.90426, 1.00000)	(0.80851, 0.98936)	(0.74441, 0.93617)	(0.63830, 0.86170)
	Gemma2 9B	(0.96809, 1.00000)	(0.84043, 0.96809)	(0.68085, 0.94681)	(0.86170, 0.97872)	(0.47872, 0.84043)
	Llama3 8B	(0.90426, 0.98936)	(0.44681, 0.78723)	(0.84043, 0.97872)	(0.85106, 0.96809)	(0.08511, 0.24468)
	Falcon 7B	(0.74468, 0.92553)	(0.96809, 1.00000)	(0.78723, 0.94681)	(0.77660, 0.92580)	(0.87234, 0.97872)
	SmolLM 135M	(0.93617, 1.00000)	(0.35106, 0.58511)	(0.81888, 0.95745)	(0.80851, 0.96809)	(0.06383, 0.21277)
	SmolLM 360M	(0.92553, 1.00000)	(0.28723, 0.51064)	(0.81915, 0.95745)	(0.88298, 0.98936)	(0.03191, 0.19149)
	SmolLM 1.7B	(0.90426, 0.98936)	(0.19149, 0.37234)	(0.82979, 0.94681)	(0.90426, 0.98936)	(0.00000, 0.09574)
	SmolLM2 135M	(0.92553, 1.00000)	(0.86170, 0.96835)	(0.79787, 0.94681)	(0.73404, 0.94681)	(0.21277, 0.39362)
	SmolLM2 360M	(0.90426, 0.98936)	(0.66995, 0.91489)	(0.82979, 0.96809)	(0.84043, 0.96809)	(0.20186, 0.38298)
	SmolLM2 1.7B	(0.94681, 1.00000)	(0.53191, 0.76596)	(0.80851, 0.96809)	(0.86170, 0.97872)	(0.15957, 0.34043)
	GPT Neo 2.7B	(0.14894, 0.43617)	(0.91489, 1.00000)	(0.60612, 0.85106)	(0.65957, 0.87234)	(0.88298, 0.98936)
	GPTJ 6B	(0.20213, 0.58511)	(0.91489, 1.00000)	(0.59574, 0.87234)	(0.71277, 0.90426)	(0.89362, 0.98936)
GB Paraphrase Paragraphs Perturbation	Gemma2 2B	(0.79775, 0.97753)	(0.86517, 0.98876)	(0.65169, 0.95506)	(0.65140, 0.85393)	(0.50562, 0.79775)
	Gemma2 9B	(0.93258, 1.00000)	(0.84270, 0.98876)	(0.51685, 0.87640)	(0.77528, 0.94382)	(0.41573, 0.83146)
	Llama3 8B	(0.73034, 0.94382)	(0.34831, 0.75281)	(0.73034, 0.96629)	(0.79775, 0.96629)	(0.07865, 0.26966)
	Falcon 7B	(0.58427, 0.85393)	(0.96629, 1.00000)	(0.62893, 0.93258)	(0.64045, 0.91011)	(0.79775, 0.94382)
	SmolLM 135M	(0.94382, 1.00000)	(0.24719, 0.60674)	(0.71910, 0.97753)	(0.78624, 0.96629)	(0.07865, 0.25843)
	SmolLM 360M	(0.92135, 1.00000)	(0.19101, 0.48315)	(0.66292, 0.95506)	(0.88764, 0.98876)	(0.06742, 0.24719)
	SmolLM 1.7B	(0.93258, 1.00000)	(0.11236, 0.37079)	(0.65140, 0.95506)	(0.92135, 1.00000)	(0.02247, 0.17978)
	SmolLM2 135M	(0.85393, 0.97753)	(0.71910, 0.94410)	(0.64017, 0.95506)	(0.67416, 0.91011)	(0.30337, 0.56180)
	SmolLM2 360M	(0.86517, 0.98876)	(0.59551, 0.82022)	(0.64045, 0.95506)	(0.77528, 0.94382)	(0.26966, 0.48315)
	SmolLM2 1.7B	(0.87640, 0.98876)	(0.41573, 0.68567)	(0.66292, 0.95506)	(0.79775, 0.97753)	(0.21348, 0.41573)
	GPT Neo 2.7B	(0.10112, 0.43820)	(0.84270, 1.00000)	(0.44944, 0.78680)	(0.60646, 0.83146)	(0.85365, 1.00000)
	GPTJ 6B	(0.07865, 0.44944)	(0.92135, 1.00000)	(0.42697, 0.79775)	(0.66292, 0.87640)	(0.86517, 0.98876)

Table 33. Detection performance of Telescope Perplexity, Binoculars, Perplexity, DetectLLM LRR, and Fast-DetectGPT across a variety of perturbation schemes and reference models. We report the confidence interval of the true positive rate at a false positive rate of 5% for each reference model on each perturbation scheme and detection technique.

Dataset	Reference Model	True Positive Rate at False Positive Rate of 5%				
		Telescope	Binoculars	Perplexity	DetectLLM LRR	Fast-DetectGPT
GB Swap Adjacent Sentences Perturbation	Gemma2 2B	(0.88298, 0.97872)	(0.88298, 0.98936)	(0.80851, 0.98936)	(0.76596, 0.94681)	(0.53165, 0.87234)
	Gemma2 9B	(0.94681, 1.00000)	(0.84043, 1.00000)	(0.72340, 0.94681)	(0.82979, 0.95745)	(0.47872, 0.89362)
	Llama3 8B	(0.89362, 0.98936)	(0.54255, 0.77686)	(0.84043, 0.96809)	(0.79787, 0.94681)	(0.12766, 0.33005)
	Falcon 7B	(0.71277, 0.89362)	(0.93617, 1.00000)	(0.72340, 0.93617)	(0.69122, 0.89362)	(0.85106, 0.97872)
	SmolLM 135M	(0.89335, 1.00000)	(0.43617, 0.71303)	(0.79787, 0.94707)	(0.84043, 0.97872)	(0.10638, 0.25532)
	SmolLM 360M	(0.91489, 0.98936)	(0.45745, 0.69149)	(0.80851, 0.95771)	(0.90426, 0.98936)	(0.11676, 0.26596)
	SmolLM 1.7B	(0.85106, 0.98936)	(0.28723, 0.57473)	(0.82979, 0.96835)	(0.90426, 0.98936)	(0.06383, 0.29787)
	SmolLM2 135M	(0.82979, 0.98936)	(0.87234, 1.00000)	(0.78697, 0.95745)	(0.68085, 0.91489)	(0.35106, 0.59574)
	SmolLM2 360M	(0.89362, 0.98936)	(0.84043, 0.96809)	(0.81915, 0.96809)	(0.85106, 0.96809)	(0.22340, 0.40426)
	SmolLM2 1.7B	(0.91489, 0.98936)	(0.61702, 0.79787)	(0.81915, 0.96809)	(0.86170, 0.98936)	(0.18085, 0.37234)
GB Paraphrase Sentences Perturbation	GPT Neo 2.7B	(0.12766, 0.36170)	(0.89362, 0.98936)	(0.68085, 0.87261)	(0.65957, 0.86170)	(0.89362, 0.98936)
	GPTJ 6B	(0.10638, 0.44707)	(0.89362, 0.98936)	(0.68085, 0.89362)	(0.70213, 0.88298)	(0.85106, 0.96809)
	Gemma2 2B	(0.91111, 1.00000)	(0.90000, 1.00000)	(0.90000, 1.00000)	(0.80000, 0.95556)	(0.57778, 0.86667)
	Gemma2 9B	(0.96667, 1.00000)	(0.87750, 1.00000)	(0.81111, 0.97778)	(0.88889, 1.00000)	(0.58889, 0.91111)
	Llama3 8B	(0.88889, 1.00000)	(0.54444, 0.75556)	(0.88889, 0.98889)	(0.86667, 0.98889)	(0.12194, 0.36667)
	Falcon 7B	(0.71111, 0.88889)	(0.96667, 1.00000)	(0.78889, 0.96667)	(0.67778, 0.91111)	(0.91111, 0.98889)
	SmolLM 135M	(0.91111, 1.00000)	(0.43333, 0.66667)	(0.78861, 0.97778)	(0.86667, 0.98889)	(0.11111, 0.27778)
	SmolLM 360M	(0.88889, 0.98889)	(0.38889, 0.64444)	(0.83333, 0.98889)	(0.90000, 0.98889)	(0.07778, 0.23333)
	SmolLM 1.7B	(0.87778, 0.98889)	(0.23333, 0.51111)	(0.86667, 0.97806)	(0.92222, 1.00000)	(0.02222, 0.20000)
	SmolLM2 135M	(0.83333, 0.95556)	(0.87778, 0.98889)	(0.81111, 0.98889)	(0.68889, 0.91111)	(0.45556, 0.67778)
GB Swap Adjacent Words Perturbation	SmolLM2 360M	(0.90000, 0.98889)	(0.74444, 0.94444)	(0.84444, 0.98889)	(0.80000, 0.98889)	(0.23333, 0.44444)
	SmolLM2 1.7B	(0.94444, 1.00000)	(0.51083, 0.74444)	(0.86667, 0.98889)	(0.91111, 1.00000)	(0.14444, 0.36667)
	GPT Neo 2.7B	(0.11111, 0.40000)	(0.93333, 1.00000)	(0.66667, 0.90000)	(0.67778, 0.87778)	(0.92222, 1.00000)
	GPTJ 6B	(0.10000, 0.48889)	(0.93333, 1.00000)	(0.70000, 0.91111)	(0.75556, 0.92222)	(0.92222, 1.00000)
	Gemma2 2B	(0.14894, 0.50000)	(0.81915, 0.96809)	(0.64894, 0.85106)	(0.62766, 0.91489)	(0.73404, 0.90426)
	Gemma2 9B	(0.40426, 0.75532)	(0.81915, 0.94681)	(0.52128, 0.86170)	(0.82979, 0.95745)	(0.70186, 0.88298)
	Llama3 8B	(0.09574, 0.36170)	(0.51064, 0.75532)	(0.71277, 0.90426)	(0.76596, 0.92553)	(0.03191, 0.25532)
	Falcon 7B	(0.00000, 0.06383)	(0.89362, 0.98936)	(0.53191, 0.82979)	(0.61702, 0.80851)	(0.74468, 0.92553)
	SmolLM 135M	(0.47872, 0.74468)	(0.17021, 0.42553)	(0.46809, 0.73404)	(0.57447, 0.80851)	(0.03191, 0.21277)
	SmolLM 360M	(0.53165, 0.76596)	(0.12766, 0.37234)	(0.43617, 0.77660)	(0.64894, 0.88298)	(0.02128, 0.18085)
GB Swap Word with Synonyms Perturbation	SmolLM 1.7B	(0.54255, 0.80851)	(0.04255, 0.19149)	(0.44681, 0.78723)	(0.73404, 0.91489)	(0.00000, 0.08511)
	SmolLM2 135M	(0.22340, 0.51064)	(0.62766, 0.86170)	(0.38298, 0.69149)	(0.46809, 0.73404)	(0.21277, 0.47872)
	SmolLM2 360M	(0.14894, 0.41516)	(0.39362, 0.74468)	(0.43617, 0.75532)	(0.50000, 0.72340)	(0.19149, 0.45745)
	SmolLM2 1.7B	(0.43617, 0.71277)	(0.24468, 0.47872)	(0.48936, 0.78723)	(0.67021, 0.88298)	(0.10638, 0.28723)
	GPT Neo 2.7B	(0.05319, 0.27660)	(0.26596, 0.64894)	(0.34043, 0.63830)	(0.41489, 0.64894)	(0.29761, 0.70239)
	GPTJ 6B	(0.00000, 0.17021)	(0.52128, 0.77660)	(0.35106, 0.67021)	(0.45745, 0.70213)	(0.53191, 0.80851)
	Gemma2 2B	(0.55319, 0.82979)	(0.82979, 0.94681)	(0.74441, 0.91489)	(0.72314, 0.89362)	(0.59574, 0.81915)
	Gemma2 9B	(0.87234, 0.98936)	(0.72340, 0.91489)	(0.55293, 0.88298)	(0.81915, 0.96809)	(0.47872, 0.81915)
	Llama3 8B	(0.41489, 0.75532)	(0.39362, 0.65957)	(0.78723, 0.93617)	(0.76596, 0.92553)	(0.06383, 0.25532)
	Falcon 7B	(0.23404, 0.46809)	(0.90426, 0.98963)	(0.67021, 0.86197)	(0.59574, 0.80851)	(0.74468, 0.91489)
GB Swap Word with Synonyms Perturbation	SmolLM 135M	(0.77660, 0.92580)	(0.31915, 0.60665)	(0.68059, 0.87234)	(0.70213, 0.90426)	(0.03191, 0.17021)
	SmolLM 360M	(0.77660, 0.92553)	(0.20213, 0.45771)	(0.68085, 0.88298)	(0.78723, 0.93644)	(0.02128, 0.14894)
	SmolLM 1.7B	(0.70213, 0.94681)	(0.07447, 0.24468)	(0.68059, 0.89362)	(0.77660, 0.96809)	(0.00000, 0.05319)
	SmolLM2 135M	(0.57447, 0.81915)	(0.71277, 0.94681)	(0.63830, 0.81915)	(0.60612, 0.87261)	(0.37234, 0.59574)
	SmolLM2 360M	(0.55319, 0.76596)	(0.69122, 0.88298)	(0.67021, 0.87234)	(0.69149, 0.89362)	(0.29787, 0.52128)
	SmolLM2 1.7B	(0.76596, 0.91489)	(0.32952, 0.60638)	(0.68085, 0.89362)	(0.74468, 0.90426)	(0.08511, 0.27660)
	GPT Neo 2.7B	(0.21277, 0.40426)	(0.73404, 0.92553)	(0.39362, 0.76596)	(0.57447, 0.79787)	(0.74468, 0.92553)
	GPTJ 6B	(0.14894, 0.34043)	(0.75532, 0.91489)	(0.47872, 0.77660)	(0.60638, 0.82979)	(0.74441, 0.90426)